Clinic Management System (Patient Account and Healthcare Management)

By

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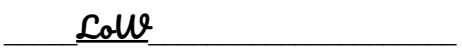
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Declaration

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Abstract

This is a **one-page** summary of the project (without any subheading) of usually not more than **300 word**s. It is often used to help the reader to quickly ascertain the documentation's purpose. Like all summaries, abstract covers the main points of a piece of writing that includes the field of study, problem statement, methodology adopted, research process, conclusion and planning of the project work, etc. It helps readers understand the project by acting as a pre-reading outline of key points.

The contents that you may include in an abstract are as follows:

***Purpose**:* Describes the main purpose of carrying the project, i.e. to justify the existence of the project by stating the existing problem to be solved. Explain why it is necessary to solve it and how the project contributes to the solution.

***Scope**:* This covers areas or size of the project in terms of function and features of the system, modules or sub-modules involved, and functional areas covered in an organization.

***Methodology**:* To describe the methodology, tools, techniques and models that are used throughout the project.

***Testing criteria used**:* To describe the various assessment areas this project has undergone.

***Results and Conclusion**:* To brief the outcome, strengths and weakness of the system or the entire project.

Note: Your abstract SHOULD NOT have bullet points or headings according to the contents listed above. Refer to abstracts of research papers for examples of abstracts.

Acknowledgement

This contains acknowledgement to those who have contributed directly or indirectly to the completion of the project. Usually the people to be acknowledged include the project supervisor(s), moderators, family, and those who have given assistance and supports to ensure the success of the project.

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Chapter 1

Introduction

1.0 Introduction

The focus of this chapter is to provide patients, parents or guardians and guests with convenient access to clinic services by developing the Clinic System Patient Account and Healthcare Management, including an Introduction to the project background, the project objectives; the advantages and contributions of the proposed system, a Project Development Plan, Project Scope, Software and Development Tools, Project Organization. The importance of a central user-side system that provides the following functions will also be explained such as appointment, medical record, billing and payment, notification reminders, personal health record, parent or guardian and AI health chatbot.

This chapter also defines the project scope in order to clarify which functions are in and out of the system. This avoids the development of the Clinic System Patient Account and Healthcare Management to extend too far into doctor side or full administrative management functions.

1.1 Project Objective(s)

The Clinic System Patient Account and Healthcare Management will be designed as a web-based system of patients' clinic services. The system facilitates equipment with web-based features such as booking appointments, accessing medical records and billing, receiving reminders, updating personal health records, and accessing basic health guidance through an AI health chatbot.

The system also includes a service for parents or guardians to manage child patients under one main account, facilitating parents or guardians to manage appointments with child patients and access medical records and other notifications regarding their children's clinic services.

The main purpose of this project is to reduce reliance on manual clinic service methods such as telephone calls, face-to-face visits, and paper tracking. In addition, it aims to enhance user convenience by providing patients and parents or guardians access to clinic services through one online system concatenated on a single platform.

1.2 Project Background

The healthcare services are essential because patients require appropriate access to the consultation, treatment, medicine records, billing and follow-up services. But many clinic services still rely on the manual approaches which include phone calls, in person visits and paper-based records. Patients are required to directly contact the clinic to book an appointment, inquire about billing or check their medical records, etc. This would be inconvenient for patients requiring fast services or more comfortable patients.

Furthermore, patients may sometimes forget about their appointed dates or about the updating of their payments if there is no suitable reminder system. Inconvenient access to medical records may also lead to patients failing to understand about their previous diagnosis, treatment and also medication. This may lower the involvement of patients in the safeguarding of their health information.

For parents or guardians, it can be more difficult to keep track of healthcare information if they must maintain appointments, medical records, and treatment information for more than one child. In the absence of a common interface, this can be confusing and difficult to administer.

With the advent of web-based systems and increasing digital healthcare services, clinic management can be enhanced by making these services available online to the patients. A web-based clinic system enables users to manage appointments, view medical records, billings and payments, notifications, and personal health record updates in a more organized form, and at the same time reduces the patients' over-reliance on phone calls and physical visits.

Therefore, this project proposes a Clinic System Patient Account and Healthcare Management, which mainly improves the experience of patients. The system can implement appointment management, medical record viewing, billing and payment management, notification reminder, parent or guardian management, personal health record management, and AI health chat support. With this system, users can use clinic services more conveniently and accumulate healthcare information on this platform.

1.3 Advantage & Contributions

For Patients:

- The system offers patients a more convenient way to get clinic service, patients can book and manage their appointment online by themselves without calling the clinic or physically visiting the clinic. This saves time, and also eases the inconvenience of booking an appointment manually.
- The system also enables patients to view their medical records, treatment history, diagnosis and prescribed medicine. Thus it would assist patients in knowing their health condition and past visits to the clinic. In addition, with the billing and payment view function, patients can easily check invoice details and payment history.
- With the notification and reminder function, patients can be reminded of upcoming appointments, payment updates and clinic announcements. Also, patients can consult the AI health chatbot to get basic health guidance when they need basic health information.

For Parents / Guardians:

- Parents or guardians of children can use the parent or guardian management function to manage child patient information in one account at a time, which is helpful because young children can’t manage their own health care information by themselves.
- Using the system, parents or guardians can manage child appointments, check child medical records, and receive reminders on child health care services so they can keep track of and manage child patient information more efficiently.

For Clinics:

- The suggested system might help clinics to reduce the manual work related to entries of appointment, follow-up of billing inquiries and communication with patients. The online availability of information would save time for the clinic staff to answer repeated questions or to find information from the file manually.
- The notification and the reminder feature would also help avoiding missed appointments, as patients receive the notification beforehand. In general, the system would facilitate the communication between the clinic and patients.

For Guests / New Users:

- Guests or new users can sign in using the system to create an account and have access to general clinic-related information. They can also use the AI health chatbot to get basic health guidance. This allows new users to get familiar with the services available before they become registered patients.

1.4 Project Plan

The project plan describes the scope, modules, and development schedule of the Clinic System Patient Account and Healthcare Management. The system will be developed based on the user-side requirements identified in the proposal. The main focus is to provide patients, parents or guardians, and guests with convenient access to clinic services through a web-based platform.

The project will be divided into several phases, including requirement analysis, system design, implementation, testing, documentation, and final submission. Each phase will help ensure that the system is developed in an organized and manageable way.

1.4.1 Project Scope

No.	Module/Function	Included in Scope	Description
1	User Account Management	Yes	The system allows patients, parents or guardians, and guests to register, log in, log out, and manage their basic profile information.
2	Appointment Management	Yes	Patients can book, view, reschedule, and cancel clinic appointments through the system.
3	Medical Record Viewing	Yes	Patients can view their consultation history, diagnosis details, treatment information, and prescribed medicine.
4	Billing and Payment Viewing	Yes	Patients can view invoices, billing details, payment status, and payment history.
5	Personal Health Record Management	Yes	Patients can update and manage their own health-related information, such as symptoms, allergies, and basic health notes.
6	Parent / Guardian Management	Yes	Parents or guardians can manage child patient accounts under one main account.
7	Notification and Reminder Management	Yes	The system provides reminders and notifications for appointments, payments, clinic announcements, and important updates.
8	AI Health Chatbot	Yes	Users can ask basic health-related questions and receive general guidance from the AI chatbot.
9	Guest Access	Yes	Guests can register an account and access limited general information or basic chatbot support.
10	Medicine Purchase Management	Yes	Patients can view prescribed medicines, purchase them through the system, and review their medicine purchase history and order status.
11	Doctor-Side Management	No	Full doctor-side functions such as diagnosis entry, treatment record creation, and doctor schedule management are not the main focus of this project.
12	Full Admin Management	No	Complete administrative functions such as staff management, full clinic operation control, and advanced report generation are not included in the main scope.
13	Inventory Management	No	Medicine stock, clinic equipment, and inventory control are not included in this project.
14	Advanced Financial Reporting	No	The system does not include full accounting reports or detailed financial analysis for clinic management.

1.4.2 Software and Development

The Clinic System Patient Account and Healthcare Management will be developed using the following technologies, tools, and development method:

Category	Technology / Tools	Description
Frontend	React.js	Used to develop an interactive and responsive user interface for patients. It allows users to access clinic services such as appointment management, medical records, billing information, notifications, personal health records, child profile management, and the AI health chatbot.
Frontend Styling	Tailwind CSS	Used to design and style the web pages efficiently and consistently across different screen sizes.
Backend	ASP.NET Core Web API	Used to handle system logic, process data, manage user authentication, provide RESTful API services, and connect the frontend application to the database.
Database	MongoDB Atlas	Used to store user profiles, appointment records, medical records, billing information, personal health records, child patient profiles, notifications, and chatbot-related data in a flexible document-based format.
Development Method	Incremental Development Model	The system is developed module by module, allowing each module to be planned, implemented, tested, evaluated, and improved before the next module is developed.

1.4.3 Incremental Development Increments

The Clinic Management System will be developed incrementally. Each increment focuses on implementing and testing a specific patient-side module before all modules are integrated into the completed system.

Activity / Increment	Module	Expected Outcome	Estimated Duration
Increment 1	Account Management	Patients can register, log in, recover their passwords, and update their profiles.	12 days
Increment 2	Appointment Management	Patients can book, reschedule, cancel, and view the history of their clinic appointments.	12 days
Increment 3	Medical Record Viewing	Patients can view their consultation history, diagnoses, treatment summaries, and prescribed medicines.	12 days
Increment 4	Billing and Payment Management	Patients can view invoices, check their payment history, and perform online payment transactions.	12 days
Increment 5	Notification Management	Patients receive appointment reminders, payment notifications, and clinic announcements.	10 days
Increment 6	Child Profile Management	Adult patients can submit child profiles and supporting information for administrator approval. Once approved, the child profiles can be managed under the adult patient's account.	14 days
Increment 7	Personal Health Record Management	Patients can add, update, and track personal health information, such as blood pressure, allergies, existing conditions, and medical history.	10 days
Increment 8	Medicine Purchase Management	Patients can view prescribed medicines, purchase them securely through the system, and review their medicine purchase history and order status.	7 days

Increment 9	AI Health Chatbot	The AI chatbot provides basic health guidance based on the patient's input and available personal health information.	20 days
Final	System Integration and Testing	All patient-side modules are integrated and tested to verify their functionality, usability, reliability, and connection with the admin-side module.	7 days

1.5 Project Team & Organization

Low Jwen Siong is responsible for the Patient Account and Healthcare Management of the Clinic Management System. This includes appointment management, medical record viewing, billing and payment information, notifications, personal health records, child profile management, and AI health chatbot integration. He is also responsible for the planning, requirement analysis, system design, implementation, testing, and documentation of these functions.

Ching Yong Le is responsible for the Doctor, pharmacist and administrative management of the Clinic Management System. This module supports the clinic's administrative operations and complements the patient-side functions developed by Low Jwen Siong.

The project supervisor, Mr Heng Jooi Huang, provides guidance and feedback throughout the project development process. His role is to review the project's progress, provide advice, and ensure that the project meets the required academic and technical standards.

This project organization allows the patient-side and admin-side modules to be developed in parallel while ensuring that both modules remain aligned with the overall project objectives, requirements, and scope.

1.6 Chapter Summary & Evaluation

This chapter introduced the Clinic System Patient Account and Healthcare Management and explained the purpose of developing the system. The chapter discussed the project objective, project background, advantages and contributions, project plan, project scope, module descriptions, task list and Gantt chart, project team and organization.

The proposed system intends to increase patient access to clinics with an easy to use centralized web-based system. This system allows patients to check in for appointments, view medical records, query billing and payment information, receive reminders, and has the ability to upload personal health records. The system includes a simple AI health chatbot to give basic health guidance. The system allows parents or guardians to manage their child patients' healthcare information under one account.

In summary, the project is expected to improve the convenience of users and clinics, organize and coordinate information between the two, and enhance communication. The project plan is limited to the user-side module, but the doctor-side and full administrative management functions are not the focus of this project. The next chapter will present the literature review, related system to support this project.

Clinic Management System (Patient Account and Healthcare Management)

Chapter 2

Literature Review

2.0 Literature Review

This chapter discusses the background of the proposed Clinic System: User Side Module and the related studies connected to patient-side clinic services. The review focuses on the main areas covered by the proposed system, including appointment management, medical record viewing, billing and payment information, notification reminders, parent or guardian healthcare management, personal health records, and AI health chatbot support.

2.1 Nature of Business

The proposed Clinic System: User Side Module is a web-based application that provides patient-side clinic services to patients, parents or guardians, and guests. The goal of this application is to help users manage clinic-related activities such as appointment booking, medical record viewing, billing and payment checking, notification reminders, personal health record updates, child patient management, and basic health enquiries through an AI health chatbot.

The current clinic service process may require patients to contact the clinic through phone calls, visit the clinic physically, or depend on staff to check appointment, billing, and medical record information. This can be inconvenient for patients, especially when they need to manage more than one healthcare service.

2.2 Existing Clinic Management Systems

1. **BookDoc (BookDoc, 2025)**

   Established in 2015, BookDoc is a digital healthcare platform that links patients with healthcare professionals through its website and mobile application. Users can locate healthcare providers, arrange medical appointments, attend virtual consultations, and access various health and wellness services. The platform uses digital technology to make healthcare services easier to access while supporting users in maintaining healthier lifestyles.

2. **Doctor Anywhere (Doctor Anywhere Malaysia, n.d.)**

   Doctor Anywhere is a digital healthcare application that brings different medical and wellness services together on one platform. Its services include appointment booking, video consultations with doctors, medical claim management, and access to healthcare providers. By offering these services online, the platform allows users to obtain healthcare support without depending entirely on physical clinic visits.

3. **DoctorOnCall (DoctorOnCall Malaysia, n.d.)**

   DoctorOnCall is a Malaysian digital healthcare platform that offers patients access to various medical services online. Users can consult doctors remotely, arrange appointments, obtain electronic prescriptions, purchase medicine through an online pharmacy, and request medicine delivery. The platform also provides services such as health screenings and vaccinations, making healthcare services more accessible to patients through a centralized digital platform.

Table 2.2.1: Comparison between existing healthcare systems and proposed system

Features	BookDoc	Doctor Anywhere	DoctorOnCall	Clinic System: User Side Module (the proposed system)
Appointment Management	- Allows users to search and book healthcare professionals. -Focuses on connecting users with healthcare providers.	- Focuses mainly on online doctor consultation. - Users can access doctors through the app.	- Focuses mainly on online doctor consultation and pharmacy services. - Appointment booking with physical clinics is not the main focus.	- Allows patients to book, reschedule, cancel, and view appointment history. - Focuses on clinic appointment management for registered patients.
Medical Record Viewing	- Provides health management features, but clinic consultation record viewing is not the main focus.	- Allows users to access medical history, records, and medical documents in the app.	-Provides consultation and prescription-related services. - Full clinic medical record viewing is not the main focus.	- Allows patients to view consultation history, diagnosis details, treatment records, and prescribed medicine.
Billing and Payment	- Provides marketplace and health-related services. - Clinic billing history is not the main focus.	- Supports consultation and medication-related payment. - Full clinic invoice tracking may depend on service type.	- Supports medicine orders and online pharmacy payment. - Clinic invoice history is not the main focus.	- Allows patients to view invoices, check payment history, and make online payment for clinic services.
Notification and Reminder	- Provides health and wellness app functions. - Reminder functions are more related to health and activity tracking.	- Provides follow-up support after consultation. - Reminder coverage depends on the service used.	- Provides consultation and medication order updates. - Appointment and clinic announcement reminders are not the main focus.	- Sends appointment reminders, payment notifications, and clinic announcements to patients.

Parent / Guardian Management	- Does not clearly focus on guardian-based child patient account management.	- Provides family health services, but child record management under one clinic account is not the main focus.	- Allows users to consult doctors, but guardian management is not the main focus.	- Allows parents or guardians to manage child patient appointments, prescriptions, and medical records under one account.
Personal Health Record	- Provides health management and health metric tracking features.	- Allows users to access medical history and health documents.	- Focuses more on consultation, prescription, and medicine delivery.	- Allows patients to update personal health records such as blood pressure, medical history, allergies, and health notes.
Medicine Purchase Management	- Does not provide a centralized clinic medicine purchasing service after physical consultations.	- Supports medicine delivery after online consultation.	- Provides online pharmacy services and medicine delivery for eligible prescriptions.	- Allows patients to view prescribed medicines, purchase them securely through the system, and review their medicine purchase history and order status.
AI Health Chatbot	- Does not clearly provide an AI health chatbot as a main feature.	- Provides online doctor consultation instead of chatbot-based basic guidance.	- Provides doctor consultation and pharmacy support instead of chatbot-based basic guidance.	- Provides an AI health chatbot for basic health guidance based on user input and personal health information.
Main Focus	- Healthcare provider search, appointment booking, wellness tracking, and health management.	- Online doctor consultation, medical history access, claims, and medication delivery.	- Online consultation, prescription, online pharmacy, and medicine delivery.	- Integrated patient-side clinic service, including appointments, records, billing, reminders, guardian management, personal health records, and AI chatbot support.

2.3 Target User

The main target users for this proposed web application would be patients, parents or guardians, guests, and clinic staff who manage patient-side clinic services. The users must have an active Internet connection and a digital device such as a smartphone, laptop, tablet, or desktop with browser accessibility to use the proposed Clinic System: User Side Module.

Patients are one of the main users because they will use the system regularly to manage their clinic-related activities. Patients can register an account, book appointments, reschedule or cancel appointments, view appointment history, check medical records, view prescribed medicine, check billing information, make online payments, receive clinic notifications, update personal health records, and use the AI health chatbot for basic health enquiries. This helps patients manage their healthcare information in a more organized way.

Parents or guardians are also target users of the system because they may need to manage healthcare services for children under their care. Through the parent or guardian account, they can manage child patient information, book appointments for child patients, view child medical records, check prescriptions, and receive notifications related to the child's healthcare. This is useful because child patients may not be able to manage their own clinic account independently.

Guests are users who have not registered an account in the system. Guests can view public clinic announcements and register as patients if they want to use the clinic services. They may also use limited AI health chatbot functions to ask general health-related questions. However, guests will not be able to access private functions such as appointment history, medical records, billing information, or personal health records until they register and log in.

Clinic staff are included as supporting users because some patient-side services still depend on clinic-side data. Clinic staff may update appointment availability, consultation records, prescriptions, billing information, payment status, and clinic announcements. Although this project focuses on the user-side module, clinic staff support is still needed to make sure the information shown to patients is accurate and updated.

2.4 Literature Review

In today's digital age, healthcare services have started to move from fully manual processes to online platforms. Patients are no longer limited to phone calls or physical visits when they need to book appointments, check health information, or communicate with healthcare providers. Digital health systems can support patient access, improve communication, and help healthcare providers manage services more efficiently. According to the World Health Organization (2021), digital health can strengthen healthcare delivery when it is designed with proper safety, privacy, and ethical considerations. This shows that a web-based clinic system can help improve the way patients access clinic services.

Patient portal systems: Improving access to healthcare services

Patient portal systems allow patients to access healthcare information and services through online platforms. These systems normally support functions such as viewing medical records, managing appointments, checking health information, and communicating with healthcare providers. Tavares and Oliveira (2016) explained that electronic health record patient portals help healthcare consumers access health information and services online. This is related to the proposed Clinic System: User Side Module because the system allows patients to manage clinic services such as appointments, medical records, billing, reminders, personal health records, and AI health chatbot support through one web-based platform.

Appointment management: Reducing scheduling problems

Appointment management is one of the main functions in a clinic system. In many clinics, patients still need to call the clinic or visit physically to book, reschedule, or cancel appointments. This process can become inconvenient when patients forget appointment dates, need to change their schedule, or cannot contact the clinic during busy hours. Dixit and Dixit (2024) stated that patient scheduling problems include missed appointments, difficulty in rescheduling, and poor appointment tracking. Therefore, the proposed system includes an appointment management module that allows patients to book appointments, reschedule appointments, cancel appointments, and view appointment history online.

Medical record access: Helping patients understand their treatment

Medical record access is important because patients need to understand their consultation history, diagnosis, treatment details, and prescribed medicine. When patients cannot access their own medical records easily, they may forget medical instructions or have difficulty tracking their treatment progress. The Asthma & Allergy Foundation of America (2025) stated that access to medical records helps patients understand and manage their own health information. In the proposed system, the medical record viewing module allows patients to view consultation records, diagnosis details, treatment summaries, and prescribed medicine after their clinic visits.

Billing and payment management: Making payment information clearer

Billing and payment management is also important in patient-side clinic services. In a manual clinic process, patients may need to ask clinic staff about invoice amounts, payment status, or payment history. This can cause confusion if the information is not organized in one place. The proposed system allows patients to view invoices, check payment history, and make online payments. This helps patients understand their payment information more clearly and reduces dependency on manual billing checks.

Health reminders and notifications: Supporting patient follow-up

Health reminders and notifications help patients remember upcoming appointments, payment updates, and clinic announcements. Opon, Tenambergen, and Njoroge (2020) found that patient reminders can help reduce missed appointments in medical settings. O'Leary, Tanghe, Pratt, and Ralston (2018) also explained that health reminders and notifications can support patient coordination and follow-up actions. In the proposed system, notification reminders are used to alert patients about appointments, payments, and clinic updates so they can manage their healthcare activities more effectively.

Parent or guardian management: Supporting child patient care

Parent or guardian management is needed because children may not be able to manage their own healthcare services independently. Parents or guardians often need to book appointments, check prescriptions, view medical records, and follow up on treatment for children under their care. Cady and Belew (2017) explained that parents play an important role in coordinating healthcare services for children. Therefore, the proposed system includes a parent or guardian management module that allows parents or guardians to manage child patient information, appointments, prescriptions, and medical records under one account.

Personal health record management: Tracking health information outside clinic visits

Personal health record management allows patients to record and update their own health information, such as blood pressure, allergies, medical history, and other health notes. This function is useful because healthcare management does not only happen during clinic visits. Patients may need to monitor their own health condition over time. In the proposed system, the personal health record module allows users to update their health information. This can also support the AI health chatbot by giving more context to the user's basic health enquiry.

Medicine purchase management: Improving access to prescribed medicines

Medicine purchase management helps patients obtain their prescribed medicines conveniently after a consultation. In many clinics, patients are required to visit the pharmacy physically or contact the clinic separately to purchase prescribed medicines. This process can be inconvenient for patients who have mobility limitations, busy schedules, or live far from the clinic. Integrating medicine purchasing into a patient portal allows patients to review their prescribed medicines, place orders, and monitor their purchase history through a single platform. Saluja et al. (2024) explained that e-pharmacy services can improve access to medicines and make medication-related services more convenient for patients. Therefore, the proposed system includes a Medicine Purchase Management module that allows patients to view prescribed medicines, purchase them securely online, and review their medicine purchase history and order status.

AI health chatbot: Providing basic health guidance

AI health chatbot support is included to provide basic health guidance to users. The chatbot can respond to general health questions based on user input and personal health information. However, it should not replace doctors or professional medical consultation. The World Health Organization (2021) stated that artificial intelligence in healthcare must be used carefully because it involves patient safety, privacy, and ethical concerns. Because of this, the AI health chatbot in the proposed system should only provide basic guidance and should advise users to seek professional medical help when symptoms are serious or unclear.

Existing healthcare platforms: BookDoc, Doctor Anywhere, and DoctorOnCall

There are several healthcare platforms available in the market, such as BookDoc, Doctor Anywhere, and DoctorOnCall. BookDoc provides healthcare provider search and appointment-related services for users who want to connect with healthcare professionals (BookDoc, 2026). Doctor Anywhere provides online doctor consultation, medication delivery, and related digital healthcare services (Doctor Anywhere Malaysia, 2026). DoctorOnCall provides online doctor consultation, prescription, and online pharmacy services (DoctorOnCall Malaysia, 2026). These systems show that digital healthcare platforms are already used in the market.

However, these systems mainly focus on online consultation, healthcare provider search, medicine delivery, or pharmacy services. The proposed Clinic System: User Side Module focuses more on patient-side clinic service management. It combines appointment management, medical record viewing, billing and payment checking, notification reminders, parent or guardian management, personal health records, and AI health chatbot support into one clinic-based system.

2.5 Chapter Summary and Evaluation

Chapter 2 focuses on the proposed **Clinic System: User Side Module**, a web-based platform that provides patient-side clinic services. The system enables patients to manage appointments, view medical records, check billing and payment history, receive notifications, update personal health records, manage child patient information, and obtain basic health guidance through an AI health chatbot.

The chapter also identifies the target users of the system, including patients, parents or guardians, guests, and clinic staff. Patients and guardians manage healthcare services through the system, guests can access public information and register for an account, while clinic staff maintain appointment schedules, consultation records, prescriptions, billing information, and clinic announcements.

The literature review examines digital healthcare and patient portal systems, highlighting key functions such as appointment management, medical record access, billing and payment, health reminders, guardian management, personal health records, and AI chatbot support. These studies show that digital healthcare systems improve accessibility, communication, and patient convenience.

Finally, the chapter compares existing healthcare systems, including BookDoc, Doctor Anywhere, and DoctorOnCall, with the proposed system. While these platforms focus on services such as healthcare provider search, online consultation, and medication delivery, the proposed system provides a more comprehensive clinic-based patient management solution by integrating appointment management, medical records, billing, notifications, guardian management, personal health records, and AI chatbot support.

Overall, Chapter 2 establishes the foundation of the proposed Clinic System: User Side Module by discussing its target users, related literature and system comparison, providing a clear basis for the system's design and development.

Clinic Management System (Patient Account and Healthcare Management)

Chapter 3

Methodology and Requirements Analysis

3.0 Methodology and Requirements Analysis

This chapter explains the methodology and requirements analysis for the proposed Clinic Management System (Patient Account and Healthcare Management). The system focuses on patient-side clinic services. It is intended to help patients, parents or guardians, and guests access clinic-related functions through one web-based platform.

The chapter describes the Incremental Development Model selected for the project, the methods used to gather information, the analysis of the questionnaire responses, the project scope, and the functional and non-functional requirements. The requirements in this chapter provide the basis for the system design, development, and testing in the following chapters.

3.1 Methodology

The development of the Clinic Management System (Patient Account and Healthcare Management) adopts the Incremental Development Model. This software development methodology divides the project into several smaller functional increments. Each increment is planned, designed, developed, tested, reviewed, and integrated before the next increment begins. By dividing the project into manageable modules, the development team can verify the correctness of each module before integrating it into the complete system.

The proposed system contains several independent but related modules, including Account Management, Appointment Management, Medical Record Viewing, Billing and Payment Management, Notification Management, Parent or Guardian Management, Personal Health Record Management, Medicine Purchase Management, AI Health Chatbot, and Guest Access. Since each module performs different responsibilities while sharing common patient information, the Incremental Development Model is suitable because each module can be completed and evaluated separately before system integration.

The incremental approach also reduces development risk because problems can be detected during the development of each module rather than after the entire system has been completed. Continuous testing and evaluation throughout every increment help improve software quality while allowing future enhancements to be incorporated without affecting completed modules significantly.

Current Challenges:

1. Patients still depend on telephone calls or physical visits to book, reschedule, or cancel appointments.
2. Consultation records, diagnosis information, prescribed medicines, and treatment history are not easily accessible through one centralized platform.
3. Billing information, invoices, payment history, and payment status are often maintained separately, making it difficult for patients to monitor their financial records.
4. Parents or guardians who manage healthcare services for multiple children cannot conveniently access all child healthcare information under one account.
5. Appointment reminders and clinic notifications are limited, increasing the possibility of patients missing appointments or important clinic announcements.
6. Patients have limited access to basic healthcare guidance outside clinic operating hours, especially for general health enquiries that do not require immediate professional consultation.

To overcome these challenges, the proposed Clinic Management System introduces several integrated patient-side modules that improve accessibility, convenience, communication, and healthcare information management.

Proposed Clinic Management Features:

1. **Appointment management**

Current	Proposed Clinic Management System
Patients contact the clinic by telephone or visit the clinic to make appointments.	Patients can book appointments online through the web-based system.
Appointment cancellation requires contacting clinic staff.	Patients can cancel appointments through their own account.
Appointment rescheduling depends on manual communication.	Patients can reschedule appointments online based on available time slots.
Appointment history is difficult to track.	Patients can view complete appointment history anytime.

2. **Medical Record Viewing**

Current	Proposed Clinic Management System
Medical records are maintained by the clinic and are not easily accessible by patients.	Patients can view consultation history through one account.
Diagnosis and treatment information are scattered across different visits.	Diagnosis details and treatment summaries are displayed in one centralized location.
Patients may forget prescribed medicines after consultation.	Prescribed medicines are available for future reference through the system.

3. Billing and Payment Management

Current	Proposed Clinic Management System
Patients need to enquire about billing information manually.	Patients can view invoices and billing information online.
Payment history is difficult to monitor.	Complete payment history is available through the patient account.
Payment status requires confirmation from clinic staff.	Payment status is updated within the system.
Online payment may not be available.	Patients can perform online payment through the system.

4. Parent or Guardian Management

Current	Proposed Clinic Management System
Parents manage children's healthcare information separately.	Parents or guardians manage multiple child profiles under one account.
Child appointments require separate communication.	Parents can manage child appointments through the same account.
Child medical records are difficult to organize.	Medical records for approved child profiles are accessible through the parent account.

5. **Notification Management**

Current	Proposed Clinic Management System
Patients depend on manual reminders.	Automated appointment reminders are provided.
Clinic announcements are communicated through separate channels.	Clinic announcements are centralized within the system.
Payment updates are communicated manually.	Payment notifications are delivered automatically.

6. **Personal Health Record Management**

Current	Proposed Clinic Management System
Patients keep personal health information separately.	Personal health information is stored securely within the system.
Health history is difficult to monitor over time.	Patients can update and review their health records continuously.
Healthcare information is not linked with other patient services.	Personal health records support healthcare management and AI health guidance.

7. **Medicine Purchase Management**

Current	Proposed Clinic Management System
Patients visit the clinic or pharmacy to obtain prescribed medicines.	Patients can purchase prescribed medicines through the system.
Medicine purchase history is not centralized.	Medicine orders and purchase history are available online.
Order progress is difficult to monitor.	Patients can track medicine order status through the system.

8. **AI Health Chatbot**

Current	Proposed Clinic Management System
Patients have limited healthcare guidance outside clinic operating hours.	AI Health Chatbot provides basic healthcare guidance at any time.
Patients search healthcare information from different sources.	Users receive basic healthcare guidance through one platform.
General enquiries increase clinic workload.	The chatbot answers basic health-related questions while directing users to seek professional consultation when necessary.

3.1.1 Fact Gathering

The requirements for the proposed system were gathered through project proposal review, literature review, existing system comparison, observation of the current workflow, user-role analysis, stakeholder interviews, and a questionnaire. These methods were selected based on the available project information and the need to understand both the existing clinic service problems and the users' preferred system functions.

1. Project Proposal Review

The project proposal was reviewed to identify the main problem, proposed solution, target users, system modules, development model, and project contribution. The proposal identifies the need for one platform where patients can manage appointments, view medical records, check billing information, receive reminders, update personal health records, purchase prescribed medicine, and receive basic health guidance through an AI health chatbot.

The proposal also identifies parent or guardian management as an important function. Parents or guardians need one account to manage the healthcare information of children under their care. This includes child appointments, medical records, prescriptions, and notifications.

2. Existing system comparison

Several existing healthcare platforms were analysed to understand their strengths, limitations, and supported features. The systems selected for comparison include **BookDoc**, **Doctor Anywhere**, and **DoctorOnCall**, as these platforms are widely used for digital healthcare services in Malaysia.

The comparison focused on features related to patient-side healthcare management, including appointment booking, medical record accessibility, billing services, notification functions, guardian management, medicine purchasing, and AI healthcare support.

The comparison revealed that these existing platforms mainly focus on online consultations, healthcare provider searching, online pharmacy services, and telemedicine. Although they provide valuable digital healthcare services, they do not fully integrate patient-side clinic management functions into a single platform.

Based on the comparison, the proposed Clinic Management System introduces a more comprehensive patient management solution by integrating appointment management, medical record viewing, billing and payment management, notification services, parent or guardian management, personal health record management, medicine purchase management, and AI Health Chatbot support within one centralized web application.

The findings from the comparison also helped identify features that could improve user convenience while avoiding unnecessary duplication of functions already available in commercial healthcare platforms.

3. Interviews

The proposed Clinic Management System was developed based on the perspective of Pee Yong Qin, who represented a potential patient during the stakeholder interview. He explained that the current appointment booking process can be inconvenient because patients often need to contact the clinic by phone or visit the clinic in person to enquire about available appointment slots. He also mentioned that rescheduling or cancelling appointments requires repeated communication with clinic staff, while accessing medical records, billing information, and payment history can be difficult because the information is stored across different processes. In addition, he suggested that the system should provide appointment reminders and clinic notifications to help patients avoid missing appointments and stay informed about important updates.

Pee Yong Qin also recommended that parents or guardians should be able to manage their children's healthcare information under a single account, making it easier to handle appointments, medical records, prescriptions, and notifications. He further suggested including a Personal Health Record module for recording allergies, chronic illnesses, blood pressure, and medical history, as well as a medicine purchase function to allow patients to order prescribed medicines online and track their order status. Finally, he supported the inclusion of an AI Health Chatbot to provide general health guidance outside clinic operating hours while reminding users that professional medical consultation should still be sought when necessary. These recommendations support the development of an integrated patient-centred platform that combines appointment management, medical record viewing, billing and payment management, notifications, parent or guardian management, personal health records, medicine purchase management, and AI health chatbot support.

4. Observation

The current patient workflow was analysed based on the problems described in the project proposal and project background. Patients may need to contact a clinic by telephone, visit the clinic physically, ask staff about billing information, request access to medical records, and visit a pharmacy separately after receiving a prescription.

This workflow separates related information across different processes. Patients may also forget appointments when the clinic does not provide suitable reminders. Parents or guardians may face additional difficulty when they need to manage appointments and medical information for more than one child.

5. User role analysis

The activities of patients, parents or guardians, guests, and supporting clinic staff were identified. Patients manage their own healthcare information. Parents or guardians manage their own information and the information of approved child profiles. Guests can view permitted public information and register for an account. Clinic staff provide supporting data such as appointment availability, consultation records, prescriptions, billing information, payment status, and clinic announcements.

6. Questionnaire

A questionnaire survey was conducted to collect quantitative data from potential users regarding their experiences with current clinic services and their expectations for the proposed Clinic Management System. The questionnaire consisted of multiple-choice questions and Likert scale questions that evaluated user opinions on appointment management, medical record accessibility, billing services, notification functions, guardian management, medicine purchasing, and AI healthcare guidance.

The questionnaire was distributed online using **Google Forms**, allowing respondents to participate conveniently through computers and mobile devices. Online distribution also enabled faster data collection and automatic generation of statistical charts for subsequent analysis.

The respondents consisted primarily of university students, working adults, and individuals who had previously visited private healthcare clinics. Their responses provided valuable insights into user expectations and common difficulties encountered during clinic service management.

3.1.2 Fact Analysis

The information gathered from the project documents, literature review, existing system comparison, observation, user-role analysis, stakeholder interviews, and questionnaire was analysed to identify the requirements of the proposed system. The questionnaire findings are presented below.

Question 1: User group of respondents

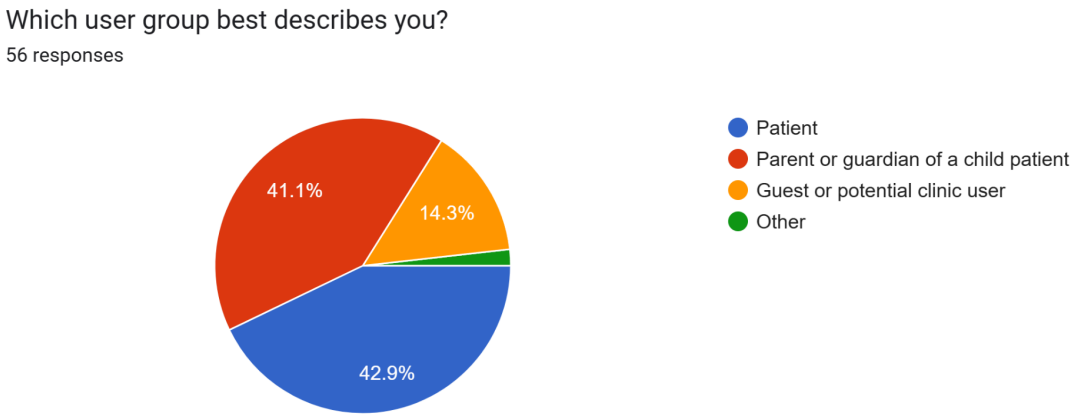


Figure 3.1.1: User group of respondents.

The questionnaire received 56 valid responses. The largest group consisted of patients, with 24 respondents or 42.9%, followed by parents or guardians of child patients, with 23 respondents or 41.1%. Eight respondents or 14.3% were guests or potential clinic users, while one respondent or 1.8% selected Other. The response distribution provides feedback from both direct patients and caregivers, which is appropriate for a system that supports individual healthcare management and parent or guardian access to child profiles.

Question 2: Frequency of clinic service use

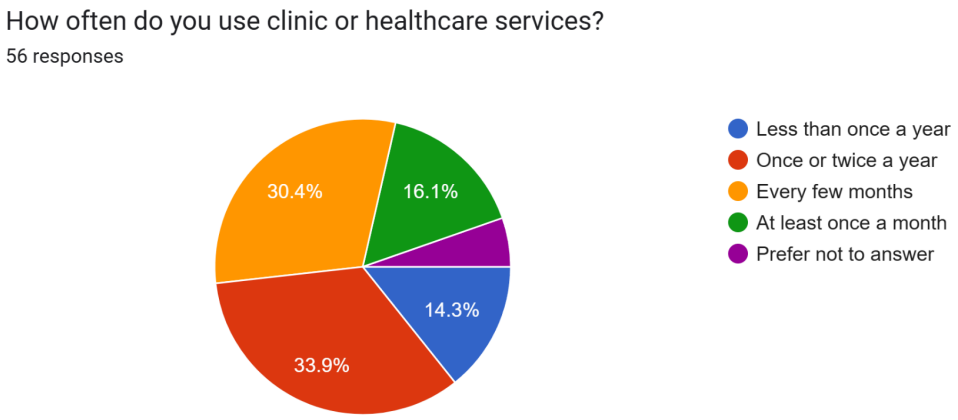


Figure 3.1.2: Frequency of clinic or healthcare service use.

The most common response was once or twice a year, selected by 19 respondents or 33.9%. This was followed by every few months, selected by 17 respondents or 30.4%. Nine respondents or 16.1% used clinic or healthcare services at least once a month, while eight respondents or 14.3% used them less than once a year. Three respondents or 5.4% preferred not to answer. The results show that the proposed system should support occasional users as well as users who require clinic services regularly. This supports the inclusion of simple navigation, appointment history, medical record access, billing information, and reminders that remain useful even when users do not access the system frequently.

Question 3: Current appointment method

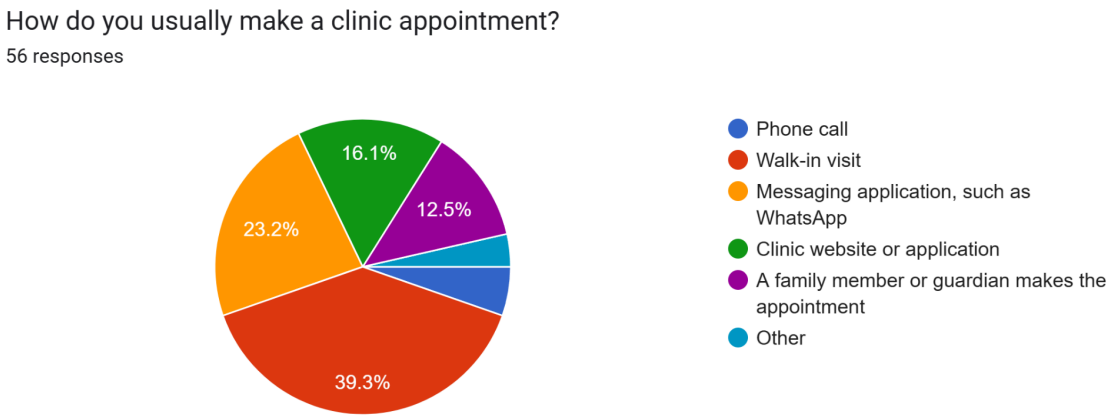


Figure 3.1.3: Current method used to make clinic appointments.

The most common method used to make a clinic appointment was a walk-in visit, selected by 22 respondents or 39.3%. This was followed by messaging applications such as WhatsApp, selected by 13 respondents or 23.2%. Nine respondents or 16.1% used a clinic website or application, seven respondents or 12.5% depended on a family member or guardian, three respondents or 5.4% used a phone call, and two respondents or 3.6% selected Other. The results indicate that many respondents still depend on physical visits or manual communication to arrange appointments. This supports the inclusion of an online appointment function where users can view available times, book appointments, reschedule appointments, cancel appointments, and view appointment history.

Question 4: Difficulty of managing appointments

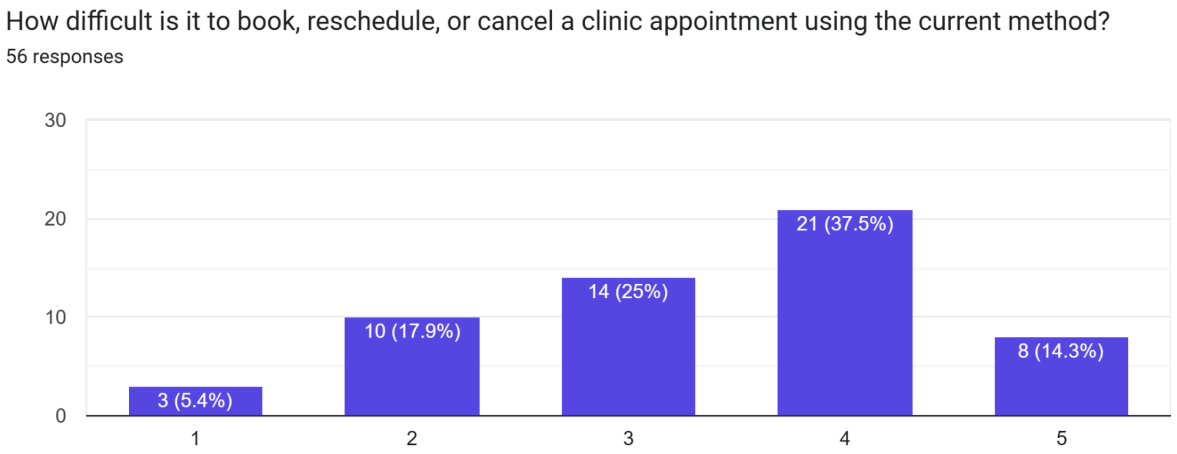


Figure 3.1.4: Difficulty of booking, rescheduling, or cancelling appointments.

The responses indicate a moderate to high level of difficulty when respondents book, reschedule, or cancel clinic appointments using the current method. Twenty-nine respondents selected a rating of 4 or 5, representing 51.8% of the responses, while 13 respondents selected a rating of 1 or 2, representing 23.2%. Fourteen respondents or 25.0% selected the middle rating of 3. The average rating was 3.38 out of 5. The result shows that more than half of the respondents experience considerable difficulty with current appointment procedures. This supports the need for a centralised appointment management function that reduces repeated communication with clinic staff.

Question 5: Clinic information that is difficult to access

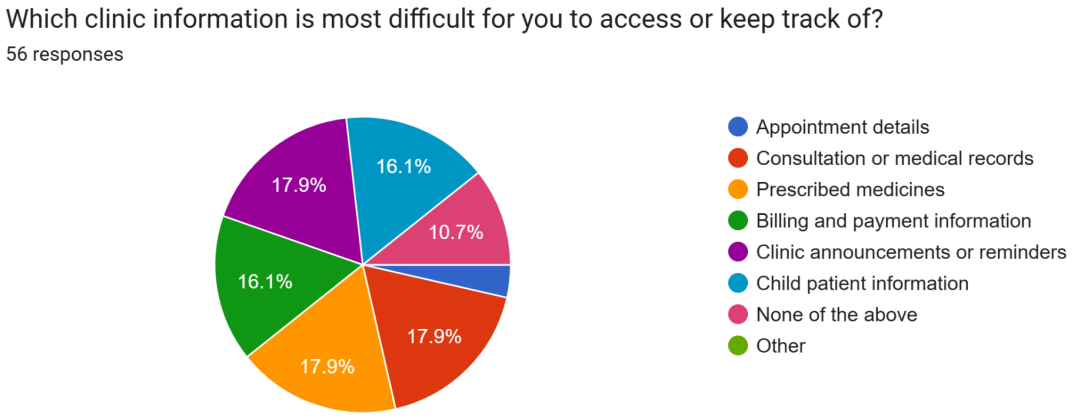


Figure 3.1.5: Clinic information that respondents find difficult to access or track.

The most selected difficulties were consultation or medical records, prescribed medicines, and clinic announcements or reminders, with 10 respondents selecting each option or 17.9% for each category. Billing and payment information and child patient information were each selected by nine respondents or 16.1%. Two respondents or 3.6% selected appointment details, six respondents or 10.7% selected None of the above, and no respondent selected Other. These results show that respondents need easier access to healthcare records, prescription information, and clinic updates. The findings support the inclusion of a patient dashboard containing appointment details, medical records, prescribed medicines, billing information, reminders, and clinic announcements.

Question 6: Online medical record access

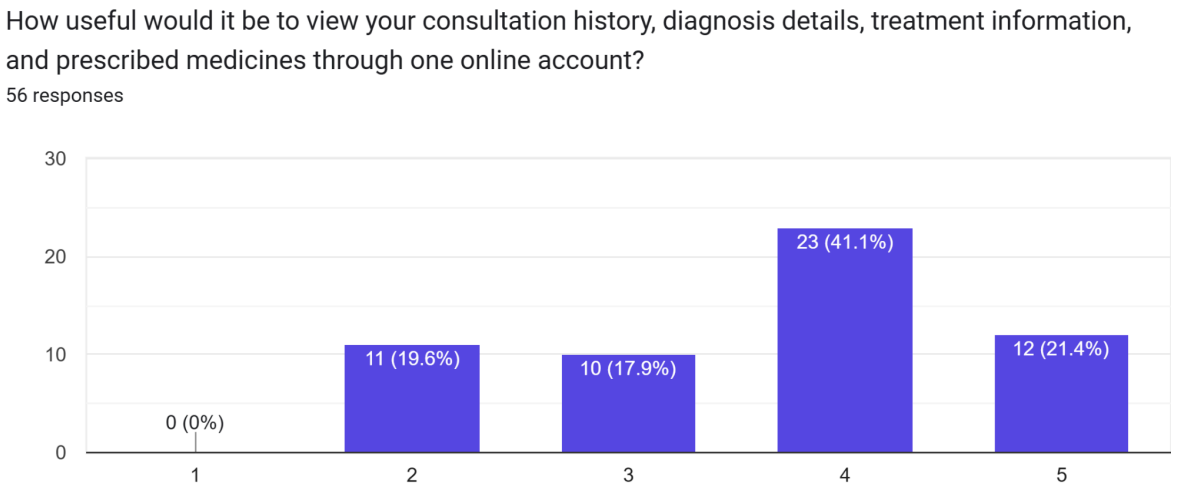


Figure 3.1.6: Usefulness of viewing medical records through one online account.

The responses show positive views about viewing consultation history, diagnosis details, treatment information, and prescribed medicines through one online account. Thirty-five respondents selected a rating of 4 or 5, representing 62.5%, while 11 respondents selected a rating of 1 or 2, representing 19.6%. Ten respondents or 17.9% selected the middle rating of 3. The average rating was 3.64 out of 5. The results indicate that most respondents consider centralised medical record access useful. This supports the Medical Record Viewing module because authorised users can access their healthcare information without requesting every record separately from clinic staff.

Question 7: Preferred appointment functions

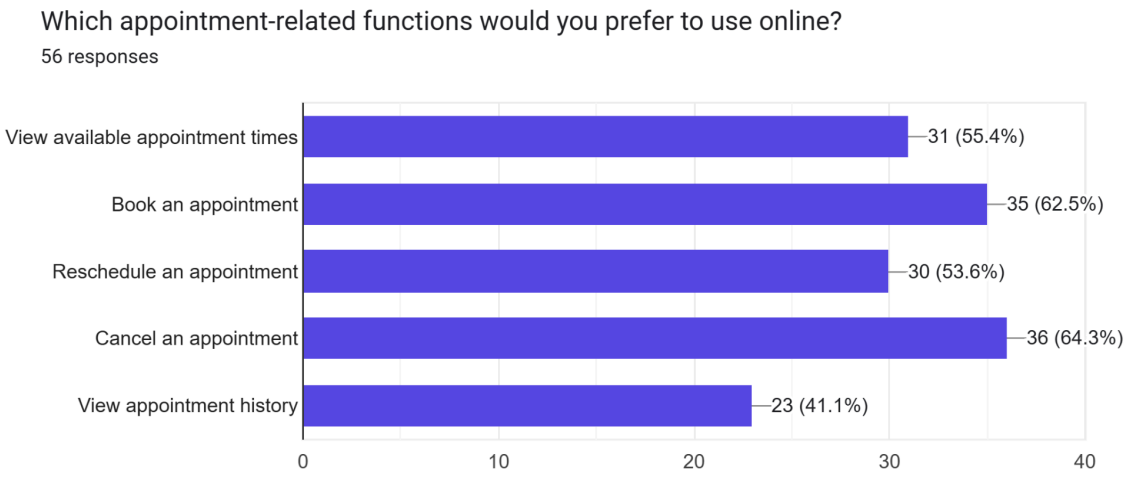


Figure 3.1.7: Appointment functions preferred by respondents.

The most selected appointment function was cancel an appointment, selected by 36 respondents or 64.3%. Book an appointment was selected by 35 respondents or 62.5%, while reschedule an appointment was selected by 30 respondents or 53.6%. Thirty-one respondents or 55.4% preferred to view available appointment times, and 23 respondents or 41.1% preferred to view appointment history. Since respondents could select more than one option, the results show that users value both appointment creation and the ability to manage changes after an appointment has been made. These findings support the inclusion of online appointment booking, appointment rescheduling, appointment cancellation, available time viewing, and appointment history.

Question 8: Preferred billing and payment functions

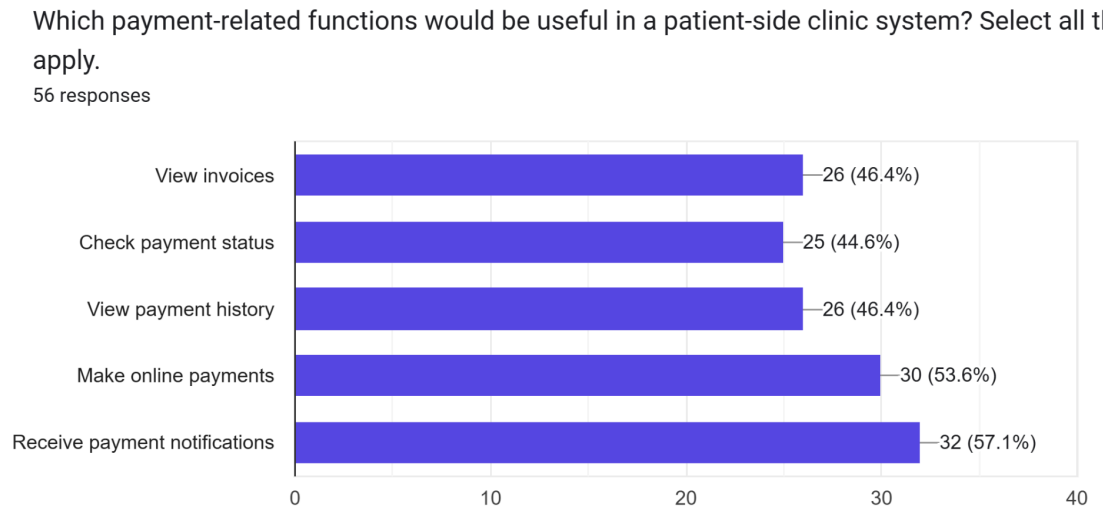


Figure 3.1.8: Billing and payment functions preferred by respondents.

The most selected billing and payment function was receive payment notifications, selected by 32 respondents or 57.1%. Make online payments was selected by 30 respondents or 53.6%. View invoices and view payment history were each selected by 26 respondents or 46.4%, while check payment status was selected by 25 respondents or 44.6%. Since respondents could select more than one option, the results indicate that users prefer to receive payment updates and complete payment-related activities through the same system. The findings support the Billing and Payment Management module, including invoice viewing, payment status, payment history, online payments, and payment notifications.

Question 9: Usefulness of reminders and notifications

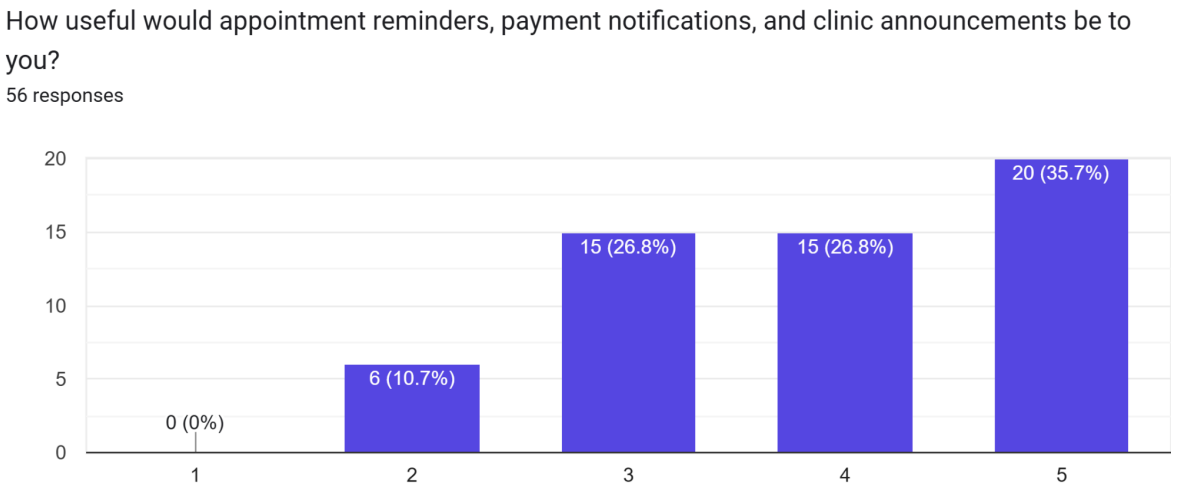


Figure 3.1.9: Usefulness of appointment reminders, payment notifications, and clinic announcements.

The responses show positive support for appointment reminders, payment notifications, and clinic announcements. Thirty-five respondents selected a rating of 4 or 5, representing 62.5%, while six respondents selected a rating of 1 or 2, representing 10.7%. Fifteen respondents or 26.8% selected the middle rating of 3. The average rating was 3.88 out of 5. This response pattern indicates that notifications are considered useful by most respondents. The result supports the inclusion of appointment reminders, payment notifications, and clinic announcements. The lower ratings also suggest that notification preferences should be considered so users can control the types of updates they receive.

Question 10: Parent or guardian management

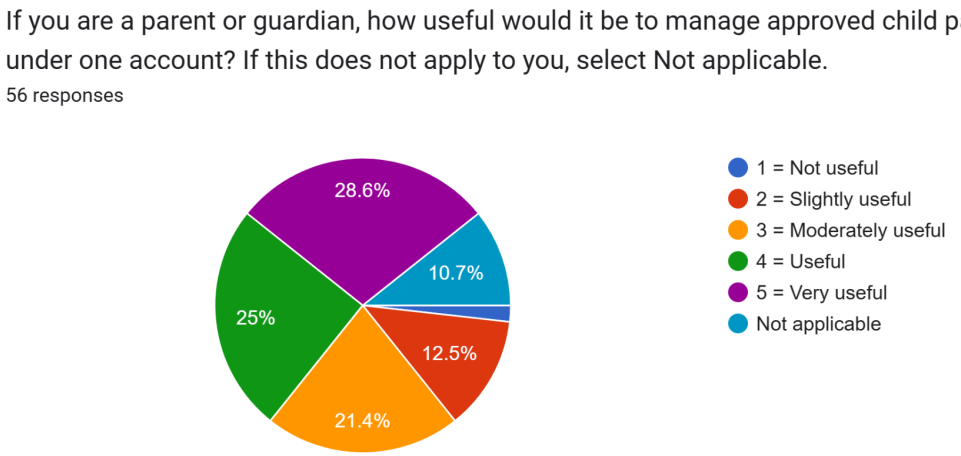


Figure 3.1.10: Usefulness of managing child patient profiles under one account.

The responses show how useful respondents consider managing approved child patient profiles under one account. Six respondents selected Not applicable and were excluded from the rating analysis, leaving 50 applicable responses. Among the applicable respondents, 30 selected a rating of 4 or 5, representing 60.0%, while eight respondents selected a rating of 1 or 2, representing 16.0%. Twelve respondents or 24.0% selected the middle rating of 3. The average rating among applicable respondents was 3.74 out of 5. This result indicates that the parent or guardian function is useful to a majority of applicable respondents. The finding supports the Parent or Guardian Management module, where an approved child profile can be managed under one adult account.

Question 11: Personal health information

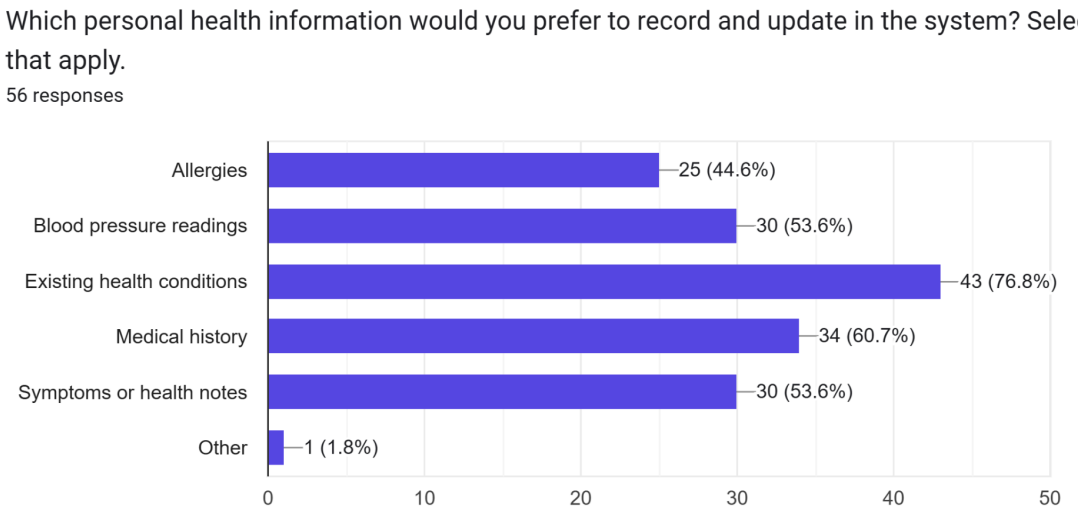


Figure 3.1.11: Personal health information preferred by respondents.

The most selected type of personal health information was existing health conditions, selected by 43 respondents or 76.8%. Medical history was selected by 34 respondents or 60.7%, while blood pressure readings and symptoms or health notes were each selected by 30 respondents or 53.6%. Allergies were selected by 25 respondents or 44.6%, and one respondent or 1.8% selected Other. Since respondents could select more than one option, the results show that users are interested in recording several types of health information, particularly existing conditions and medical history. This supports the Personal Health Record Management module, which includes fields for allergies, blood pressure readings, existing health conditions, medical history, symptoms, and health notes.

Question 12: Medicine purchase management

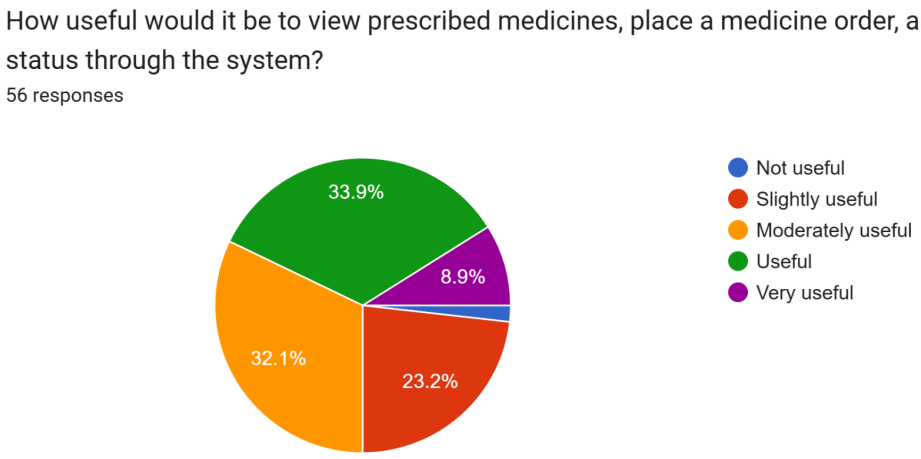


Figure 3.1.12: Usefulness of medicine purchase and order tracking.

The responses show a mixed level of interest in viewing prescribed medicines, placing medicine orders, and tracking order status through the system. Twenty-four respondents selected Useful or Very useful, representing 42.9%. The most common response was Useful, selected by 19 respondents or 33.9%, followed by Moderately useful, selected by 18 respondents or 32.1%. Fourteen respondents selected Not useful or Slightly useful, representing 25.0%. This included one respondent or 1.8% who selected Not useful and 13 respondents or 23.2% who selected Slightly useful. The average rating was 3.25 out of 5. This result indicates that the medicine purchase function is supported by a considerable group but may not be equally important to every user. The finding supports the Medicine Purchase Management module because patients can review prescribed medicines and monitor their order information through the same patient-side system.

Question 13: AI health chatbot

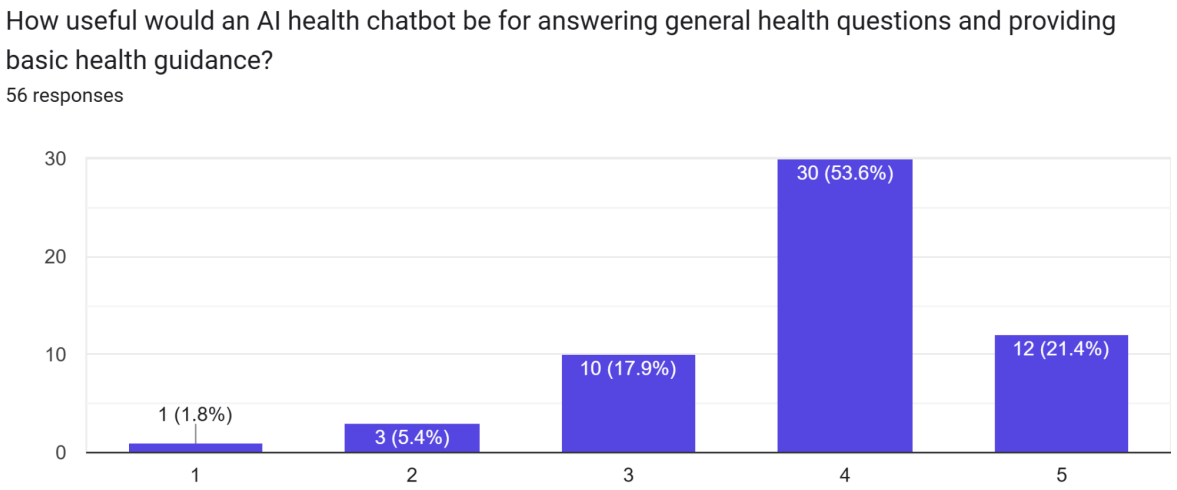


Figure 3.1.13: Usefulness of the AI health chatbot.

The responses show strong support for using an AI health chatbot for general health questions and basic health guidance. Forty-two respondents selected a rating of 4 or 5, representing 75.0%, while four respondents selected a rating of 1 or 2, representing 7.1%. Ten respondents or 17.9% selected the middle rating of 3. The average rating was 3.88 out of 5. The results indicate that respondents are interested in receiving basic health guidance through the proposed system. The response supports the inclusion of the AI Health Chatbot module, but the chatbot must remain limited to general guidance. It must not present its response as a diagnosis, prescription, or replacement for professional medical consultation.

Question 14: Intention to use the proposed system

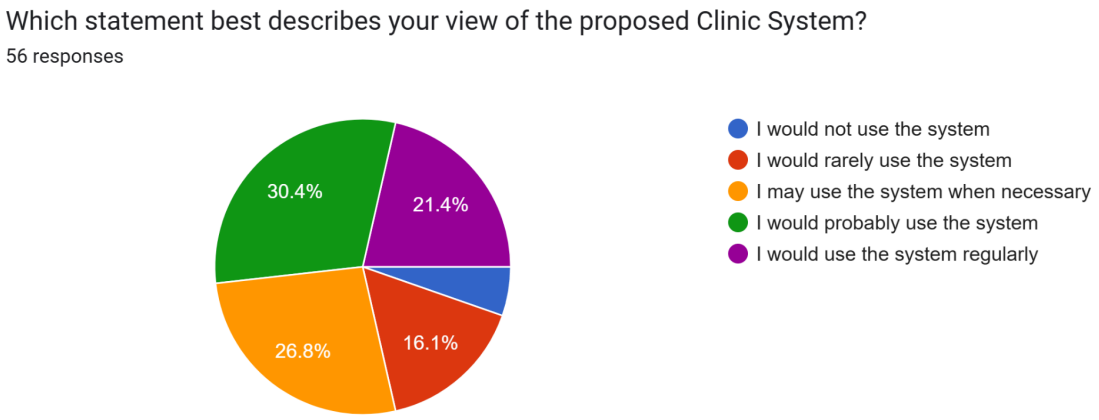


Figure 3.1.14: Respondents' intention to use the proposed Clinic System.

The responses show a generally positive intention to use the proposed Clinic System. Seventeen respondents selected I would probably use the system and 12 selected I would use the system regularly, giving a combined total of 29 respondents or 51.8%. Fifteen respondents or 26.8% selected I may use the system when necessary. The remaining 12 respondents selected I would not use the system or I would rarely use the system, representing 21.4%. The most common response was I would probably use the system, selected by 17 respondents or 30.4%. The result indicates that more than half of the respondents show a clear intention to use the proposed system, while a further group may use it when required. This supports the development of a central patient-side system that combines the clinic services identified in the earlier questions.

Question 15: Additional functions and improvements

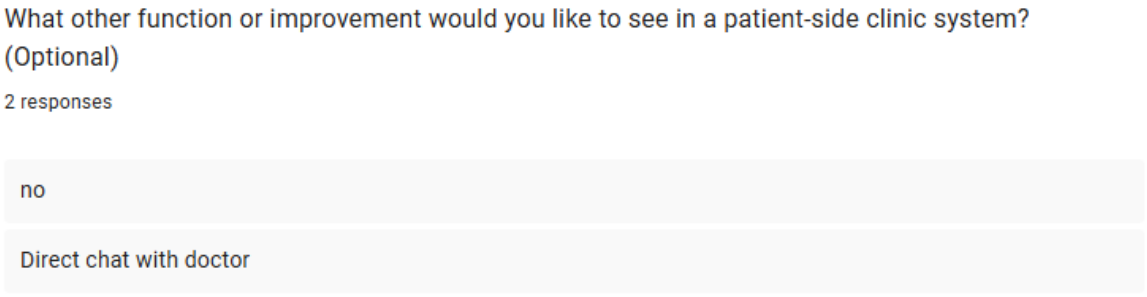


Figure 3.1.15: Suggestions for additional patient-side clinic functions.

The optional question received two responses from the 56 respondents. One respondent entered no, while another suggested direct chat with a doctor. Since most respondents did not provide an additional suggestion, no dominant additional requirement could be identified from this question. The suggestion for direct communication with a doctor may be considered as a possible future enhancement, but it is outside the main scope of the current patient-side module. The responses also show that the functions already included in the proposed system cover the main needs identified by the questionnaire.

The questionnaire findings provide primary feedback from 56 potential users of the proposed Clinic Management System. The results show strong support for viewing medical records online, receiving appointment reminders and clinic notifications, recording personal health information, and using an AI health chatbot for general health guidance. The AI health chatbot received the strongest rating, with 42 respondents or 75.0% selecting a rating of 4 or 5. Existing health conditions were the most selected personal health record item, chosen by 43 respondents or 76.8%. Online medical record viewing and appointment notifications were each rated 4 or 5 by 35 respondents or 62.5%.

The questionnaire also showed mixed support for medicine purchase management. Although 24 respondents or 42.9% selected Useful or Very useful, 18 respondents selected Moderately useful and 14 selected Not useful or Slightly useful. This result will be considered when prioritising the development increments. Functions with strong support can be given a higher development priority, while functions with mixed responses can be reviewed during interface testing and later user feedback. The questionnaire findings were used together with the project scope and literature review to finalise the functional and non-functional requirements in Section 3.2.

3.2 Requirement Analysis

Requirements analysis was used to define the services the proposed Clinic Management System must provide and the way the system must respond to user actions. The requirements are divided into functional and non-functional requirements. Functional requirements describe the services provided by the system. Non-functional requirements describe the expected quality, security, performance, and operational characteristics of the system.

3.2.1 Project Scope

The scope of this project is the patient-side module of the Clinic Management System (Patient Account and Healthcare Management). The system focuses on providing patients, parents or guardians, and guests with access to clinic-related services through one web-based platform. The overall project scope is shown in Figure 3.2.1 below.

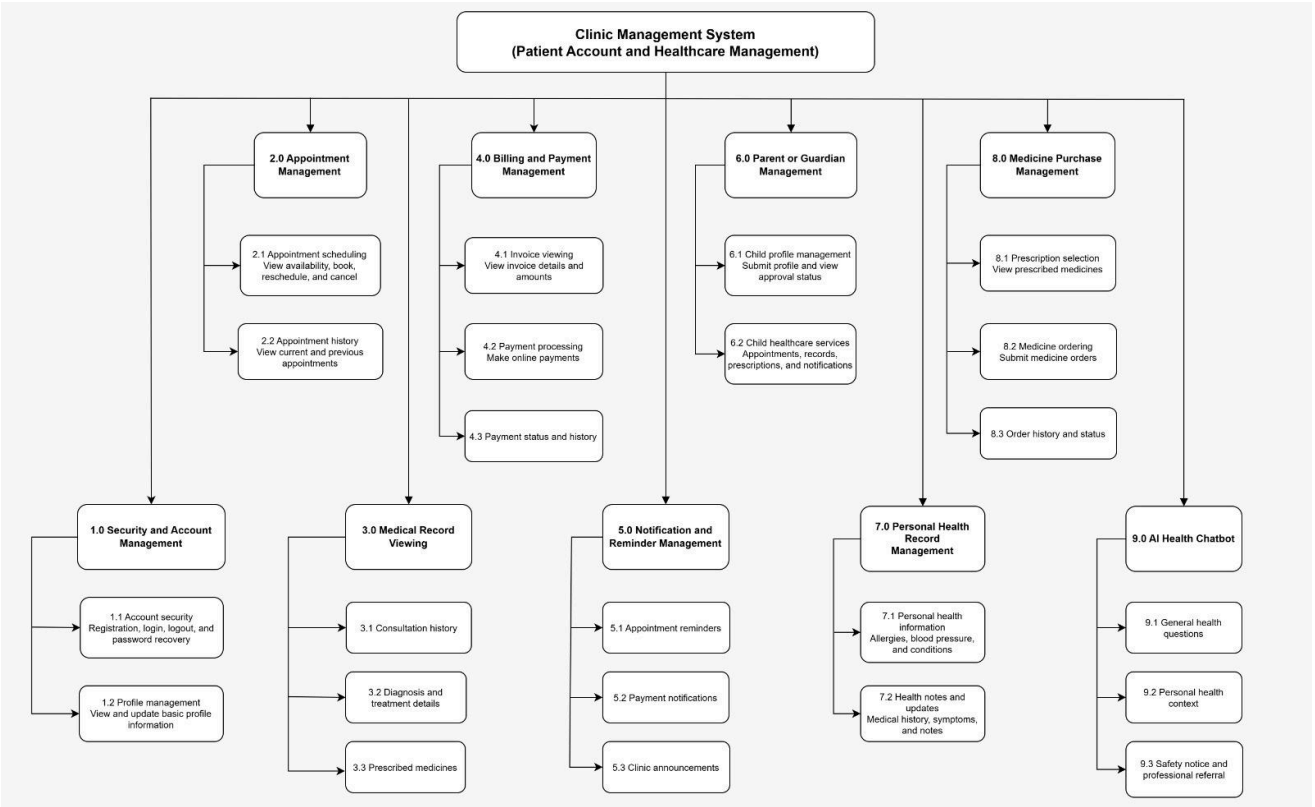


Figure 3.2.1: Scope of the Clinic Management System (Patient Account and Healthcare Management).

1. Security and Account Management

- **1.1 Account Security:**
  - Supports registration, login, logout, and password recovery for authorised users.
- **1.2 Profile Management:**
  - Allows users to view and update their basic profile information.

2. Appointment Management

- **2.1 Appointment Scheduling:**
  - Allows users to view available appointment times, book appointments, reschedule appointments, and cancel appointments.
- **2.2 Appointment History:**
  - Allows users to review current and previous appointment information.

3. Medical Record Viewing

- **3.1 Consultation History:**
  - Allows authorised users to view previous consultation information.
- **3.2 Diagnosis and Treatment Details:**
  - Displays authorised diagnosis and treatment information.
- **3.3 Prescribed Medicines:**
  - Displays medicines prescribed to the patient or approved child profile.

4. Billing and Payment Management

- **4.1 Invoice Viewing:**
  - Allows users to view invoices and billing details.
- **4.2 Payment Processing:**
  - Supports online payment where the supporting payment function is available.
- **4.3 Payment Status and History:**
  - Allows users to view payment status and previous payment information.

5. Notification and Reminder Management

- **5.1 Appointment Reminders:**
    - Provides reminders about upcoming appointments.
- **5.2 Payment Notifications:**
    - Provides updates about payment-related activities.
- **5.3 Clinic Announcements:**
    - Displays relevant clinic announcements and updates.

6. Parent or Guardian Management

- **6.1 Child Profile Management:**
    - Allows a parent or guardian to submit and manage approved child patient profiles.
- **6.2 Child Healthcare Services:**
    - Allows access to the child's appointments, medical records, prescriptions, billing information, and notifications according to the granted access rights.

7. Personal Health Record Management

- **7.1 Personal Health Information:**
    - Allows users to record allergies, blood pressure readings, existing health conditions, and medical history.
- **7.2 Health Notes and Updates:**
    - Allows users to record symptoms and update personal health notes.

8. Medicine Purchase Management

- **8.1 Prescription Selection:**
    - Allows patients to select eligible prescribed medicines.
- **8.2 Medicine Ordering:**
    - Allows patients to submit medicine orders through the system.
- **8.3 Order History and Status:**
    - Allows patients to review purchase history and order status.

9. AI Health Chatbot

- **9.1 General Health Questions:**
    - Allows users to ask general health-related questions.
- **9.2 Personal Health Context:**
    - Provides basic guidance using permitted personal health information where authorised.
- **9.3 Safety Notice and Professional Referral:**
    - Reminds users that chatbot guidance does not replace professional medical consultation.

3.2.2 Functional Requirements

1. Security and Account Management

User requirements:

1. New patients, parents or guardians, and guests can register for an account.
2. Registered users can log in and log out securely.
3. Users can recover or reset a forgotten password.
4. Users can view and update their basic profile information.

System requirements:

1. The system shall validate registration and profile information before storing it.
2. The system shall authenticate users before allowing access to private healthcare functions.
3. The system shall provide password recovery and password update functions.
4. The system shall restrict each user to the information and functions allowed by the user's role.
5. The system shall display a suitable success or error message after account-related actions.

2. Appointment Management

User requirements:

1. Patients can view available appointment options.
2. Patients can book an appointment.
3. Patients can reschedule or cancel an existing appointment.
4. Patients can view current and previous appointment information.
5. Parents or guardians can manage appointments for an approved child profile.

System requirements:

1. The system shall display appointment availability received from the supporting clinic-side module.
2. The system shall record the selected patient, date, time, appointment status, and relevant appointment details.
3. The system shall prevent users from selecting unavailable appointment slots.
4. The system shall update the appointment status when an appointment is booked, rescheduled, or cancelled.
5. The system shall show a confirmation or error message after each appointment action.
6. The system shall generate relevant notification information for confirmed or changed appointments.

3. Medical Record Viewing

User requirements:

1. Patients can view their consultation history.
2. Patients can view diagnosis details, treatment summaries, and prescribed medicines.
3. Parents or guardians can view the medical records of approved child profiles.

System requirements:

1. The system shall retrieve medical records associated with the authenticated patient or approved child profile.
2. The system shall display the consultation date, diagnosis, treatment information, and prescribed medicine details in an organised format.
3. The system shall prevent a user from viewing records not associated with the user or an approved linked child profile.
4. The patient-side module shall display records supplied by the supporting clinic-side functions without allowing unauthorised modification.

4. Billing and Payment Management

User requirements:

1. Patients can view invoices and billing details.
2. Patients can check the payment status of an invoice.
3. Patients can view their payment history.
4. Patients can make online payments through the available payment function.
5. Parents or guardians can view permitted billing information for an approved child profile.

System requirements:

1. The system shall display invoice details, amounts, payment status, and payment dates where available.
2. The system shall associate each invoice with the correct patient or approved child profile.
3. The system shall validate payment requests before processing them.
4. The system shall update and display the payment result after a successful or unsuccessful transaction.
5. The system shall retain payment history for later viewing by the authorised user.
6. The system shall display an appropriate message when a payment cannot be completed.

5. Notification and Reminder Management

User requirements:

1. Patients can receive notifications about upcoming appointments.
2. Patients can receive payment updates and clinic announcements.
3. Parents or guardians can receive notifications related to approved child profiles.
4. Users can view their available notifications.

System requirements:

1. The system shall create or receive notification information for relevant appointment, payment, and clinic events.
2. The system shall display notifications to the correct patient or parent or guardian account.
3. The system shall distinguish unread and read notifications where this function is available.
4. The system shall prevent private notification information from being displayed to unauthorised users.

6. Parent or Guardian Management

User requirements:

1. A parent or guardian can submit the details of a child patient.
2. A parent or guardian can view the status of a child profile request.
3. A parent or guardian can select an approved child profile when managing healthcare services.
4. A parent or guardian can manage the child's appointments, medical records, prescriptions, billing information, and notifications according to the granted access rights.

System requirements:

1. The system shall store the child profile and the relationship between the adult account and the child profile.
2. The system shall support the required approval or verification status for a child profile.
3. The system shall allow access only after the child profile has been correctly linked and approved.
4. The system shall show whether the user is viewing personal information or child patient information.
5. The system shall keep the data of different child profiles separate to prevent incorrect record access.

7. Personal Health Record Management

User requirements:

1. Patients can add personal health information.
2. Patients can update existing information when their health condition changes.
3. Patients can view their recorded information over time.
4. Users can record information such as blood pressure, allergies, existing conditions, medical history, symptoms, and health notes.

System requirements:

1. The system shall provide fields for recording supported personal health information.
2. The system shall validate the format of information before storing it where validation applies.
3. The system shall associate each health record with the correct patient profile.
4. The system shall allow authorised users to view and update their own personal health information.
5. The system shall protect personal health records from unauthorised access.

8. Medicine Purchase Management

User requirements:

1. Patients can view medicines prescribed to them.
2. Patients can select eligible prescribed medicines for purchase.
3. Patients can submit a medicine order through the system.
4. Patients can view medicine purchase history and order status.

System requirements:

1. The system shall display prescribed medicines associated with the correct patient or approved child profile.
2. The system shall allow an order to be created only from available prescription information.
3. The system shall store the order details, order date, and order status.
4. The system shall show an order confirmation or error message after an order is submitted.
5. The system shall display order status updates supplied by the supporting pharmacy or administrative functions.

9. AI Health Chatbot Support

User requirements:

1. Guests and registered users can ask general health-related questions.
2. Registered users can receive basic guidance using permitted personal health information as context.
3. Users must be informed that the chatbot does not replace a doctor or professional medical consultation.

System requirements:

1. The system shall accept and process the user's health-related question.
2. The system shall provide general and non-diagnostic guidance based on the available input.
3. The system shall use personal health information only when the user is authorised and the information is permitted for chatbot support.
4. The system shall display a safety notice advising users to contact a qualified healthcare professional when symptoms are serious, unclear, or urgent.
5. The system shall not present chatbot output as a confirmed diagnosis, prescription, or replacement for professional advice.
6. The system shall display a suitable message when a chatbot response is unavailable or unsuitable.

3.2.3 Non-Functional Requirements

Usability requirement

The system should provide a clear and consistent interface for patients, parents or guardians, and guests. Navigation, labels, forms, and messages should be easy to understand for users with different levels of technical knowledge. Common actions, such as booking an appointment or checking a medical record, should require only the necessary steps.

Efficiency and performance requirement

The system should respond to normal user actions within a reasonable time under the expected project usage conditions. Pages and patient information should load efficiently. The system should reduce repeated data entry and use suitable API calls and database queries to retrieve the information required by the current user and function.

Reliability requirement

The system should perform its main functions consistently. It should display a clear error message when an operation cannot be completed. Data should not be lost when a form submission, payment request, appointment action, or medicine order fails. Each increment should be tested before final integration, and the integrated system should be tested again after all modules are connected.

Security and privacy requirement

The system handles healthcare, account, billing, and personal information. Users must log in before private data is displayed. Access must follow the user's role and patient relationship. Passwords and sensitive information must be handled securely. The system must prevent one patient from viewing or changing another patient's information.

Data integrity requirement

Patient profiles, appointments, medical records, prescriptions, invoices, payments, notifications, personal health records, and medicine orders must be associated with the correct user. The system must validate submitted data and maintain consistent statuses when appointments, payments, or orders are updated.

Maintainability requirement

The source code should be organised into understandable components and services. The React.js frontend, ASP.NET Core Web API backend, and MongoDB Atlas database operations should follow consistent naming and coding practices. The modular structure should allow one increment to be corrected or updated without affecting unrelated modules unnecessarily.

Portability and compatibility requirement

The web-based system should be accessible through current web browsers on desktop computers, laptops, tablets, and smartphones. The interface should remain usable on different screen sizes.

Scalability requirement

The system should be structured so additional patient-side functions, notification types, and healthcare services can be added later. Separating the frontend, backend API, and database supports future expansion of the system.

AI safety requirement

The AI health chatbot must be limited to basic health guidance. Its output must not be treated as a medical diagnosis, prescription, or emergency service. The system must communicate this limitation clearly and direct users to professional medical help when the situation is serious or uncertain.

3.3 Chapter Summary and Evaluation

This chapter described the methodology and requirements analysis for the proposed Clinic Management System (Patient Account and Healthcare Management). The Incremental Development Model was selected because the system contains several patient-side modules. Each module can be analysed, developed, tested, reviewed, and connected to the other modules in stages.

The requirements were gathered through review of the project proposal, literature, existing healthcare systems, current patient workflows, user roles, and questionnaire responses. The questionnaire provided feedback about appointment management, medical record access, billing and payments, reminders, parent or guardian management, personal health records, medicine purchases, AI health chatbot support, and the intention to use the proposed system.

The questionnaire findings were analysed using the response numbers and percentages from Google Forms. The results showed strong support for online medical record access, appointment reminders, personal health record management, and basic AI health guidance. The chapter also defined the target users, project scope, functional requirements, non-functional requirements, constraints, and assumptions.

Clinic Management System (Patient Account and Healthcare Management)

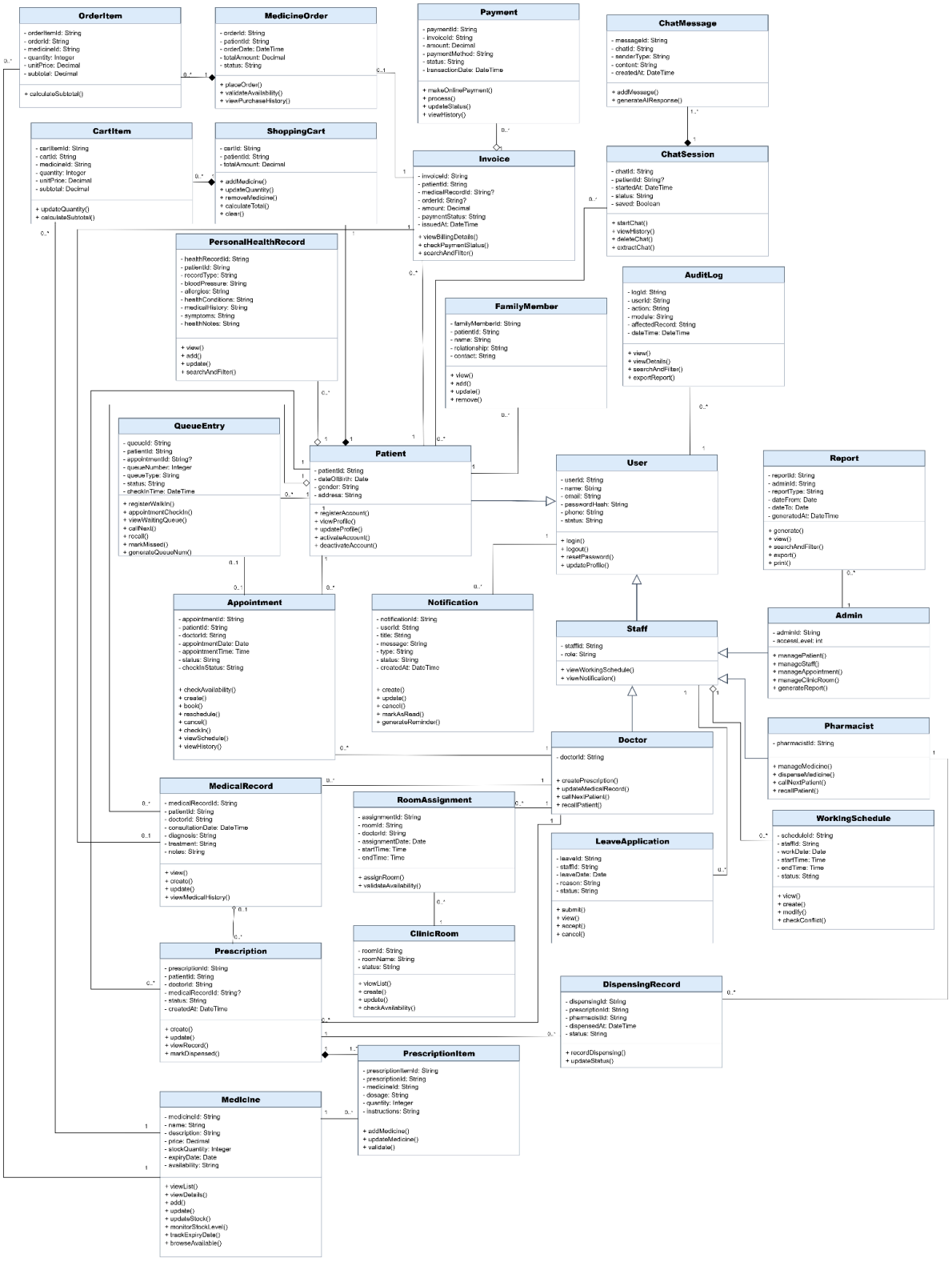
Chapter 4

System Design

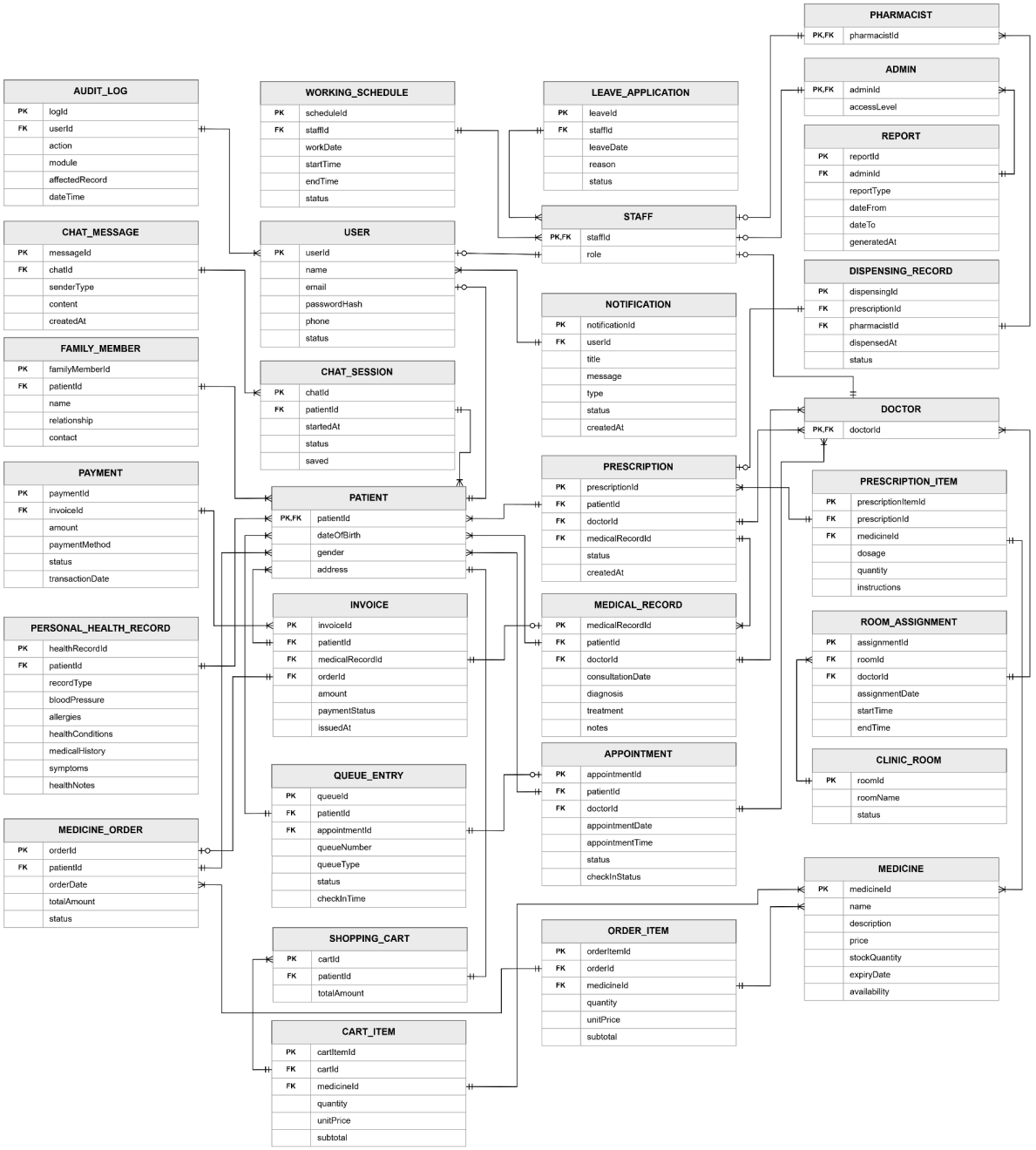
4. System Design

This chapter presents the detailed design of the proposed **Clinic Management System**, with particular emphasis on the **Patient Account and Healthcare Management**. Various diagrams are used to clearly illustrate the system's structure, functions, and interactions. The collaboration and class diagrams describe the relationships among system components, patient-related objects, and stored data, while the sequence diagrams demonstrate how patients and guests interact with the system when performing tasks such as managing accounts and profiles, booking and managing appointments, viewing billing information and making online payments, receiving notifications and reminders, managing personal health records, purchasing medicines, using the AI health chatbot, and managing family information. The overview and detailed use case diagrams further describe the available patient-side functions at different levels of detail, supported by use case descriptions that explain user actions, system responses, conditions, and expected outcomes. These designs are subsequently translated into a user interface prototype that focuses on clear navigation, ease of use, efficient task completion, readable information, and consistent system feedback. Therefore, this chapter provides a structured foundation for the implementation and testing of the **Patient Account and Healthcare Management** in the subsequent development phases.

4.1 Class Diagram



4.2 ERD Diagram



4.3 Use Case Diagram

Overview Use Case Diagram



Figure 4.3.1: Overview use case diagram

Detailed Use Case Diagram

Security & Account Management

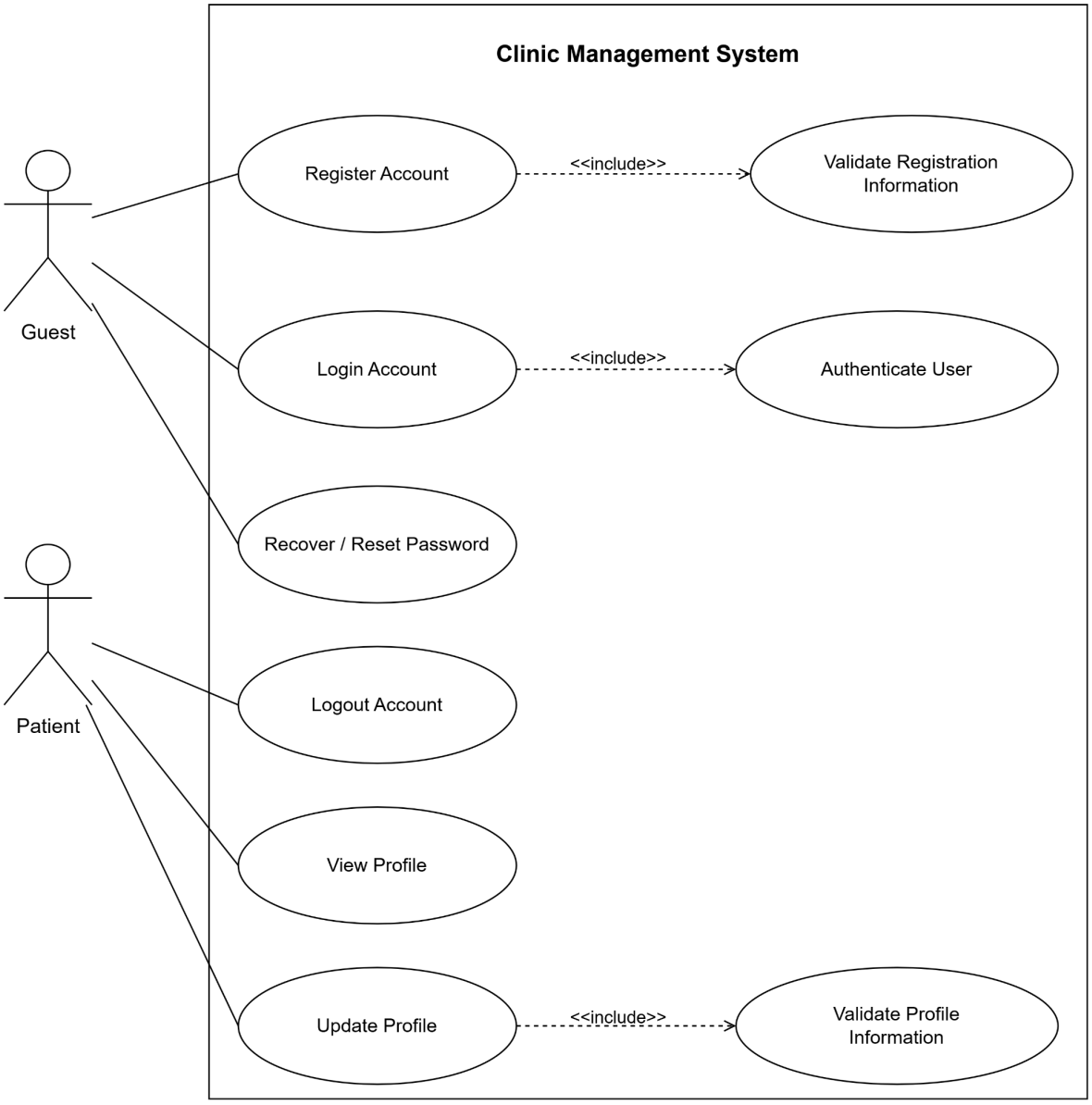


Figure 4.3.2: Detailed Use Case for Security & Account Management

Appointment Management

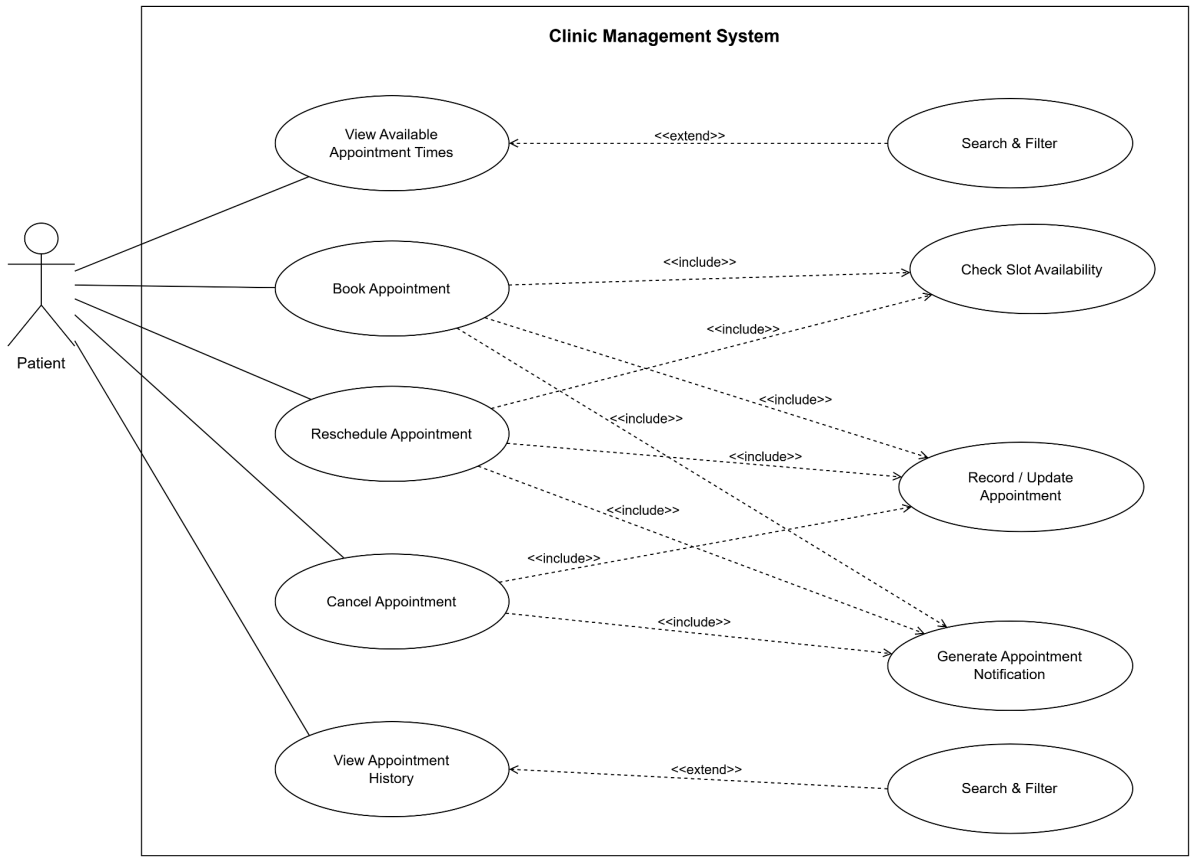


Figure 4.3.3: Detailed Use Case for Appointment Management

Billing & Payment Management

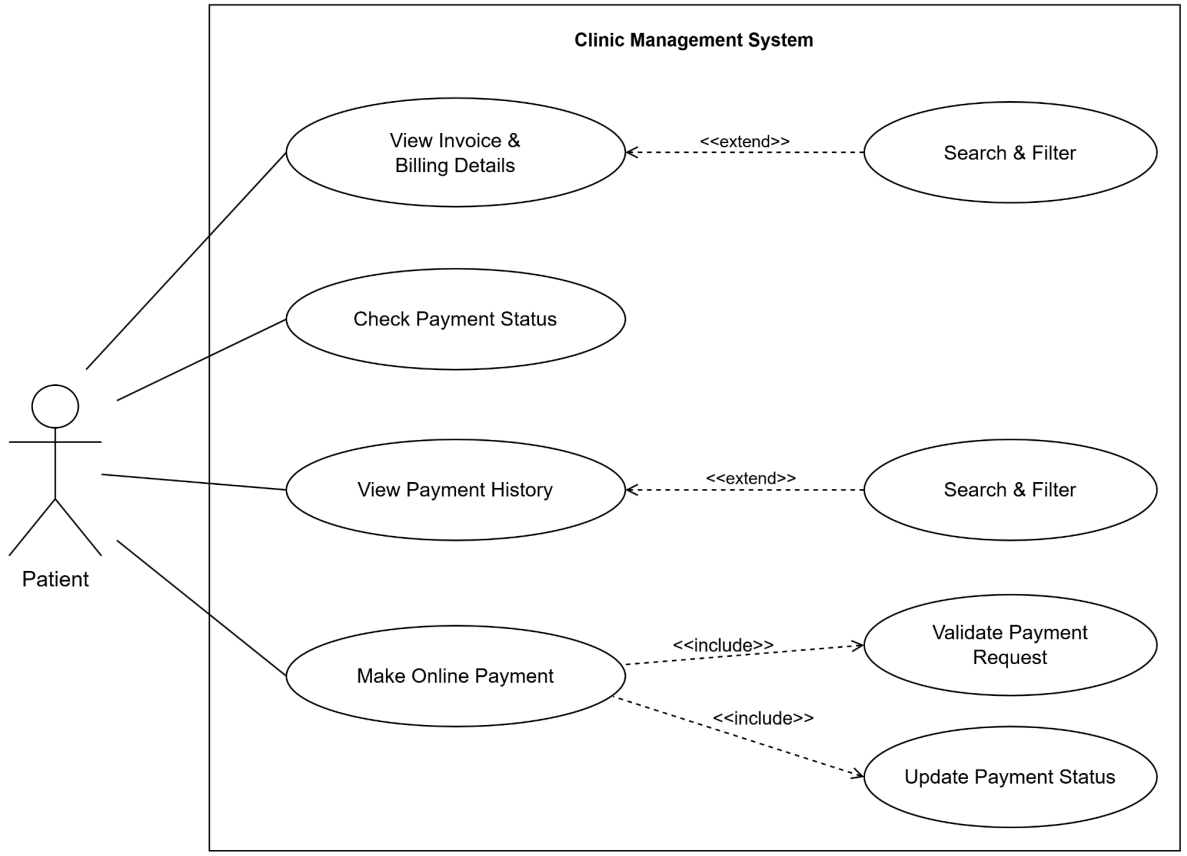


Figure 4.3.4: Detailed Use Case for Billing & Payment Management

Notification & Reminder Management

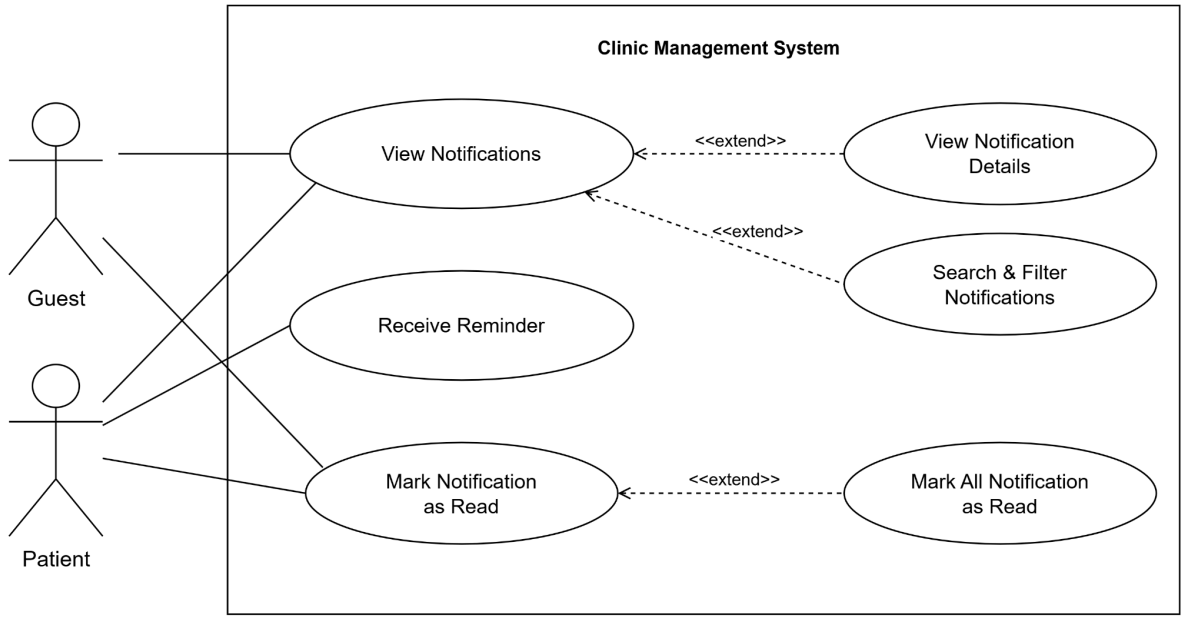


Figure 4.3.5: Detailed Use Case for Notification & Reminder Management

Personal Health Record Management

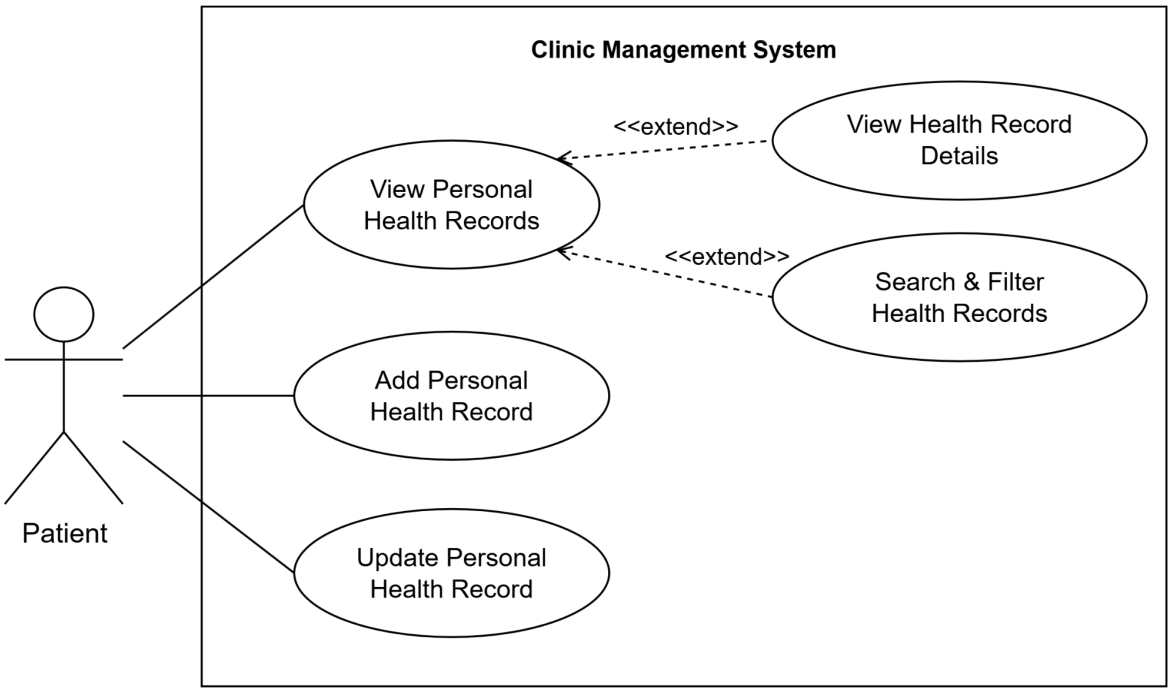


Figure 4.3.6: Detailed Use Case For Personal Health Record Management

Medicine Purchase Management

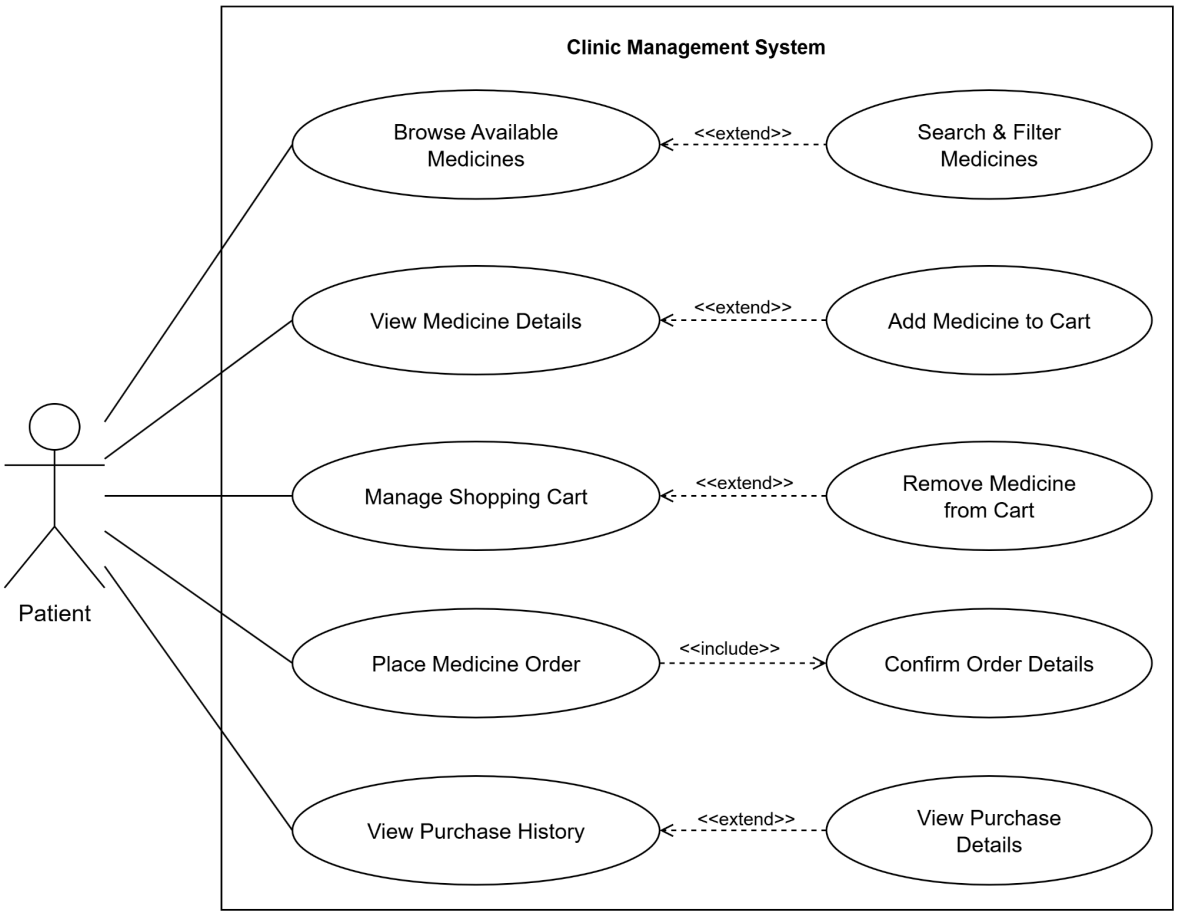


Figure 4.3.7: Detailed Use Case for Medicine Purchase Management

AI Health Chatbot Management

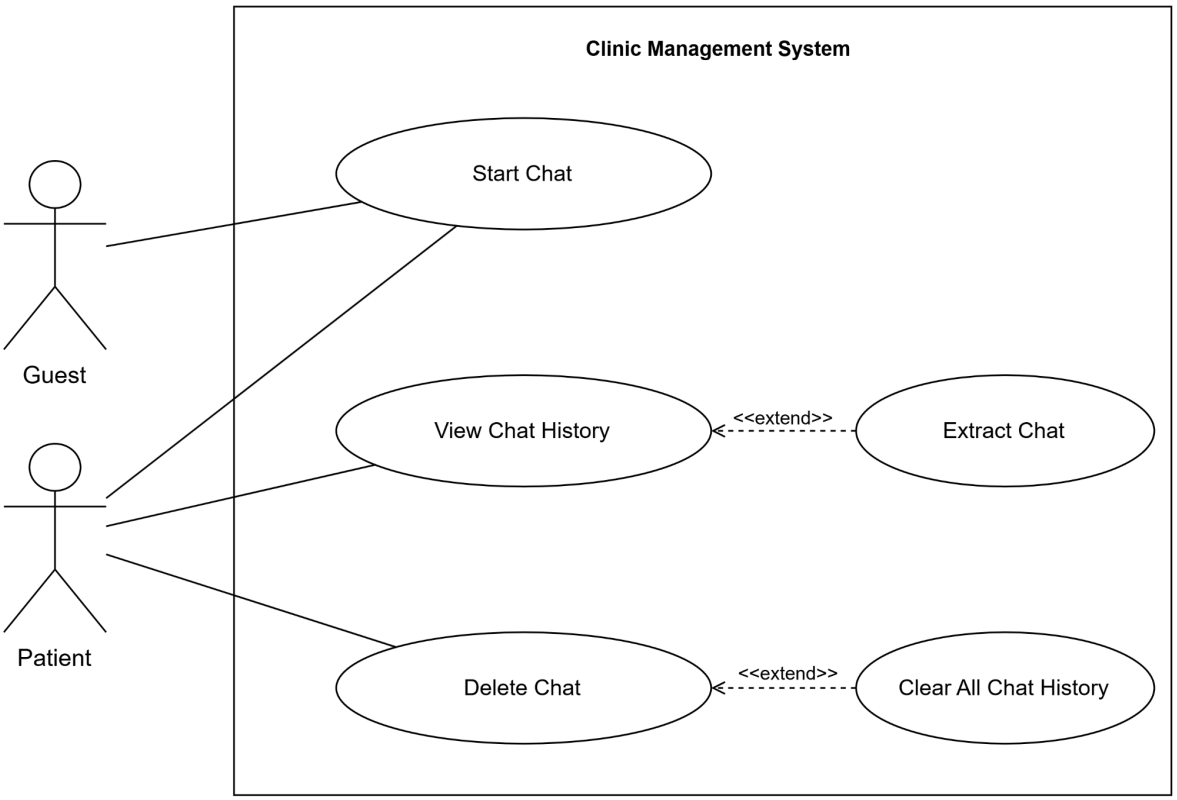


Figure 4.3.8: Detailed Use Case For AI Health Chatbot Management

Parent/Guardian Management

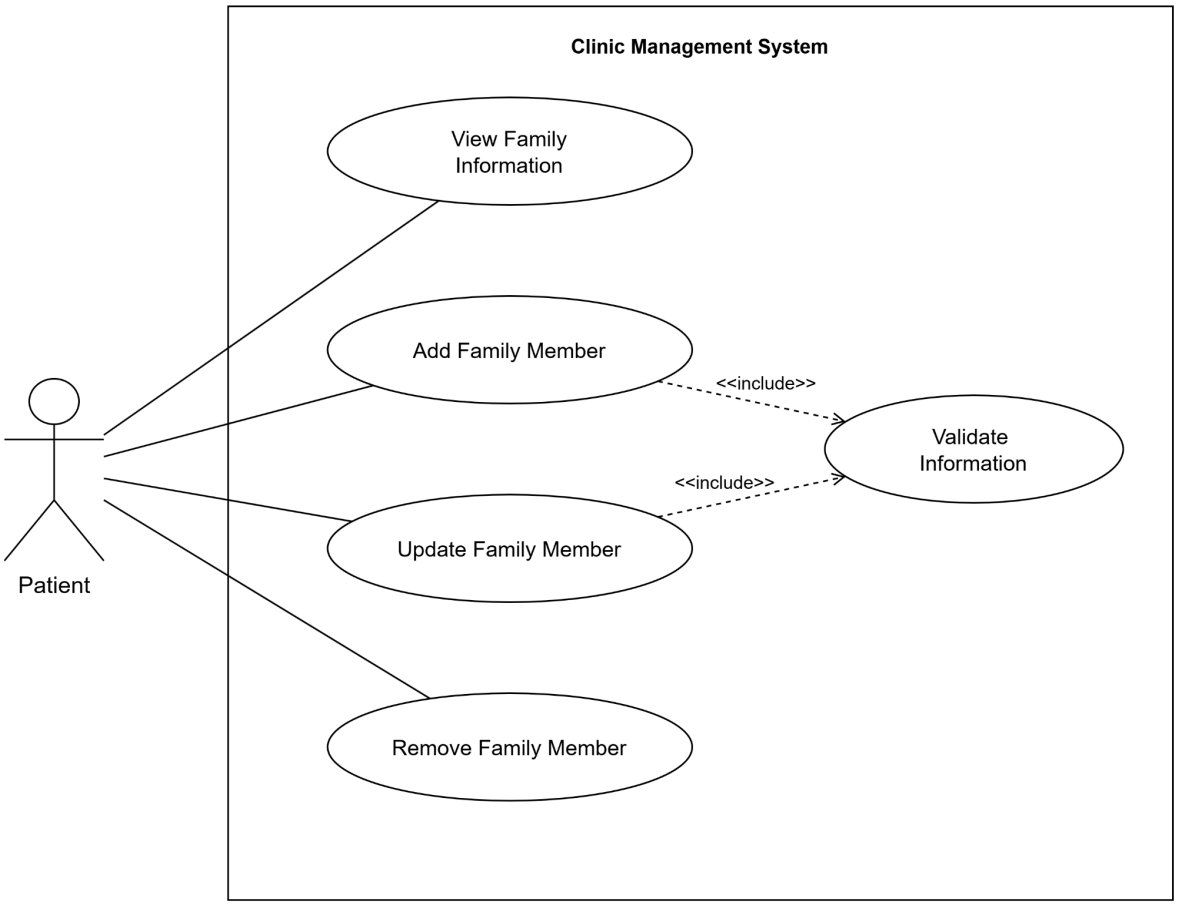


Figure 4.3.9: Detailed Use Case For Parent/Guardian Management

4.4 Use Case Description

Use Case Name: Register Account	
Actor: Guest	
Brief Description: This use case allows a guest to register a new patient account by providing the required personal and account information.	
Precondition: The guest has accessed the registration page and does not have an existing patient account.	
Main Flow of Events	
Actor Action	System Responses
1. The guest clicks the register button.	2. The system displays the registration page.
3. Guest enters the required personal and account information.	4. The system validates the entered information.
5. Guests click the register button to submit the registration form.	6. The system checks whether the account information has already been registered.
	7. The system creates a new patient account.
	8. The system stores the patient account information.
	9. The system displays a successful registration message.
	10. The system redirects the guest to the login page.
Alternative flow: A1 step 4: If required information is missing or entered in an invalid format, the system displays an error message and prompts the guest to correct the information. A2 step 6: If the account information has already been registered, the system displays a message indicating that the account already exists.	
Postcondition: The system creates and stores a new patient account successfully. The patient can use the registered account to log in to the system.	

Table 4.4.1: Detailed Use Case Description for Register Account

Use Case Name: Login Account	
Actor: Guest	
Brief Description: This use case allows a registered patient to log in to the system using the registered account credentials to access the available patient functions.	
Precondition: The user has registered an account and is currently not logged in to the system.	
Main Flow of Events	
Actor Action	System Responses
1. The user accesses the login page.	2. The system displays the login page.
3. The user enters the registered email address and password.	
4. The user clicks the login button.	5. The system validates the entered login credentials.
	6. The system authenticates the patient account.
	7. The system redirects the patient to the main page after successful authentication.
Alternative flow: A1 step 5: If the email address or password is incorrect, the system displays an invalid login credentials message and prompts the patient to enter the credentials again. A2 step 5: If any required login field is empty, the system displays a validation message and prompts the patient to fill in the required field.	
Postcondition: The patient is successfully authenticated and logged in to the system.	

Table 4.4.2: Detailed Use Case Description for Login Account

Use Case Name: Logout Account	
Actor: Patient	
Brief Description: This use case allows a logged-in patient to log out from the system and end the current authenticated session securely.	
Precondition: The patient is currently logged in to the system.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the logout button.	2. The system receives the logout request.
	3. The system ends the patient's current authenticated session.
	4. The system clears the related login session information.
	5. The system redirects the patient to the login page or public page.
Alternative flow: A1 step 2: If the logout request cannot be processed due to a temporary system error, the system displays an error message and keeps the patient on the current page.	
Postcondition: The patient is logged out successfully and can no longer access functions that require authentication until logging in again.	

Table 4.4.3: Detailed Use Case Description for Logout Account

Use Case Name: Recover/Reset Password	
Actor: Guest	
Brief Description: This use case allows a patient who has forgotten the account password to recover access to the account by resetting the password.	
Precondition: The user has a registered account and has accessed the password recovery function.	
Main Flow of Events	
Actor Action	System Responses
1. The user clicks the forgot password option on the login page.	2. The system displays the password recovery page.
3. The user enters the registered email address.	4. The system validates the entered email address.
5. The user submits the password recovery request.	6. The system verifies that the email address is associated with a registered account.
	7. The system provides the patient with the password reset process.
8. The user enters a new password and confirms the new password.	9. The system validates the new password.
10. The user submits the new password.	11. The system updates the patient's account with the new password.
	12. The system displays a successful password reset message and redirects the patient to the login page.
Alternative flow: A1 step 4: If the email address is entered in an invalid format or is empty, the system displays a validation message and prompts the patient to enter a valid email address. A2 step 6: If the email address is not associated with a registered account, the system displays a message indicating that the account cannot be found. A3 step 9: If the new password does not meet the password requirements, the system displays a validation message and prompts the patient to enter a valid password. A4 step 9: If the new password and confirmation password do not match, the system displays a message and prompts the patient to re-enter the passwords.	
Postcondition: The patient's password is successfully updated, and the patient can log in to the account using the new password.	

Table 4.4.4: Detailed Use Case Description for Recover/Reset Password

Use Case Name: View Profile	
Actor: Patient	
Brief Description: This use case allows a patient to view the personal and account information stored in the patient profile.	
Precondition: The patient is logged in to the system.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the profile option.	2. The system receives the request to view the patient profile.
	3. The system retrieves the patient's stored profile information.
	4. The system displays the patient profile page with the available personal and account information.
5. The patient reviews the displayed profile information.	
Alternative flow: A1 step 3: If the patient profile information cannot be retrieved, the system displays an error message and asks the patient to try again. A2 step 3: If some optional profile information has not been provided, the system displays the available information and leaves the unavailable fields empty or indicates that the information has not been provided.	
Postcondition: The patient's profile information is displayed successfully without changing the stored profile data.	

Table 4.4.5: Detailed Use Case Description for View Profile

Use Case Name: Update Profile	
Actor: Patient	
Brief Description: This use case allows a patient to update the personal and account information stored in the patient profile.	
Precondition: The patient is logged in to the system and has accessed the profile page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the edit or update profile option.	2. The system displays the editable profile information.
3. The patient modifies the required profile information.	
4. The patient clicks the save or update button.	5. The system validates the updated profile information.
	6. The system updates the patient profile with the new information.
	7. The system saves the updated profile information.
	8. The system displays a successful profile update message.
Alternative flow: A1 step 5: If any required information is missing or entered in an invalid format, the system displays the relevant validation message and prompts the patient to correct the information. A2 step 4: If the patient cancels the update process, the system does not save any changes and returns to the profile page. A3 step 6: If the profile information cannot be updated due to a system error, the system displays an error message and keeps the previously stored profile information unchanged.	
Postcondition: The patient's profile information is successfully updated and the latest information is stored in the system.	

Table 4.4.6: Detailed Use Case Description for Update Profile

Use Case Name: View Available Appointment Time	
Actor: Patient	
Brief Description: This use case allows a patient to view the available appointment dates and time slots before making or rescheduling an appointment.	
Precondition: The patient is logged in to the system and has accessed the appointment management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the appointment management page.	2. The system displays the appointment management page.
3. The patient selects the option to view available appointment times.	4. The system retrieves the available appointment dates and time slots.
	5. The system checks the availability of the appointment time slots.
	6. The system displays the available appointment dates and time slots to the patient.
7. Patient reviews the available appointment times.	
Alternative flow: A1 step 4: If there are no available appointment time slots, the system displays a message indicating that no appointment times are currently available. A2 step 4: If the appointment availability cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again.	
Postcondition: The available appointment dates and time slots are successfully displayed to the patient without making any changes to the appointment records.	

Table 4.4.7: Detailed Use Case Description for View Available Appointment Time

Use Case Name: Book Appointment	
Actor: Patient	
Brief Description: This use case allows a patient to book an appointment by selecting an available appointment date and time slot.	
Precondition: The patient is logged in to the system and has accessed the appointment management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the book appointment option.	2. The system displays the available appointment dates and time slots.
3. The patient selects the preferred appointment date and time slot.	4. The system checks whether the selected appointment time slot is still available.
5. The patient confirms the appointment booking.	6. The system creates the appointment record.
	7. The system stores the appointment information.
	8. The system displays a successful appointment booking message.
	9. The system generates an appointment confirmation notification for the patient.
Alternative flow: A1 step 4: If the selected appointment time slot is no longer available, the system displays a message indicating that the selected time slot is unavailable and prompts the patient to select another available time slot. A2 step 5: If the patient cancels the booking before confirmation, the system does not create the appointment and returns the patient to the appointment management page. A3 step 6: If the appointment cannot be created due to a system error, the system displays an error message and does not store the appointment record.	
Postcondition: A new appointment is successfully created and stored in the system. The selected appointment time slot is reserved for the patient, and an appointment confirmation notification is generated.	

Table 4.4.8: Detailed Use Case Description for Book Appointment

Use Case Name: Reschedule Appointment	
Actor: Patient	
Brief Description: This use case allows a patient to reschedule an existing appointment by selecting a new available appointment date and time slot.	
Precondition: The patient is logged in to the system and has an existing upcoming appointment.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the appointment management page.	2. The system displays the patient's existing upcoming appointment.
3. The patient selects the appointment to be rescheduled.	4. The system displays the appointment details and the reschedule option.
5. The patient clicks the reschedule appointment option.	6. The system displays the available appointment dates and time slots.
7. The patient selects a new appointment date and time slot.	8. The system checks whether the selected appointment time slot is still available.
9. The patient confirms the appointment rescheduling.	10. The system updates the existing appointment with the newly selected date and time.
	11. The system stores the updated appointment information.
	12. The system displays a successful appointment rescheduling message.
	13. The system generates an appointment rescheduling notification for the patient.
Alternative flow: A1 step 8: If the selected appointment time slot is no longer available, the system displays a message indicating that the selected time slot is unavailable and prompts the patient to select another available time slot. A2 step 9: If the patient cancels the rescheduling process before confirmation, the system does not update the appointment and retains the original appointment date and time. A3 step 10: If the appointment cannot be updated due to a system error, the system displays an error message and retains the original appointment information.	
Postcondition: The existing appointment is successfully updated with the newly selected appointment date and time. The previous appointment time slot becomes available again, and an appointment rescheduling notification is generated for the patient.	

Table 4.4.9: Detailed Use Case Description for Reschedule Appointment

Use Case Name: Cancel Appointment	
Actor: Patient	
Brief Description: This use case allows a patient to cancel an existing upcoming appointment that is no longer required.	
Precondition: The patient is logged in to the system and has an existing upcoming appointment.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the appointment management page.	2. The system displays the patient's existing upcoming appointment.
3. The patient selects the appointment to be cancelled.	4. The system displays the selected appointment details and the cancel option.
5. The patient clicks the cancel appointment option.	6. The system displays a confirmation message for the appointment cancellation.
7. Patient confirms the appointment cancellation.	8. The system cancels the selected appointment.
	9. The system updates the appointment status as cancelled.
	10. The system releases the cancelled appointment time slot for future booking.
	11. The system displays a successful appointment cancellation message.
	12. The system generates an appointment cancellation notification for the patient.
Alternative flow: A1 step 7: If the patient decides not to proceed with the cancellation, the system closes the confirmation message and keeps the appointment unchanged. A2 step 8: If the appointment cannot be cancelled due to a system error, the system displays an error message and retains the existing appointment information.	
Postcondition: The selected appointment is successfully cancelled, the appointment status is updated, the corresponding appointment time slot becomes available for future booking, and an appointment cancellation notification is generated for the patient.	

Table 4.4.10: Detailed Use Case Description for Cancel Appointment

Use Case Name: View Appointment History	
Actor: Patient	
Brief Description: This use case allows a patient to view previous and current appointment records stored in the system.	
Precondition: The patient is logged in to the system and has accessed the appointment management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the appointment history option.	2. The system receives the request to view the patient's appointment history.
	3. The system retrieves the appointment records associated with the patient.
	4. The system displays the patient's appointment history.
5. The patient reviews the appointment records.	
Alternative flow: A1 step 3: If the patient does not have any appointment records, the system displays a message indicating that no appointment history is available. A2 step 3: If the appointment history cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again.	
Postcondition: The patient's appointment history is successfully displayed without making any changes to the appointment records.	

Table 4.4.11: Detailed Use Case Description for View Appointment History

Use Case Name: View Invoice & Billing Details	
Actor: Patient	
Brief Description: This use case allows a patient to view invoices and billing details associated with the patient's clinic services and transactions.	
Precondition: The patient is logged in to the system and has accessed the billing and payment management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the invoice and billing option.	2. The system receives the request to view the patient's invoices and billing information.
	3. The system retrieves the invoice and billing records associated with the patient.
	4. The system displays the available invoices with the relevant billing information and payment status.
5. The patient selects an invoice to view.	6. The system retrieves the selected invoice details.
	7. The system displays the detailed billing information of the selected invoice.
Alternative flow: A1 step 3: If the patient does not have any invoice or billing records, the system displays a message indicating that no billing records are available. A2 step 3: If the invoice and billing records cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A3 step 4: Patient may search or filter the displayed invoices to locate a specific invoice. The system displays the invoice records according to the entered search criteria or selected filter.	
Postcondition: The selected invoice and its billing details are successfully displayed without making any changes to the billing records.	

Table 4.4.12: Detailed Use Case Description for View Invoice & Billing Details

Use Case Name: Check Payment Status	
Actor: Patient	
Brief Description: This use case allows a patient to check the payment status of an invoice to determine whether the payment is pending, successful, or failed.	
Precondition: The patient is logged in to the system and has an existing invoice or payment record.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the billing and payment management page.	2. The system displays the patient's available invoices and payment information.
3. The patient selects an invoice or payment record to check its payment status.	4. The system retrieves the payment information associated with the selected record.
	5. The system checks the current payment status.
	6. The system displays the payment status of the selected record to the patient.
7. The patient reviews the displayed payment status.	
Alternative flow: A1 step 4: If the selected invoice does not have a payment record, the system displays the invoice as unpaid or pending payment. A2 step 4: If the payment information cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again.	
Postcondition: The current payment status of the selected invoice or payment record is successfully displayed without making any changes to the payment information.	

Table 4.4.13: Detailed Use Case Description for Check Payment Status

Use Case Name: View Payment History	
Actor: Patient	
Brief Description: This use case allows a patient to view the history of payments made through the system, including the relevant payment details and payment status.	
Precondition: The patient is logged in to the system and has accessed the billing and payment management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the payment history option.	2. The system receives the request to view the patient's payment history.
	3. The system retrieves the payment records associated with the patient.
	4. The system displays the patient's payment history with the relevant payment information and status.
5. The patient selects a payment record to view.	6. The system retrieves the details of the selected payment record.
	7. The system displays the detailed payment information to the patient.
Alternative flow: A1 step 3: If the patient does not have any payment records, the system displays a message indicating that no payment history is available. A2 step 3: If the payment history cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A3 step 4: Patient may search or filter the payment history to locate a specific payment record. The system displays the payment records according to the entered search criteria or selected filter.	
Postcondition: The patient's payment history and the selected payment details are successfully displayed without making any changes to the payment records.	

Table 4.4.14: Detailed Use Case Description for View Payment History

Use Case Name: Make Online Payment	
Actor: Patient	
Brief Description: This use case allows a patient to make an online payment for an outstanding invoice through the system.	
Precondition: The patient is logged in to the system and has an unpaid invoice that is eligible for online payment.	
Main Flow of Events	
Actor Action	System Responses
1. The patient selects an unpaid invoice.	2. The system displays the invoice details and available payment option.
3. The patient clicks the online payment option.	4. The system validates the payment request and the selected invoice amount.
5. The patient confirms the payment.	6. The system processes the payment request.
	7. The system receives the payment result.
	8. The system updates the payment status of the invoice.
	9. The system stores the payment transaction record.
	10. The system displays a successful payment message and the updated payment status.
Alternative flow: A1 step 4: If the selected invoice has already been paid or is not eligible for payment, the system displays a message and prevents the payment from proceeding. A2 step 6: If the payment is unsuccessful or rejected, the system displays a payment failure message and keeps the invoice unpaid. A3 step 6: If the payment process is interrupted or cancelled, the system stops the payment process and retains the current payment status. A4 step 8: If the payment result is still pending, the system displays the payment status as pending until the final result is received.	
Postcondition: The online payment transaction is recorded and the invoice payment status is updated according to the payment result.	

Table 4.4.15: Detailed Use Case Description for Make Online Payment

Use Case Name: View Notification	
Actor: Guest, Patient	
Brief Description: This use case allows a guest or patient to view notifications available in the system, such as appointment reminders, payment notifications, and clinic announcements.	
Precondition: The guest or patient has accessed the notification section of the system.	
Main Flow of Events	
Actor Action	System Responses
1. Guest or patient accesses the notification section.	2. The system retrieves the notifications available to the current user.
	3. The system displays the list of available notifications.
4. Guest or patient selects a notification to view.	5. The system retrieves the selected notification details.
	6. The system displays the notification details.
7. Patients may mark the notification as read after viewing it.	8. The system updates the notification status as read for the patient.
Alternative flow: A1 step 2: If no notifications are available, the system displays a message indicating that there are no notifications to show. A2 step 2: If the notifications cannot be retrieved due to a system error, the system displays an error message and prompts the user to try again. A3 step 3: Guest or patient may search or filter the notification list to locate a specific notification. The system displays notifications according to the selected search criteria or filter.	
Postcondition: The selected notification is successfully displayed. For a logged-in patient, the notification status may be updated to read after viewing.	

Table 4.4.16: Detailed Use Case Description for View Notification

Use Case Name: Receive Reminder	
Actor: Patient	
Brief Description: This use case allows a patient to receive reminders from the system for upcoming appointments and other relevant clinic activities.	
Precondition: The patient is logged in to the system and has an upcoming appointment or activity that requires a reminder.	
Main Flow of Events	
Actor Action	System Responses
	1. The system checks for upcoming appointments or activities associated with the patient.
	2. The system identifies an upcoming appointment or activity that requires a reminder.
	3. The system generates a reminder for the patient.
4. The patient receives the reminder.	
5. The patient selects the reminder to view.	6. The system displays the details of the reminder.
Alternative flow: A1 step 2: If there are no upcoming appointments or activities that require a reminder, the system does not generate a reminder. A2 step 3: If the reminder cannot be generated due to a system error, the system does not send the reminder and records the unsuccessful reminder process.	
Postcondition: The reminder is successfully generated and delivered to the patient for the relevant upcoming appointment or activity.	

Table 4.4.17: Detailed Use Case Description for Receive Reminder

Use Case Name: Mark Notification as Read	
Actor: Guest, Patient	
Brief Description: This use case allows a guest or patient to mark a notification as read after viewing the notification.	
Precondition: The guest or patient has accessed the notification section and an unread notification is available.	
Main Flow of Events	
Actor Action	System Responses
1. Guest or patient accesses the notification section.	2. The system displays the available notifications and their read or unread status.
3. Guest or patient selects an unread notification.	4. The system displays the details of the selected notification.
5. Guest or patient marks the notification as read.	6. The system updates the selected notification status from unread to read.
	7. The system displays the updated notification status.
Alternative flow: A1 step 3: If the selected notification has already been marked as read, the system displays the notification without changing its status. A2 step 6: If the notification status cannot be updated due to a system error, the system displays an error message and retains the existing notification status.	
Postcondition: The selected notification is successfully marked as read and its updated status is displayed in the system.	

Table 4.4.18: Detailed Use Case Description for Mark Notification as Read

Use Case Name: View Personal Health Records	
Actor: Patient	
Brief Description: This use case allows a patient to view the personal health records stored in the system, including blood pressure, allergies, existing health conditions, medical history, symptoms, and health notes.	
Precondition: The patient is logged in to the system and has accessed the personal health record management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the personal health record option.	2. The system receives the request to view the patient's personal health records.
	3. The system retrieves the personal health records associated with the patient.
	4. The system displays the patient's available personal health records.
5. The patient selects a personal health record to view.	6. The system retrieves the details of the selected personal health record.
	7. The system displays the details of the selected personal health record.
Alternative flow: A1 step 3: If the patient does not have any personal health records, the system displays a message indicating that no personal health records are available. A2 step 3: If the personal health records cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A3 step 4: Patients may search or filter the personal health records to locate a specific record. The system displays the personal health records according to the entered search criteria or selected filter.	
Postcondition: The patient's personal health records and the selected record details are successfully displayed without making any changes to the stored health information.	

Table 4.4.19: Detailed Use Case Description for View Personal Health Records

Use Case Name: Add Personal Health Record	
Actor: Patient	
Brief Description: This use case allows a patient to add personal health information, such as blood pressure, allergies, existing health conditions, medical history, symptoms, and health notes into the system.	
Precondition: The patient is logged in to the system and has accessed the personal health record management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the add personal health record option.	2. The system displays the personal health record form.
3. The patient selects the type of personal health information to be added.	4. The system displays the relevant fields based on the selected health information type.
5. The patient enters the required personal health information.	
6. The patient clicks the save button.	7. The system validates the entered personal health information.
	8. The system creates a new personal health record.
	9. The system stores the personal health record under the patient's account.
	10. The system displays a successful record creation message.
Alternative flow: A1 step 7: If required information is missing or entered in an invalid format, the system displays the relevant validation message and prompts the patient to correct the information. A2 step 6: If the patient cancels the process before saving the record, the system does not create a new personal health record and returns the patient to the personal health record management page. A3 step 8: If the personal health record cannot be created due to a system error, the system displays an error message and does not store the new record.	
Postcondition: A new personal health record is successfully created and stored under the patient's account.	

Table 4.4.20: Detailed Use Case Description for Add Personal Health Record

Use Case Name: Update Personal Health Record	
Actor: Patient	
Brief Description: This use case allows a patient to update existing personal health information, such as blood pressure, allergies, existing health conditions, medical history, symptoms, and health notes stored in the system.	
Precondition: The patient is logged in to the system and has an existing personal health record.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the personal health record management page.	2. The system displays the patient's existing personal health records.
3. The patient selects a personal health record to update.	4. The system retrieves and displays the details of the selected personal health record.
5. The patient clicks the update or edit option.	6. The system displays the selected personal health record in editable form.
7. The patient modifies the personal health information.	
8. The patient clicks the save button.	9. The system validates the updated personal health information.
	10. The system updates the selected personal health record.
	11. The system stores the updated personal health information.
	12. The system displays a successful record update message.
Alternative flow: A1 step 9: If required information is missing or entered in an invalid format, the system displays the relevant validation message and prompts the patient to correct the information. A2 step 8: If the patient cancels the update process before saving, the system does not save the changes and retains the existing personal health record. A3 step 10: If the personal health record cannot be updated due to a system error, the system displays an error message and retains the previously stored information.	
Postcondition: The selected personal health record is successfully updated, and the latest personal health information is stored under the patient's account.	

Table 4.4.21: Detailed Use Case Description for Update Personal Health Record

Use Case Name: Browse Available Medicines	
Actor: Patient	
Brief Description: This use case allows a patient to browse the medicines that are currently available for purchase in the system.	
Precondition: The patient is logged in to the system and has accessed the medicine purchase management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the medicine purchase management page.	2. The system retrieves the medicines that are currently available for purchase.
	3. The system displays the available medicines with relevant information such as medicine name, price, and availability.
4. The patient browses the available medicine list.	
5. The patient selects a medicine to view.	6. The system retrieves the selected medicine information.
	7. The system displays the details of the selected medicine.
Alternative flow: A1 step 2: If no medicines are currently available, the system displays a message indicating that no medicines are available for purchase. A2 step 2: If the medicine information cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A3 step 3: Patients may search or filter the available medicines to locate a specific medicine. The system displays the medicines according to the entered search criteria or selected filter.	
Postcondition: The available medicines and the selected medicine details are successfully displayed without making any changes to the medicine records.	

Table 4.4.22: Detailed Use Case Description for Browse Available Medicine

Use Case Name: View Medicine Details	
Actor: Patient	
Brief Description: This use case allows a patient to view detailed information about a selected medicine before deciding whether to add the medicine to the shopping cart.	
Precondition: The patient is logged in to the system and has accessed the medicine purchase management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient browses the available medicines.	2. The system displays the list of available medicines.
3. The patient selects a medicine to view its details.	4. The system retrieves the information of the selected medicine.
	5. The system displays the medicine details, such as medicine name, description, price, and availability.
6. The patient reviews the displayed medicine details.	
7. Patients may choose to add the medicine to the shopping cart.	8. The system adds the selected medicine to the patient's shopping cart.
Alternative flow: A1 step 4: If the selected medicine information cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A2 step 5: If the selected medicine is unavailable, the system displays the medicine as unavailable and prevents it from being added to the shopping cart. A3 step 7: If the patient does not wish to add the medicine to the shopping cart, the patient may return to the available medicine list and continue browsing.	
Postcondition: The selected medicine details are successfully displayed. If the patient chooses to add the medicine to the shopping cart, the selected medicine is added to the cart.	

Table 4.4.23: Detailed Use Case Description for View Medicine Details

Use Case Name: Manage Shopping Cart	
Actor: Patient	
Brief Description: This use case allows a patient to manage the medicines added to the shopping cart before placing an order. The patient can review the selected medicines, update the medicine quantity, or remove medicines from the cart.	
Precondition: The patient is logged in to the system and has at least one medicine added to the shopping cart.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the shopping cart.	2. The system retrieves the medicines currently added to the patient's shopping cart.
	3. The system displays the shopping cart with the selected medicines, quantities, prices, and total amount.
4. The patient reviews the medicines in the shopping cart.	
5. The patient updates the quantity of a selected medicine.	6. The system validates the updated quantity and recalculates the total amount.
7. Patients may remove medicine from the shopping cart.	8. The system removes the selected medicine and updates the shopping cart total.
9. Patient reviews the updated shopping cart.	10. The system displays the latest shopping cart information.
Alternative flow: A1 step 5: If the updated quantity is invalid or exceeds the available medicine quantity, the system displays a validation message and retains the previous quantity. A2 step 7: If the patient cancels the removal action, the system keeps the medicine in the shopping cart. A3 step 2: If the shopping cart is empty, the system displays a message indicating that no medicines have been added to the cart. A4 step 2: If the shopping cart information cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again.	
Postcondition: The patient's shopping cart is updated according to the selected actions, and the latest medicine quantities and total amount are displayed.	

Table 4.4.24: Detailed Use Case Description for Manage Shopping Cart

Use Case Name: Place Medicine Order	
Actor: Patient	
Brief Description: This use case allows a patient to place an order for the medicines added to the shopping cart by confirming the order details.	
Precondition: The patient is logged in to the system and has at least one medicine added to the shopping cart.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the shopping cart.	2. The system displays the medicines added to the shopping cart and the total amount.
3. The patient clicks the place order button.	4. The system displays the order details for confirmation.
5. The patient reviews the medicine, quantity, price, and total amount of the order.	
6. The patient confirms the medicine order.	7. The system validates the order information and medicine availability.
	8. The system creates a new medicine order.
	9. The system stores the medicine order details.
	10. The system updates the availability of the ordered medicines.
	11. The system clears the successfully ordered medicines from the shopping cart.
	12. The system displays a successful order confirmation message with the order details.
Alternative flow: A1 step 7: If one or more medicines are no longer available or the requested quantity exceeds the available quantity, the system informs the patient and prompts the patient to update the shopping cart. A2 step 7: If the order information is invalid or incomplete, the system displays a validation message and prevents the order from being submitted.	
Postcondition: A new medicine order is successfully created and stored in the system. The ordered medicines are removed from the shopping cart and the patient can view the order in the purchase history.	

Table 4.4.25: Detailed Use Case Description for Place Medicine Order

Use Case Name: View Purchase History	
Actor: Patient	
Brief Description: This use case allows a patient to view previous medicine orders and their relevant purchase details.	
Precondition: The patient is logged in to the system and has accessed the medicine purchase management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient clicks the purchase history option.	2. The system receives the request to view the patient's purchase history.
	3. The system retrieves the medicine order records associated with the patient.
	4. The system displays the patient's purchase history.
5. The patient selects a medicine order to view.	6. The system retrieves the details of the selected medicine order.
	7. The system displays the purchase details, including the ordered medicines, quantities, total amount, and order status.
Alternative flow: A1 step 3: If the patient does not have any previous medicine orders, the system displays a message indicating that no purchase history is available. A2 step 3: If the purchase history cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A3 step 5: If the selected purchase record cannot be retrieved, the system displays an error message and returns the patient to the purchase history.	
Postcondition: The patient's purchase history and the details of the selected medicine order are successfully displayed without making any changes to the purchase records.	

Table 4.4.26: Detailed Use Case Description for View Purchase History

Use Case Name: Start Chat	
Actor: Guest, Patient	
Brief Description: This use case allows a guest or patient to start a conversation with the AI health chatbot by entering health-related questions or symptoms. A guest's chat is session-based and is not saved, while a patient's chat can be saved under the patient account.	
Precondition: The guest or patient has accessed the AI health chatbot page. The patient must be logged in for the chat to be saved under the patient account.	
Main Flow of Events	
Actor Action	System Responses
1. Guest or patient accesses the AI health chatbot page.	2. The system displays the AI health chatbot interface.
3. Guest or patient starts a new chat.	4. The system creates a new chat session.
5. Guest or patient enters a health-related question or symptom.	6. The system receives and processes the entered message.
	7. The system generates an AI response based on the entered message.
	8. The system displays the AI-generated response in the chat.
9. Guests or patients may continue the conversation by entering another message.	10. The system processes the new message and displays the corresponding AI-generated response.
	11. If the actor is a logged-in patient, the system saves the chat under the patient account.
Alternative flow: A1 step 5: If the guest or patient submits an empty message, the system does not process the request and prompts the user to enter a message. A2 step 6: If the message cannot be processed due to a system or AI service error, the system displays an error message and prompts the user to try again. A3 step 11: If the actor is a guest, the system keeps the chat within the current session only and does not save the chat to an account.	
Postcondition: A chat session is created and the AI-generated response is displayed. For a patient, the chat is saved under the patient account. For a guest, the chat remains session-based and is not stored as a saved chat.	

Table 4.4.27: Detailed Use Case Description for Start Chat

Use Case Name: View Chat History	
Actor: Patient	
Brief Description: This use case allows a patient to view previous AI health chatbot conversations that have been saved under the patient account.	
Precondition: The patient is logged in to the system and has accessed the AI health chatbot page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the AI health chatbot page.	2. The system displays the AI health chatbot interface.
3. The patient clicks the chat history option.	4. The system retrieves the saved chat records associated with the patient account.
	5. The system displays the patient's saved chat history.
6. The patient selects a saved chat to view.	7. The system retrieves the messages associated with the selected chat.
	8. The system displays the complete conversation of the selected chat.
9. Patient reviews the previous conversation.	
Alternative flow: A1 step 4: If the patient does not have any saved chat history, the system displays a message indicating that no chat history is available. A2 step 4: If the chat history cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again. A3 step 6: The patient may choose to extract the selected chat. The system extracts the selected chat conversation for the patient.	
Postcondition: The selected saved chat conversation is successfully displayed without making any changes to the stored chat history.	

Table 4.4.28: Detailed Use Case Description for View Chat History

Use Case Name: Delete Chat	
Actor: Patient	
Brief Description: This use case allows a patient to delete a saved AI health chatbot conversation that is no longer required.	
Precondition: The patient is logged in to the system and has at least one saved chat under the patient account.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the AI health chatbot page.	2. The system displays the AI health chatbot interface.
3. The patient accesses the chat history.	4. The system retrieves and displays the patient's saved chat history.
5. The patient selects the chat to be deleted.	6. The system displays the selected chat and the delete option.
7. The patient clicks the delete chat option.	8. The system displays a confirmation message before deleting the chat.
9. The patient confirms the deletion.	10. The system deletes the selected chat from the patient's saved chat history.
	11. The system displays a successful deletion message.
	12. The system refreshes the chat history and displays the remaining saved chats.
Alternative flow: A1 step 9: If the patient cancels the deletion, the system closes the confirmation message and keeps the selected chat unchanged. A2 step 10: If the selected chat no longer exists, the system displays a message indicating that the chat cannot be found and refreshes the chat history. A3 step 10: If the chat cannot be deleted due to a system error, the system displays an error message and retains the selected chat in the patient's chat history.	
Postcondition: The selected chat is successfully removed from the patient's saved chat history and can no longer be viewed by the patient.	

Table 4.4.29: Detailed Use Case Description for Delete Chat

Use Case Name: View Family Information	
Actor: Patient	
Brief Description: This use case allows a patient to view the family information associated with the patient account, including the registered parent or guardian information.	
Precondition: The patient is logged in to the system and has accessed the parent/guardian management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the parent/guardian management page.	2. The system receives the request to view the patient's family information.
	3. The system retrieves the family information associated with the patient account.
	4. The system displays the available parent or guardian information.
5. The patient selects a parent or guardian record to view.	6. The system retrieves the details of the selected parent or guardian record.
	7. The system displays the selected parent or guardian information to the patient.
Alternative flow: A1 step 3: If no parent or guardian information has been registered, the system displays a message indicating that no family information is available. A2 step 3: If the family information cannot be retrieved due to a system error, the system displays an error message and prompts the patient to try again.	
Postcondition: The patient's family information and the selected parent or guardian details are successfully displayed without making any changes to the stored information.	

Table 4.4.30: Detailed Use Case Description for View Family Information

Use Case Name: Add Family Member	
Actor: Patient	
Brief Description: This use case allows a patient to add a family member, such as a parent or guardian, by providing the required family member information.	
Precondition: The patient is logged in to the system and has accessed the parent/guardian management page.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the parent/guardian management page.	2. The system displays the patient's existing family information.
3. The patient clicks the add family member option.	4. The system displays the family member information form.
5. The patient enters the required family member information.	
6. The patient clicks the save button.	7. The system validates the entered family member information.
	8. The system creates a new family member record.
	9. The system associates the new family member with the patient account.
	10. The system stores the family member information.
	11. The system displays a successful family member addition message.
Alternative flow: A1 step 7: If required information is missing or entered in an invalid format, the system displays the relevant validation message and prompts the patient to correct the information. A2 step 6: If the patient cancels the process before saving, the system does not create a new family member record and returns the patient to the parent/guardian management page. A3 step 8: If the family member cannot be added due to a system error, the system displays an error message and does not store the new family member information.	
Postcondition: A new family member is successfully added, associated with the patient account, and stored in the system.	

Table 4.4.31: Detailed Use Case Description for Add Family Member

Use Case Name: Update Family Member	
Actor: Patient	
Brief Description: This use case allows a patient to update the information of an existing family member, such as a parent or guardian, associated with the patient account.	
Precondition: The patient is logged in to the system and has at least one family member associated with the patient account.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the parent/guardian management page.	2. The system retrieves and displays the patient's existing family information.
3. The patient selects a family member to update.	4. The system retrieves and displays the information of the selected family member.
5. The patient clicks the edit or update option.	6. The system displays the selected family member information in editable form.
7. The patient modifies the family member information.	
8. The patient clicks the save button.	9. The system validates the updated family member information.
	10. The system updates the selected family member record.
	11. The system stores the updated family member information.
	12. The system displays a successful family member update message.
Alternative flow: A1 step 9: If required information is missing or entered in an invalid format, the system displays the relevant validation message and prompts the patient to correct the information. A2 step 8: If the patient cancels the update process before saving, the system does not save any changes and retains the existing family member information. A3 step 10: If the family member information cannot be updated due to a system error, the system displays an error message and retains the previously stored family member information.	
Postcondition: The selected family member information is successfully updated and the latest information is stored under the patient account.	

Table 4.4.32: Detailed Use Case Description for Update Family Member

Use Case Name: Remove Family Member	
Actor: Patient	
Brief Description: This use case allows a patient to remove an existing family member, such as a parent or guardian, associated with the patient account.	
Precondition: The patient is logged in to the system and has at least one family member associated with the patient account.	
Main Flow of Events	
Actor Action	System Responses
1. The patient accesses the parent/guardian management page.	2. The system retrieves and displays the patient's existing family information.
3. The patient selects a family member to remove.	4. The system retrieves and displays the information of the selected family member.
5. The patient clicks the remove family member option.	6. The system displays a confirmation message before removing the selected family member.
7. The patient confirms the removal.	8. The system removes the selected family member from the patient account.
	9. The system updates the patient's family information.
	10. The system displays a successful family member removal message.
Alternative flow: A1 step 7 If the patient cancels the removal, the system closes the confirmation message and retains the selected family member information. A2 step 8 If the selected family member no longer exists, the system displays a message indicating that the family member cannot be found and refreshes the family information. A3 step 8 If the family member cannot be removed due to a system error, the system displays an error message and retains the existing family member information.	
Postcondition: The selected family member is successfully removed from the patient account and is no longer displayed in the patient's family information.	

Table 4.4.33: Detailed Use Case Description for Remove Family Member

4.5 Sequence Diagram

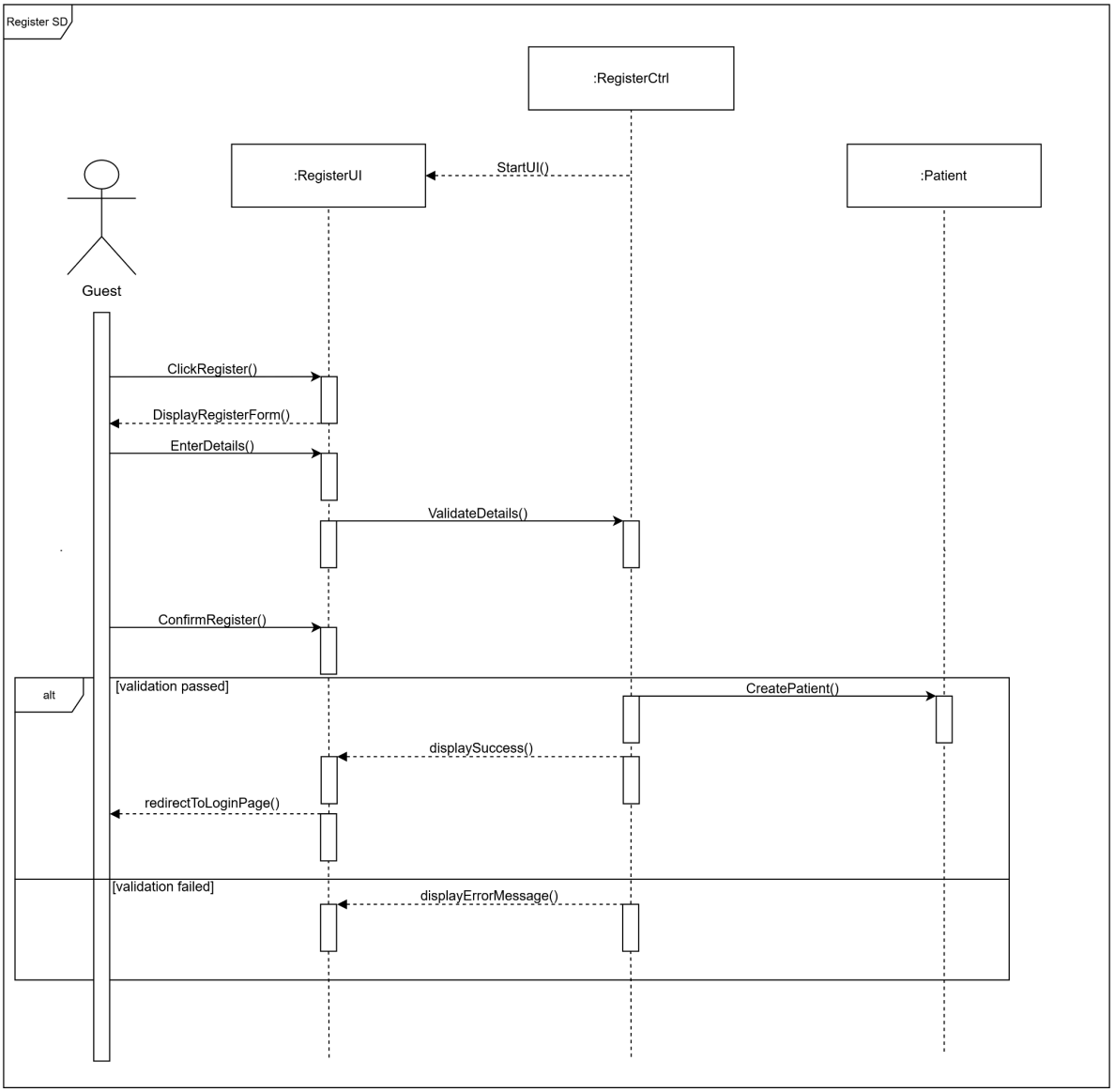


Figure 4.5.1: Sequence Diagram for Register Account

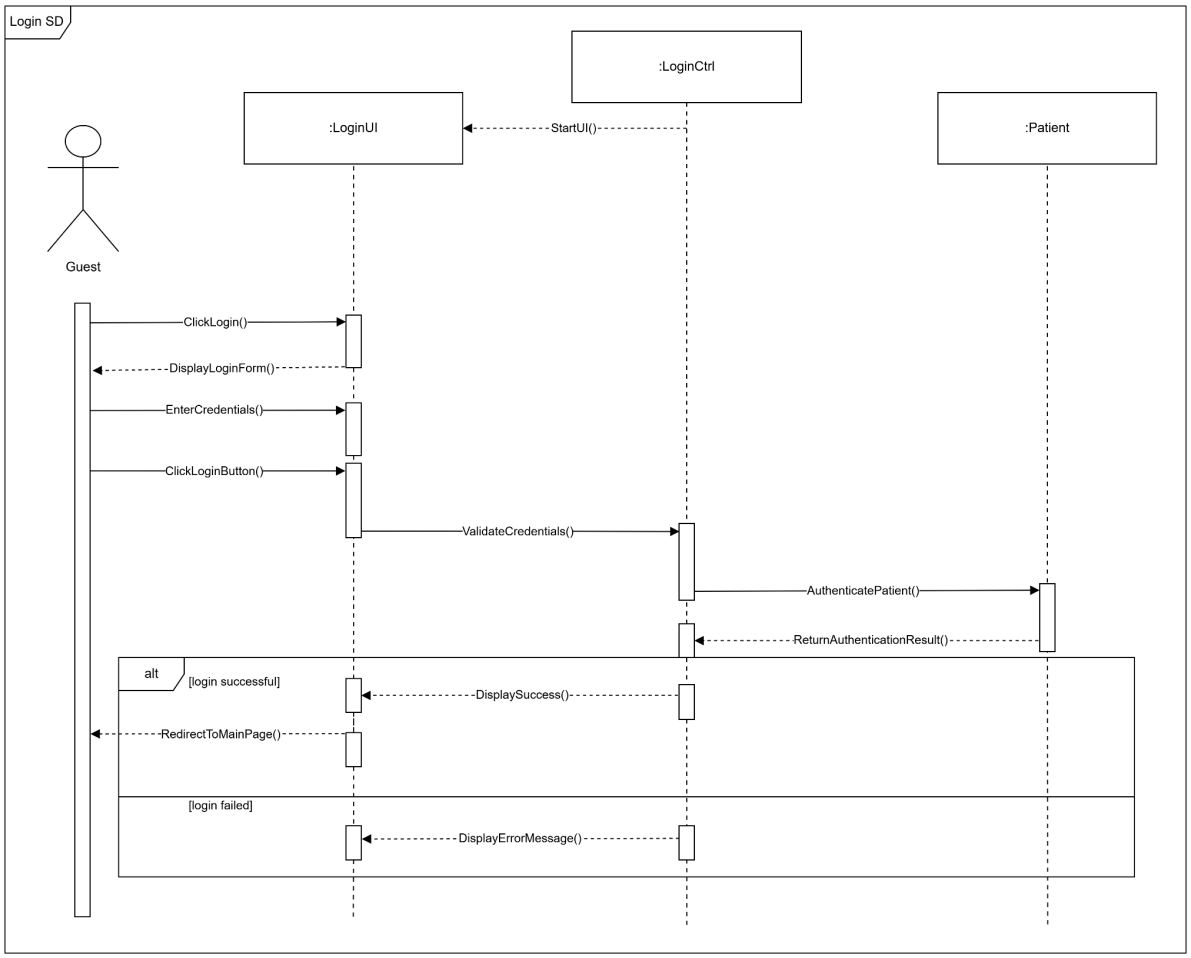


Figure 4.5.2: Sequence Diagram for Login Account

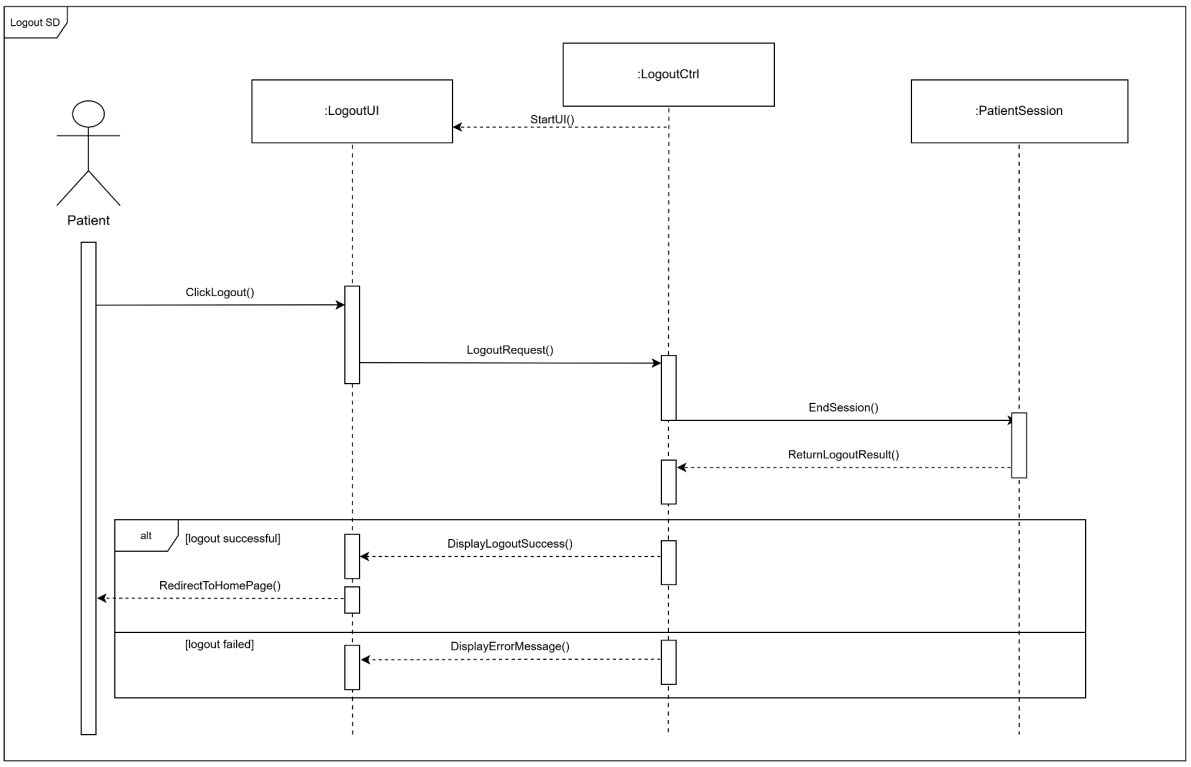


Figure 4.5.3: Sequence Diagram for Logout Account

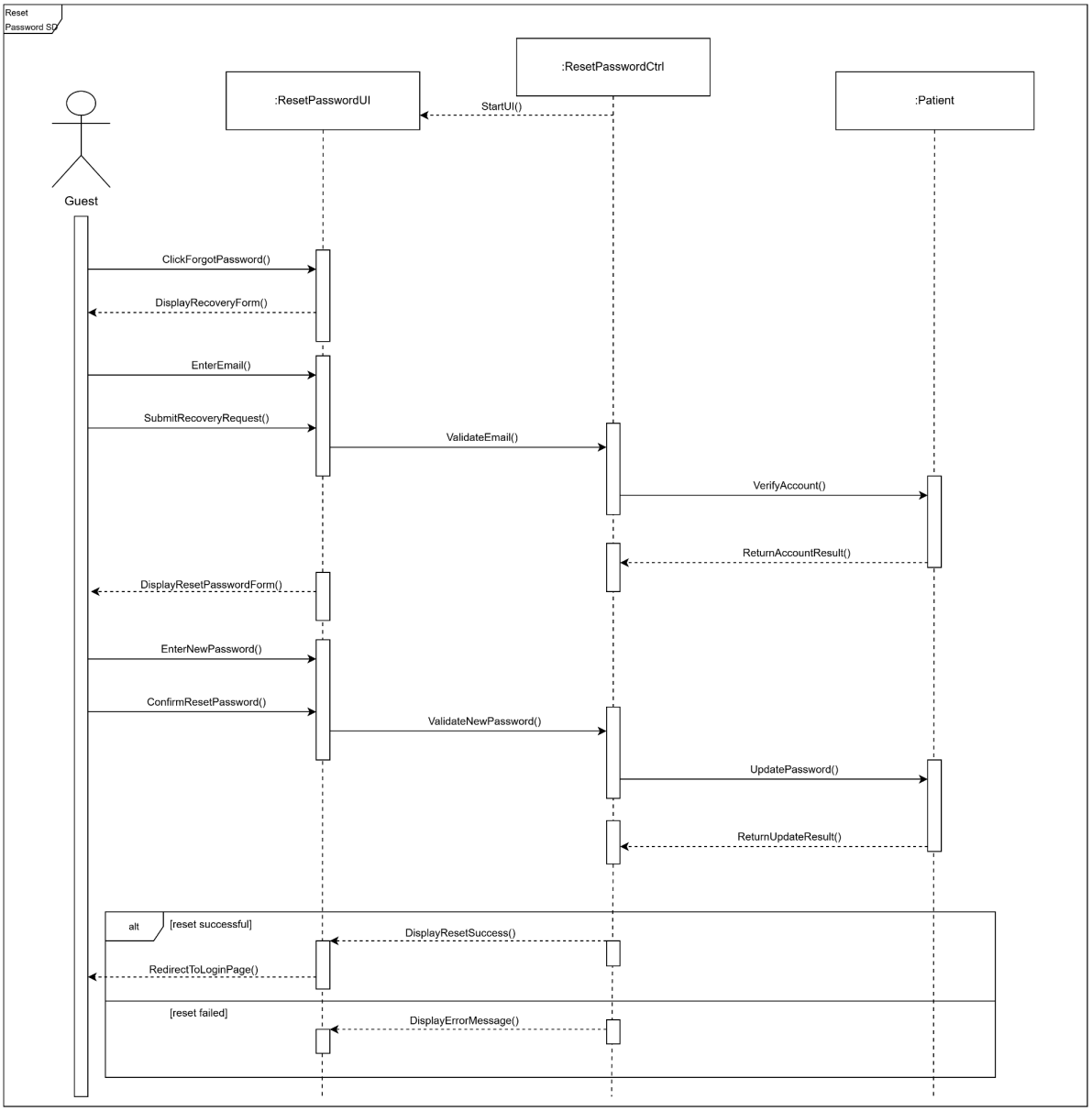


Figure 4.5.4: Sequence Diagram for Recover/Reset Password

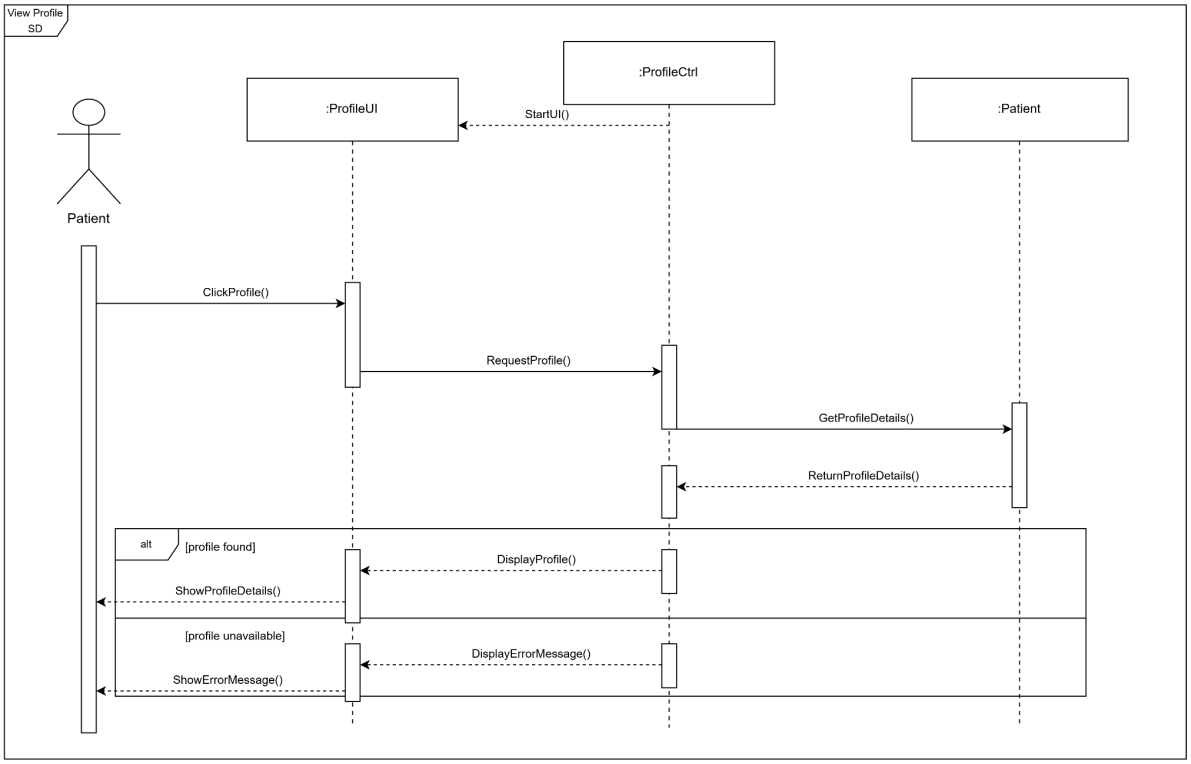


Figure 4.5.5: Sequence Diagram for View Profile

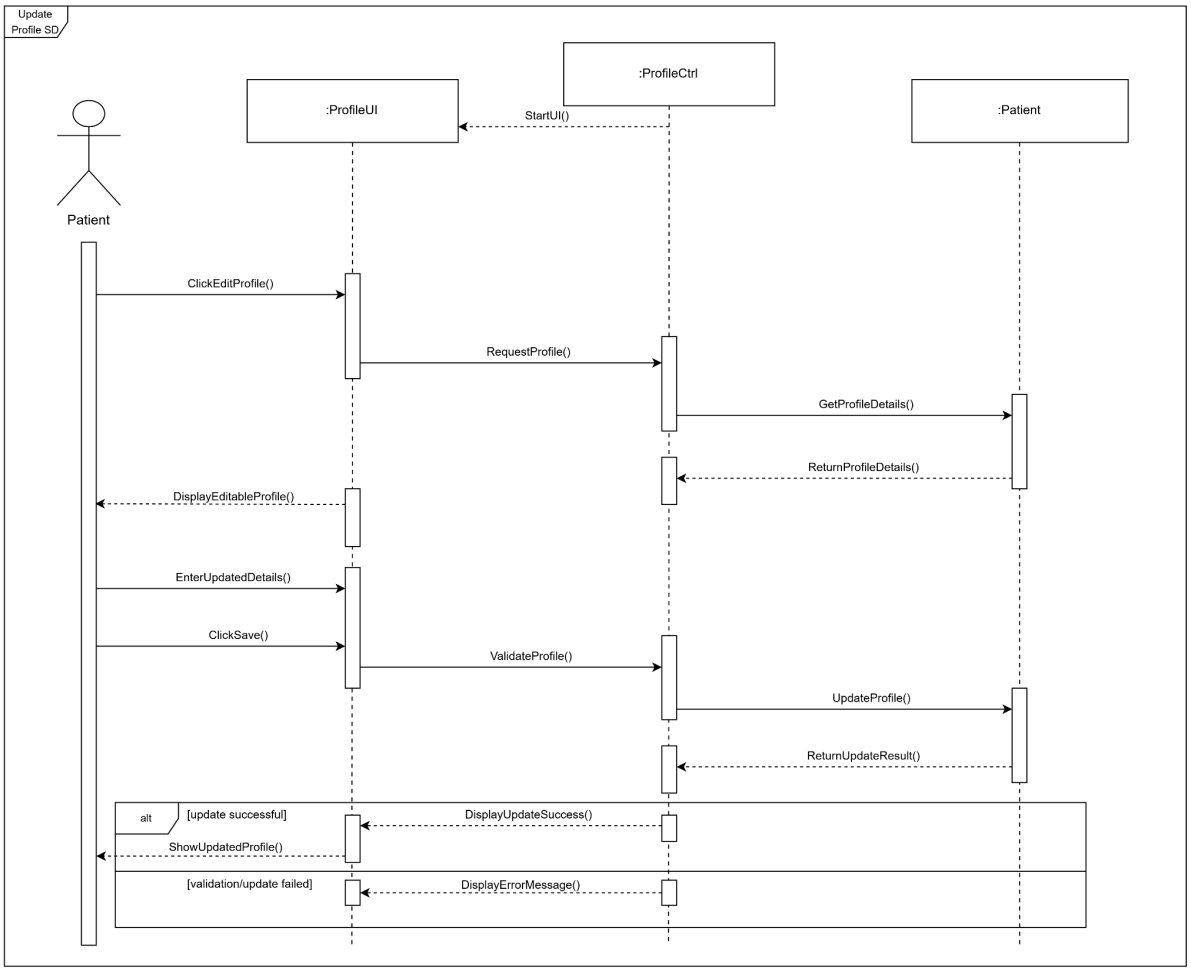


Figure 4.5.6: Sequence Diagram for Update Profile

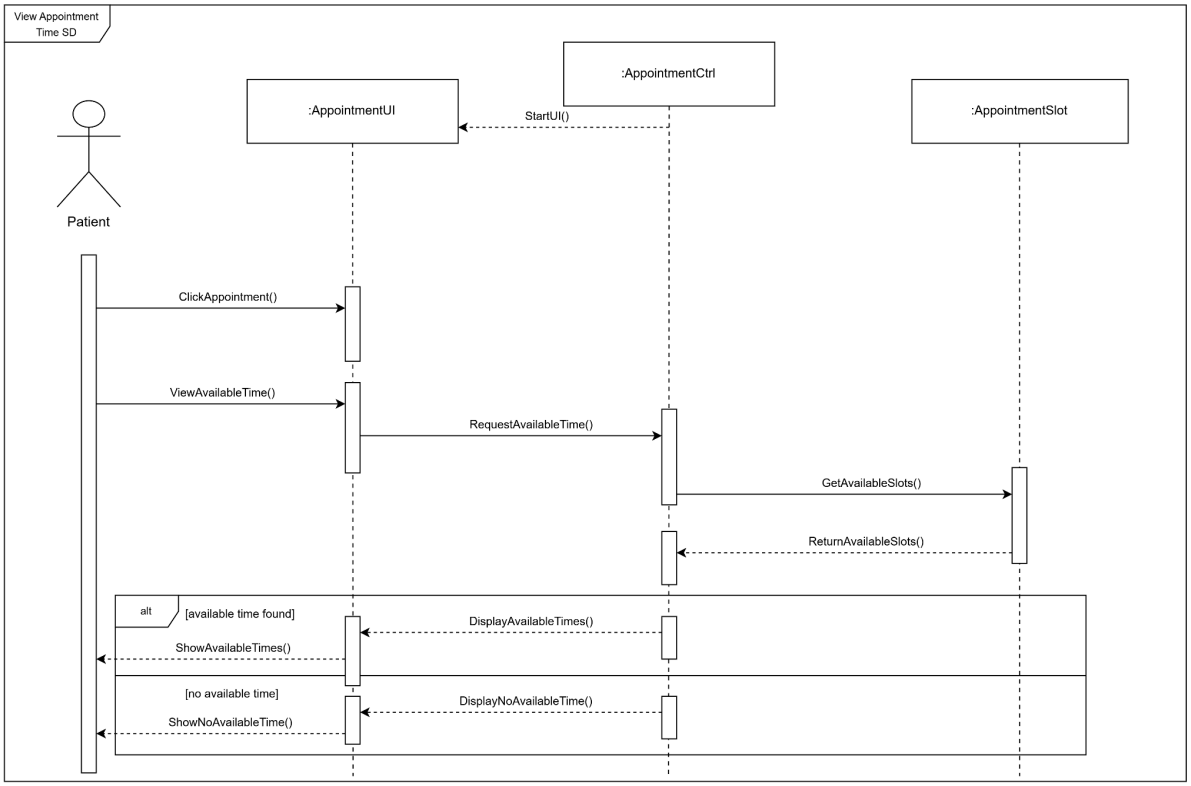


Figure 4.5.7: Sequence Diagram for View Appointment Time

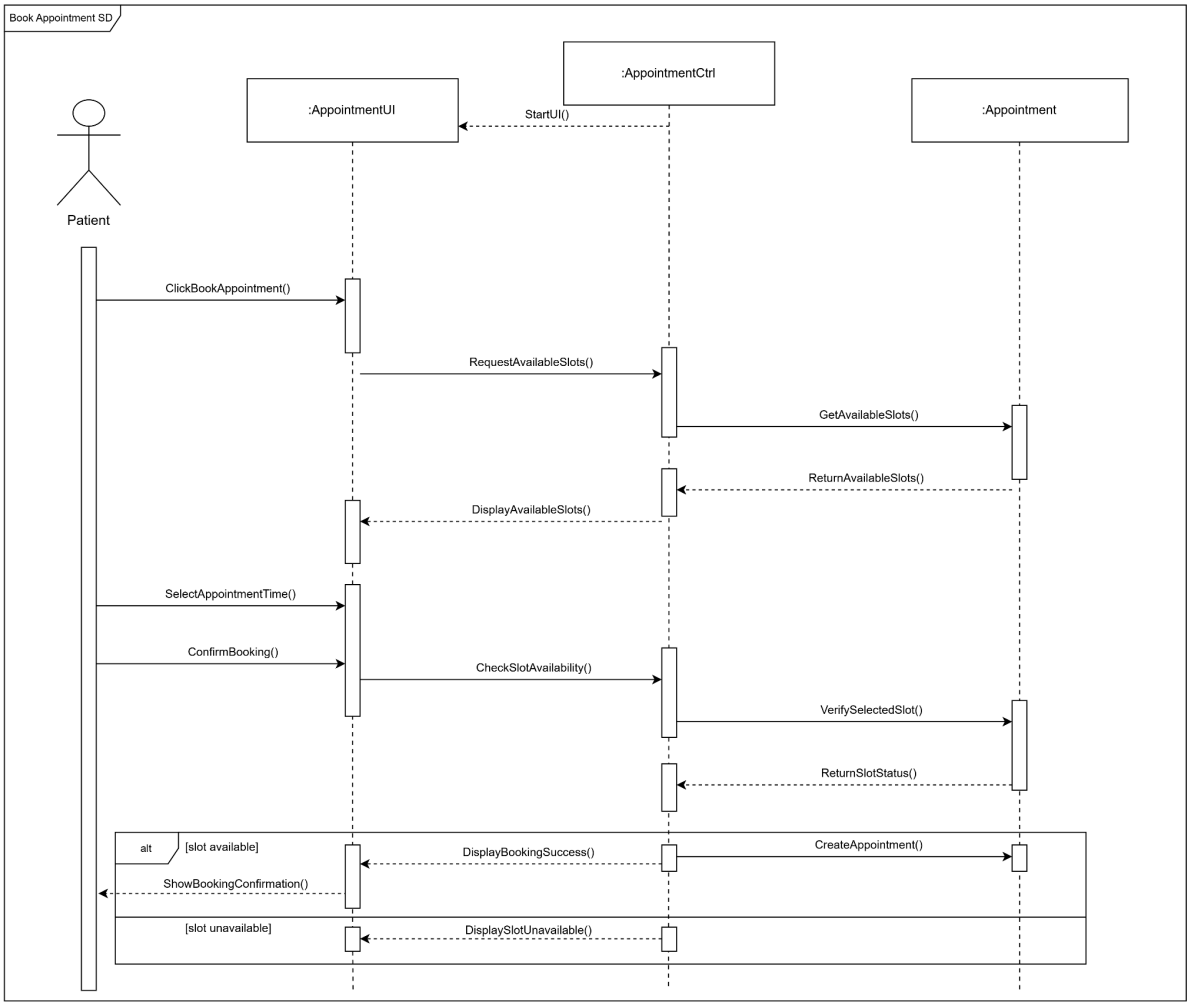


Figure 4.5.8: Sequence Diagram for Book Appointment

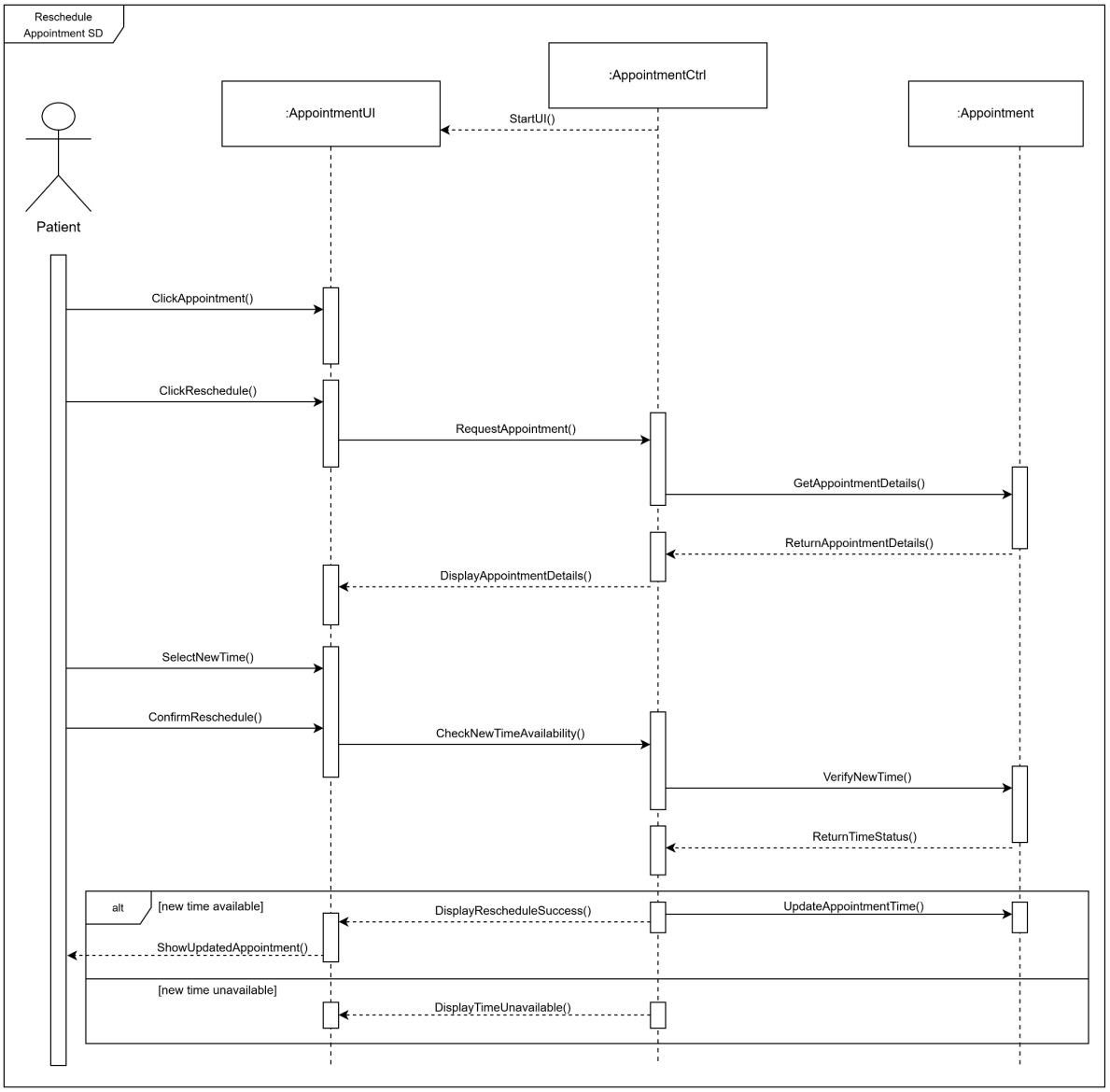


Figure 4.5.9: Sequence Diagram for Reschedule Appointment

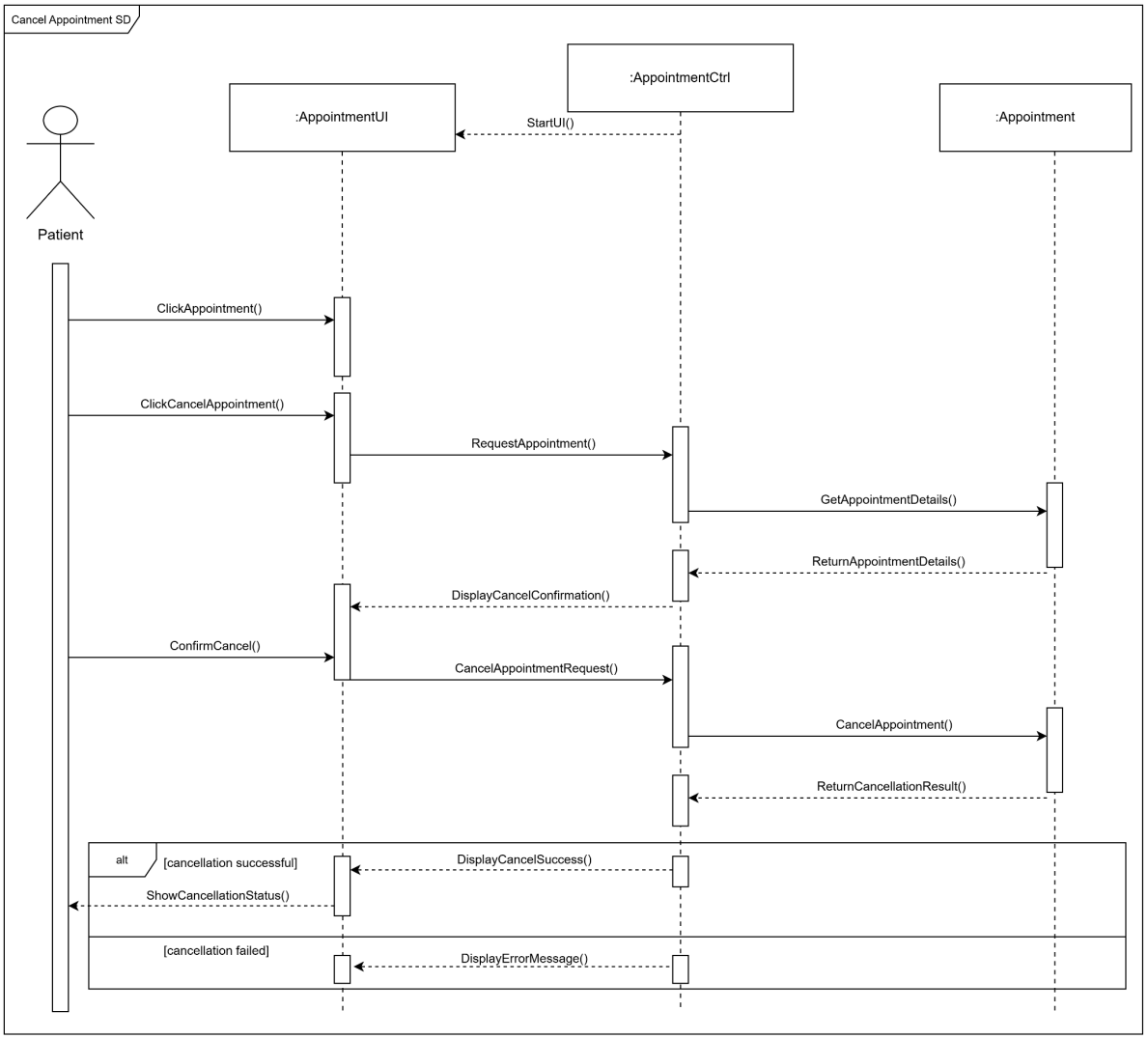


Figure 4.5.10: Sequence Diagram for Cancel Appointment

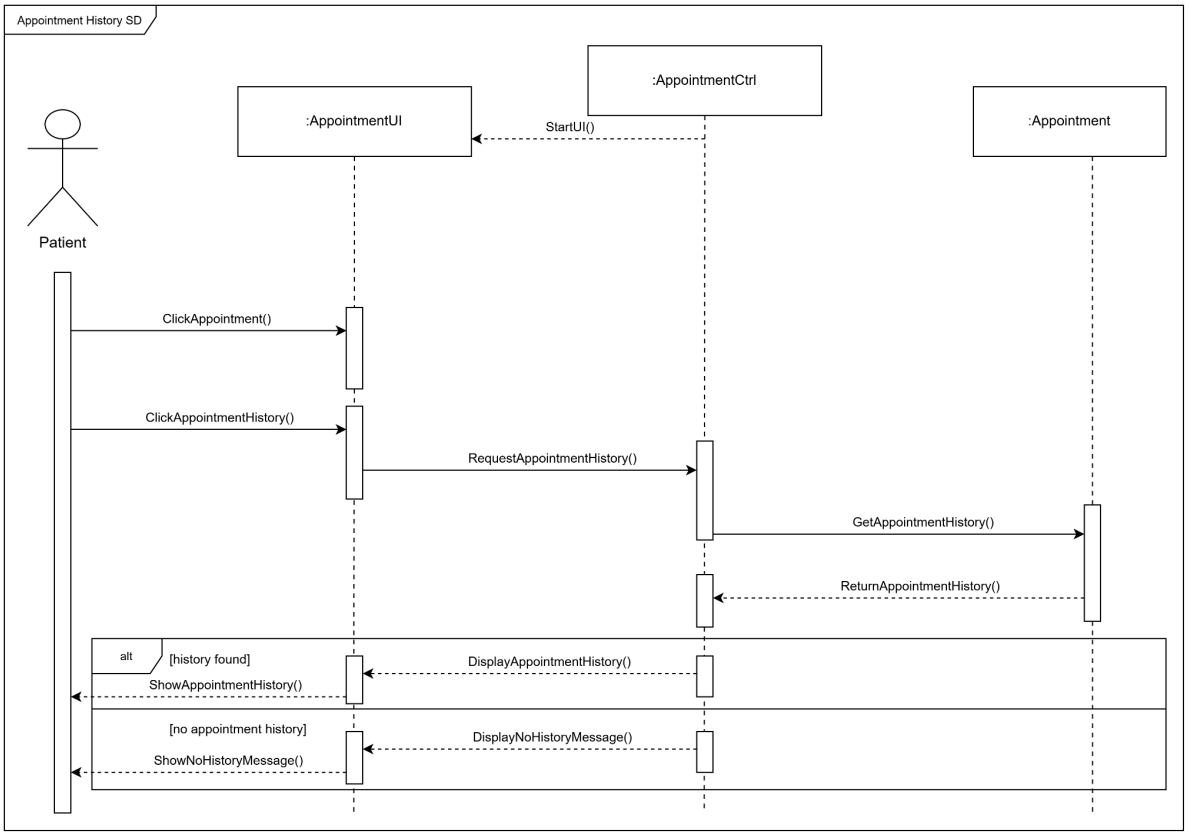


Figure 4.5.11: Sequence Diagram for Appointment History

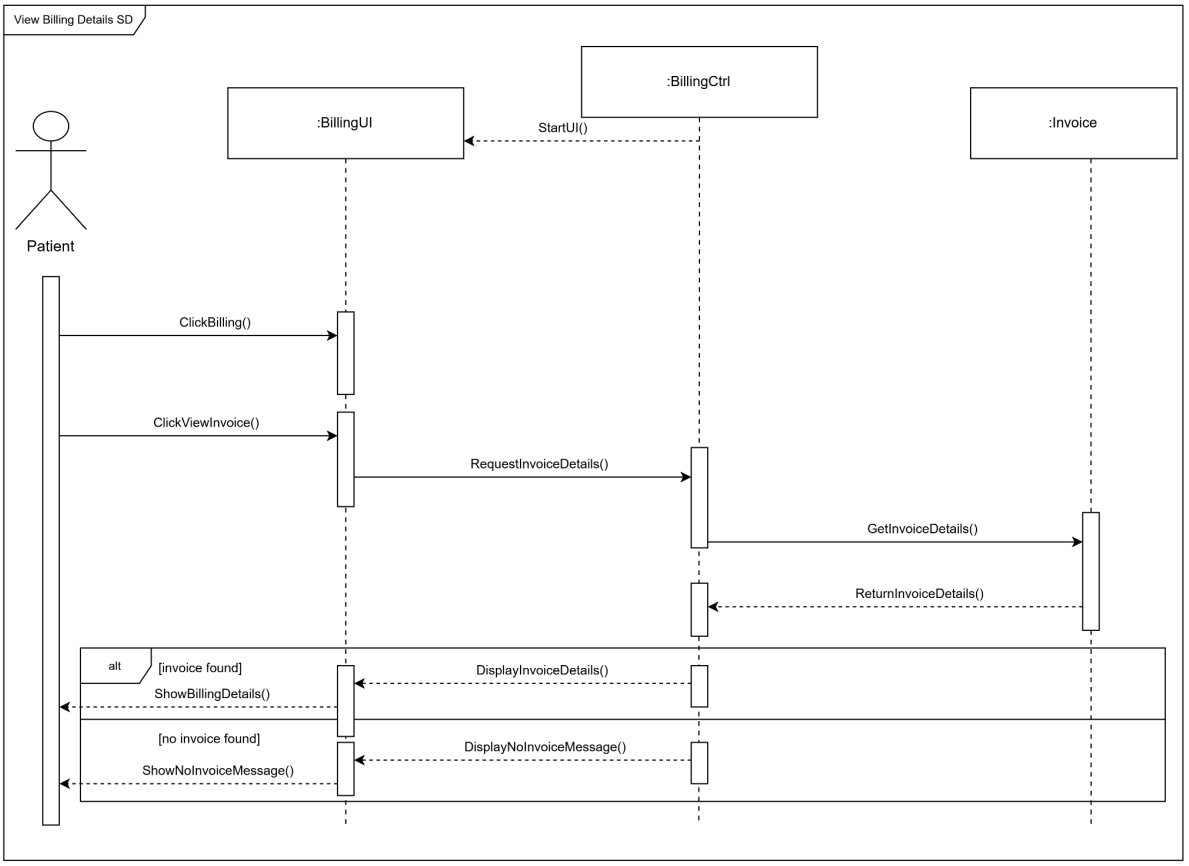


Figure 4.5.12: Sequence Diagram for View Invoice & Billing Details

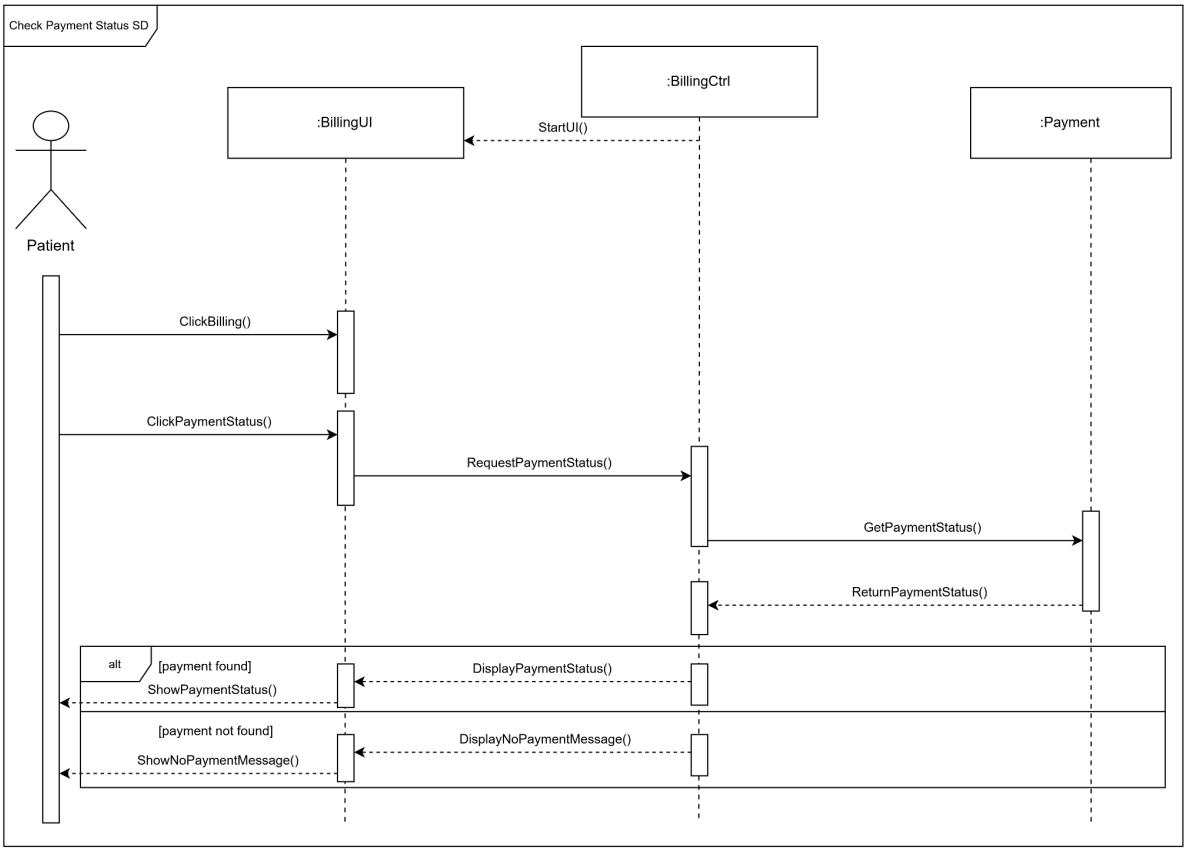


Figure 4.5.13: Sequence Diagram for Check Payment Status

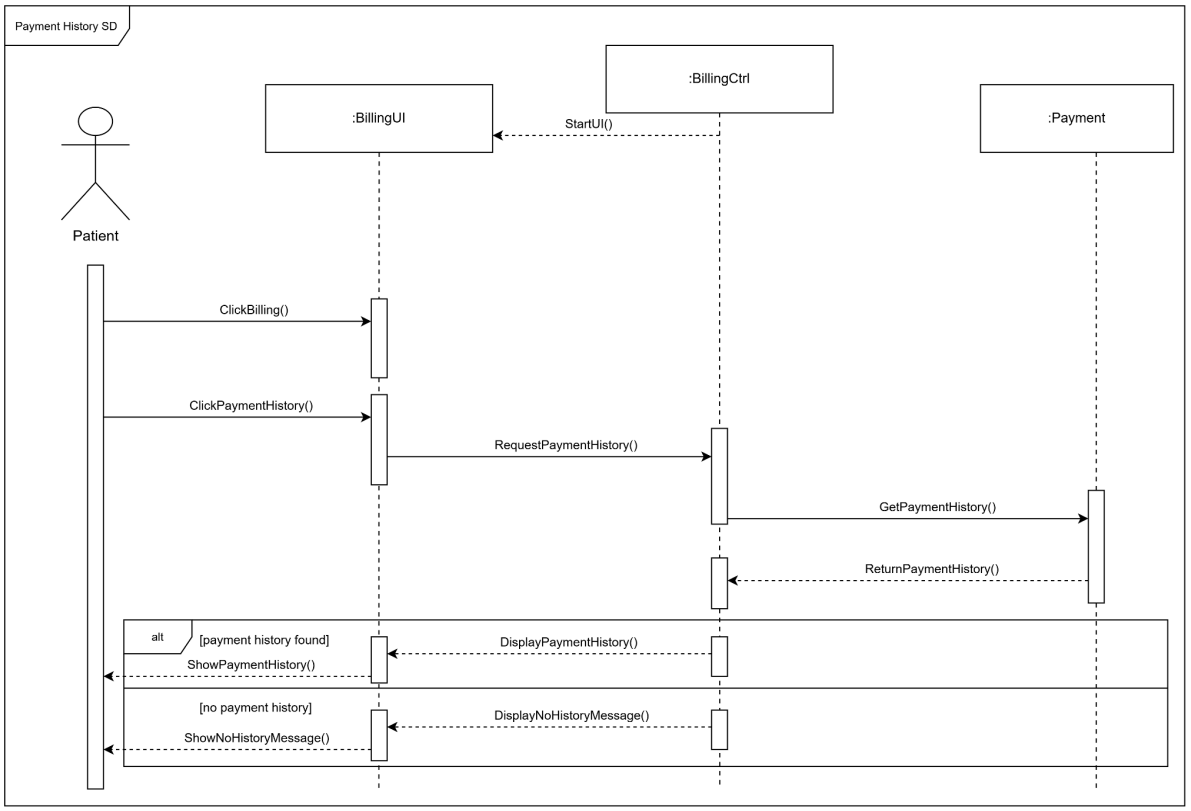


Figure 4.5.14: Sequence Diagram for View Payment History

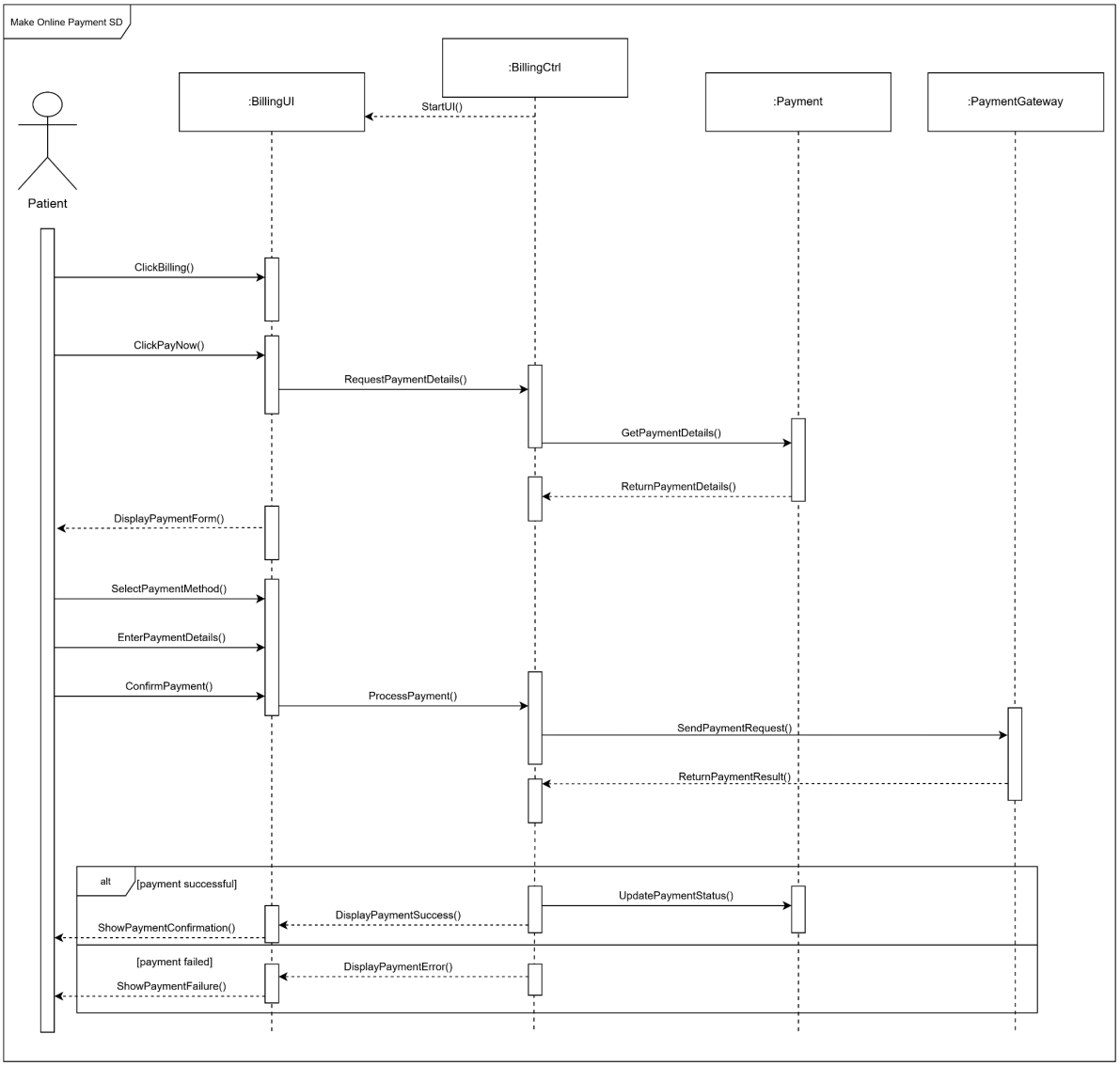


Figure 4.5.15: Sequence Diagram for Make Online Payment

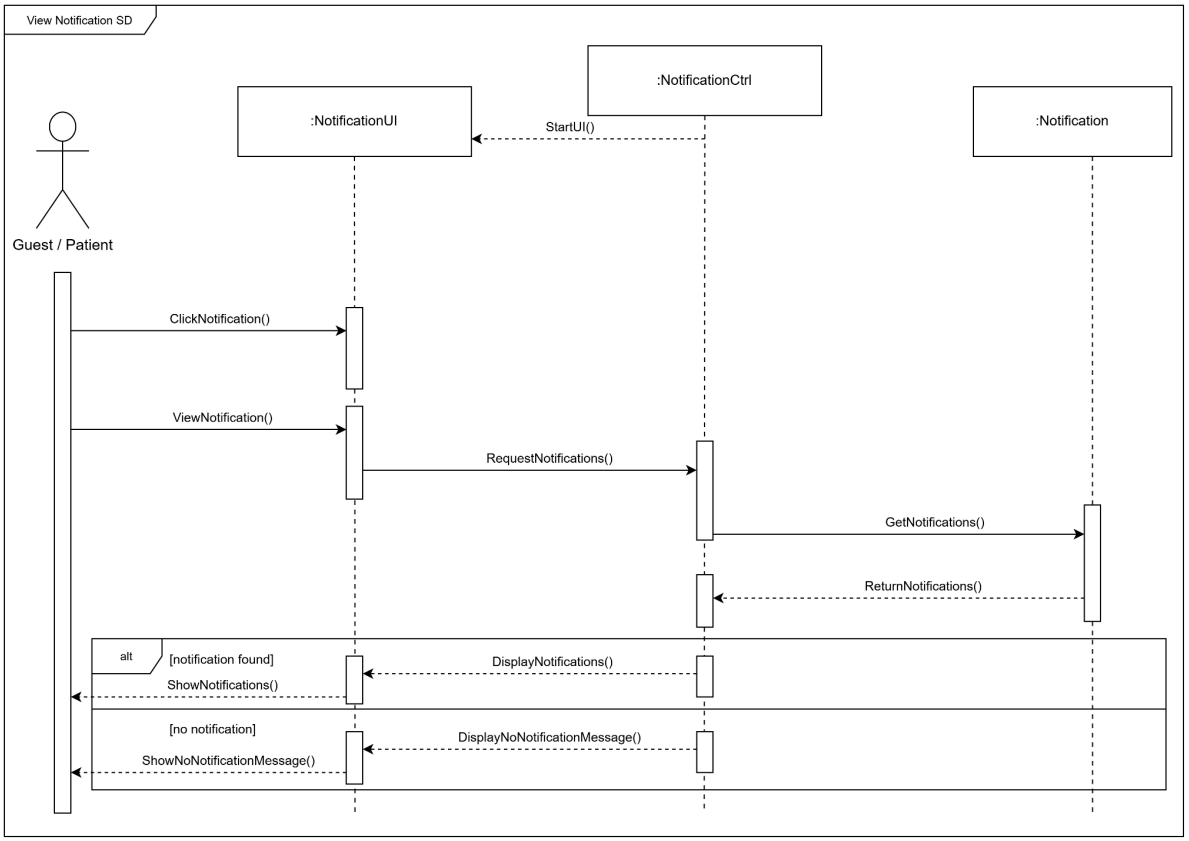


Figure 4.5.16: Sequence Diagram for View Notifications

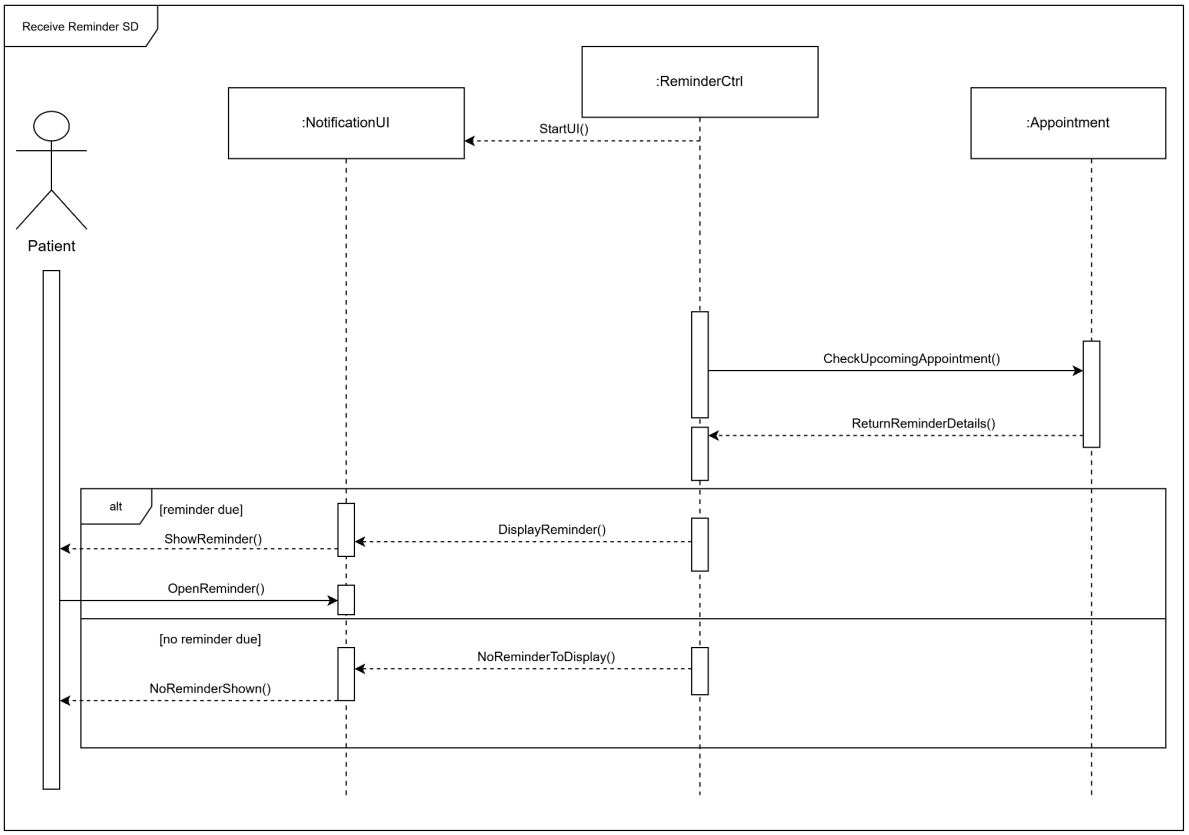


Figure 4.5.17: Sequence Diagram for Receive Reminder

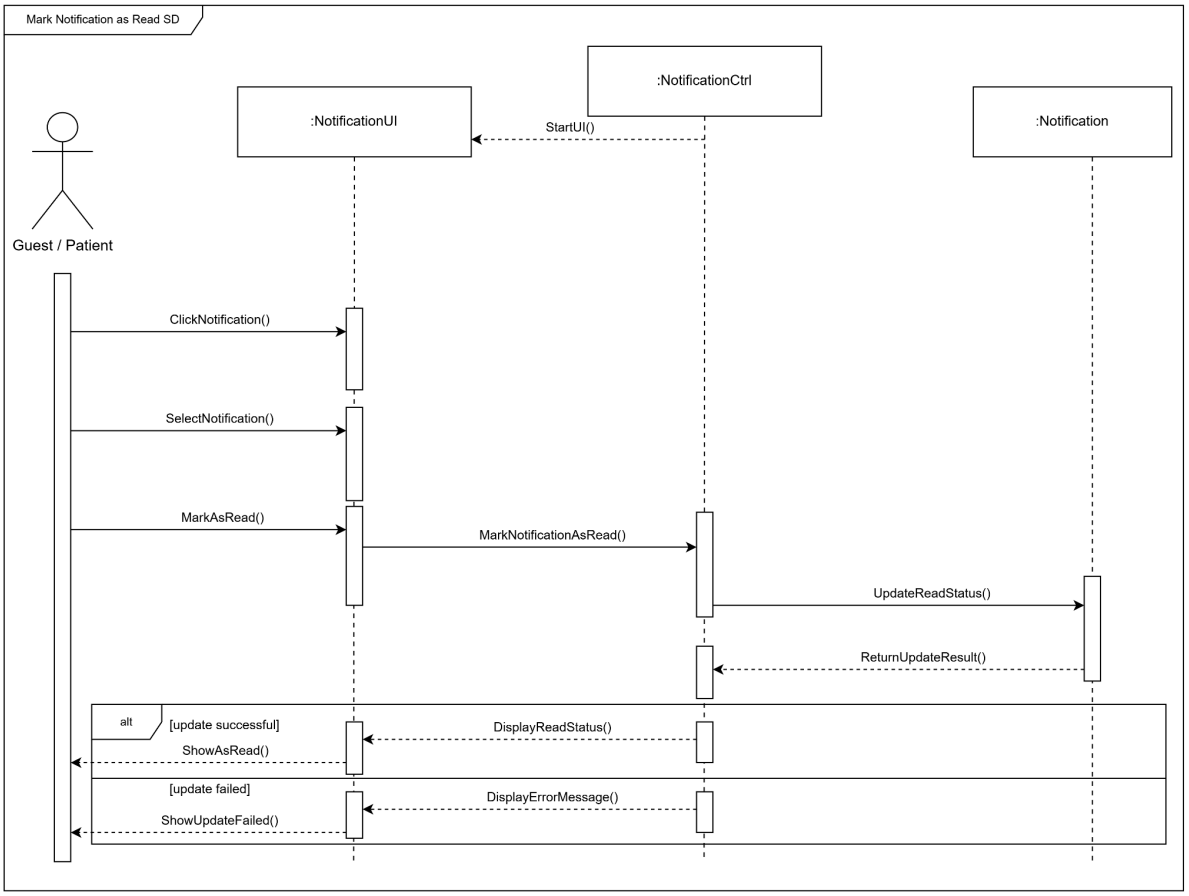


Figure 4.5.18: Sequence Diagram for Mark Notification as Read

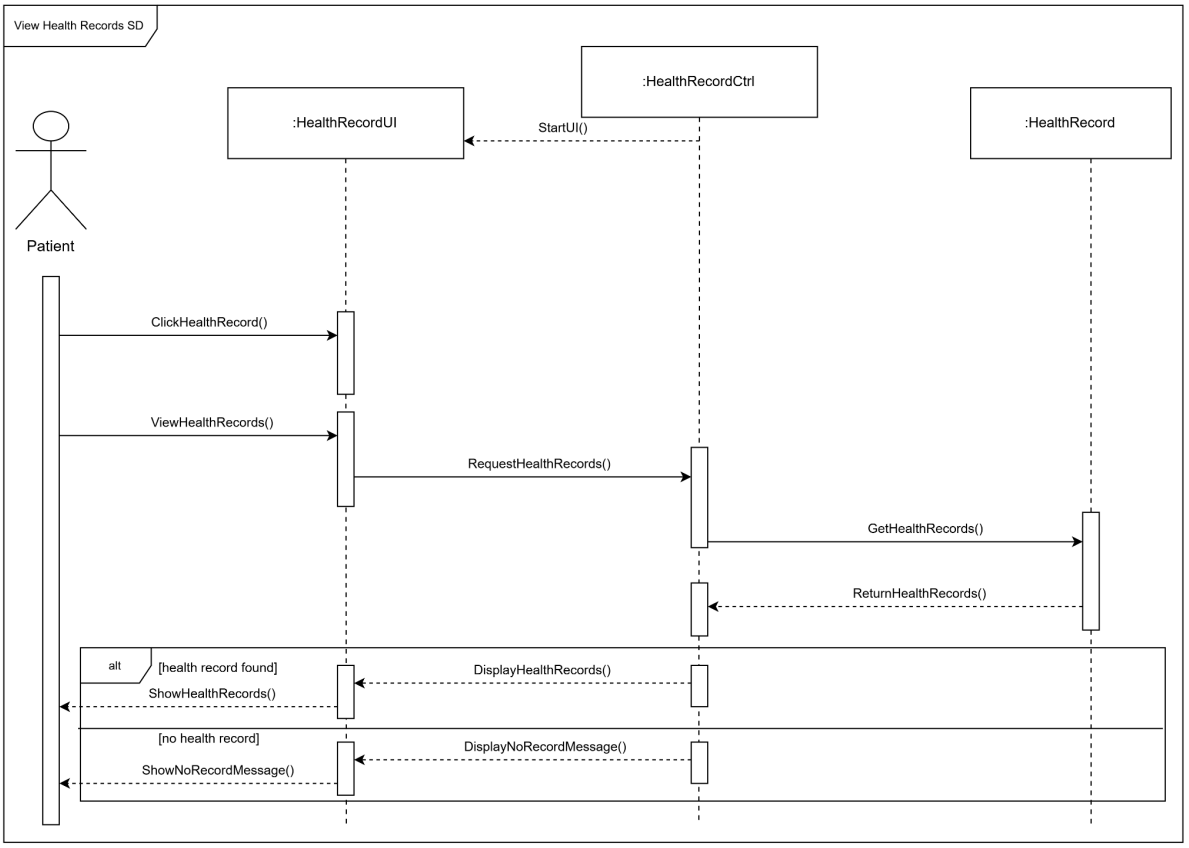


Figure 4.5.19: Sequence Diagram for View Personal Health Records

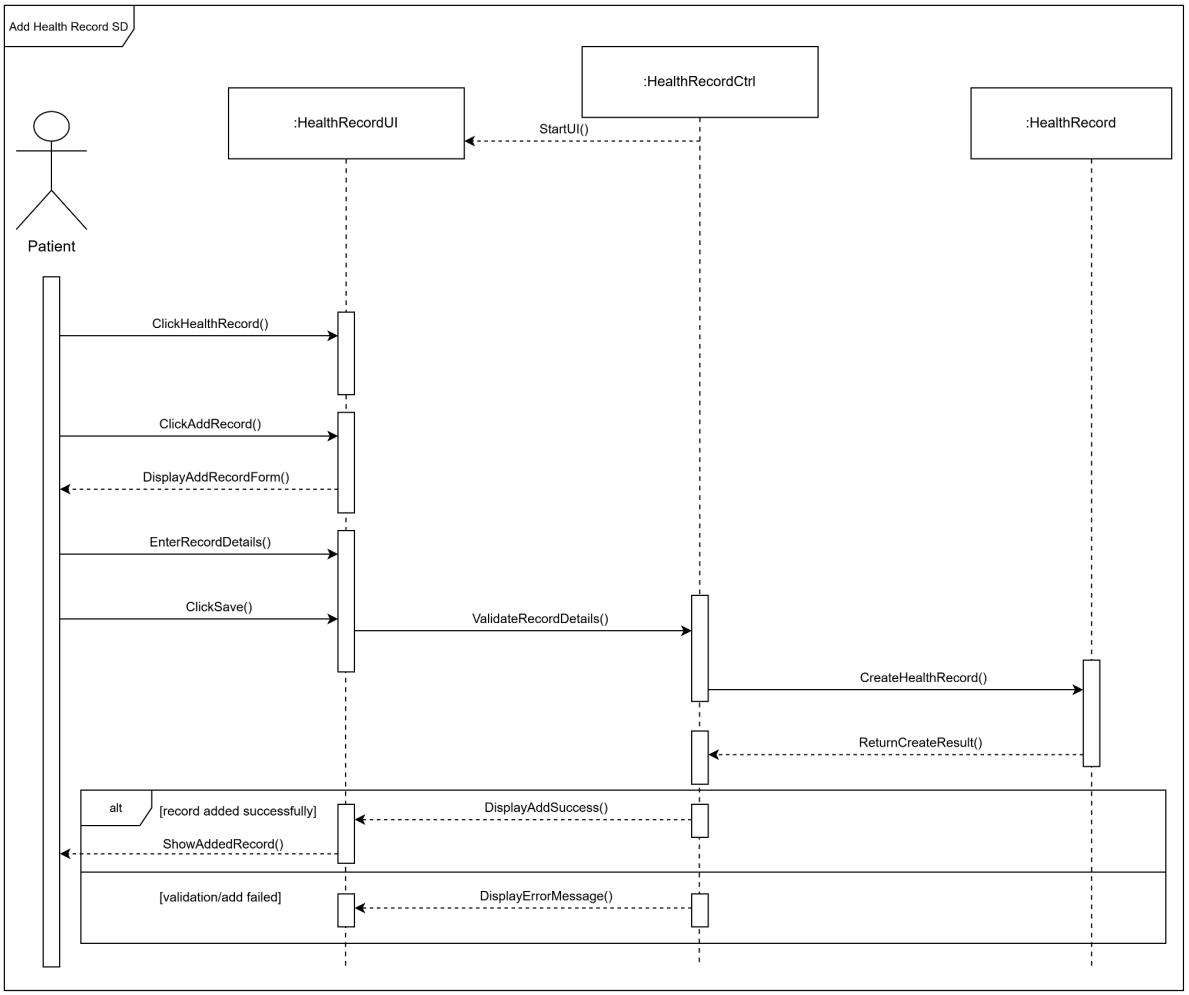


Figure 4.5.20: Sequence Diagram for Add Personal Health Record

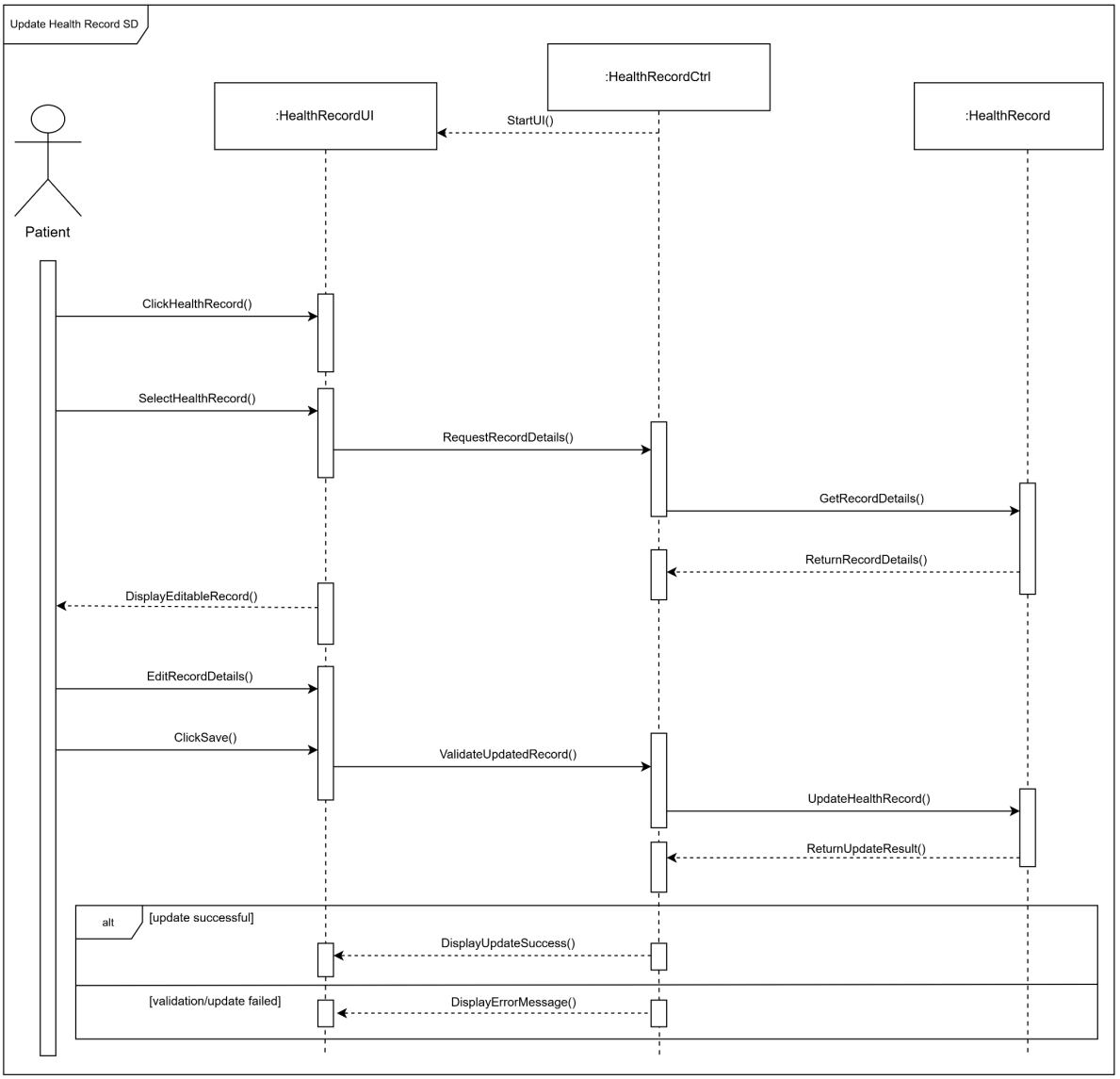


Figure 4.5.21: Sequence Diagram for Update Personal Health Record

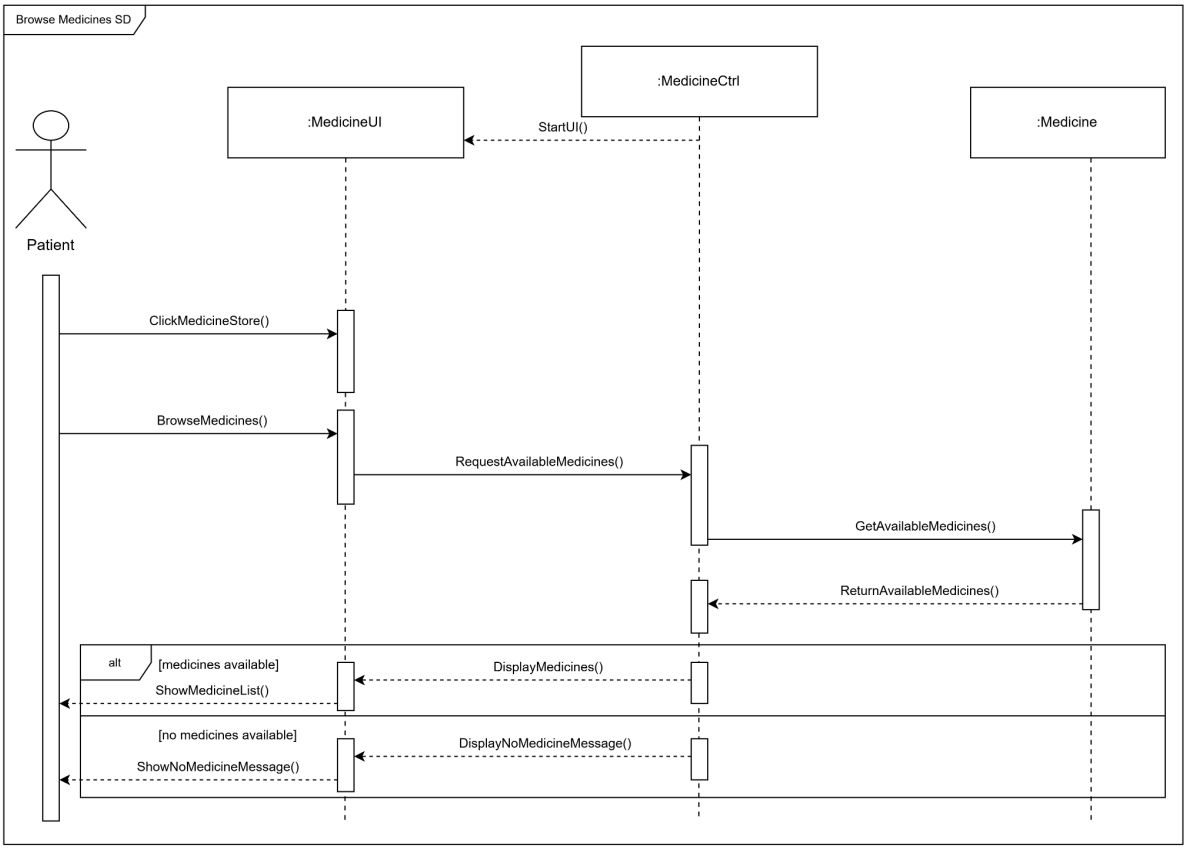


Figure 4.5.22: Sequence Diagram for Browse Available Medicines

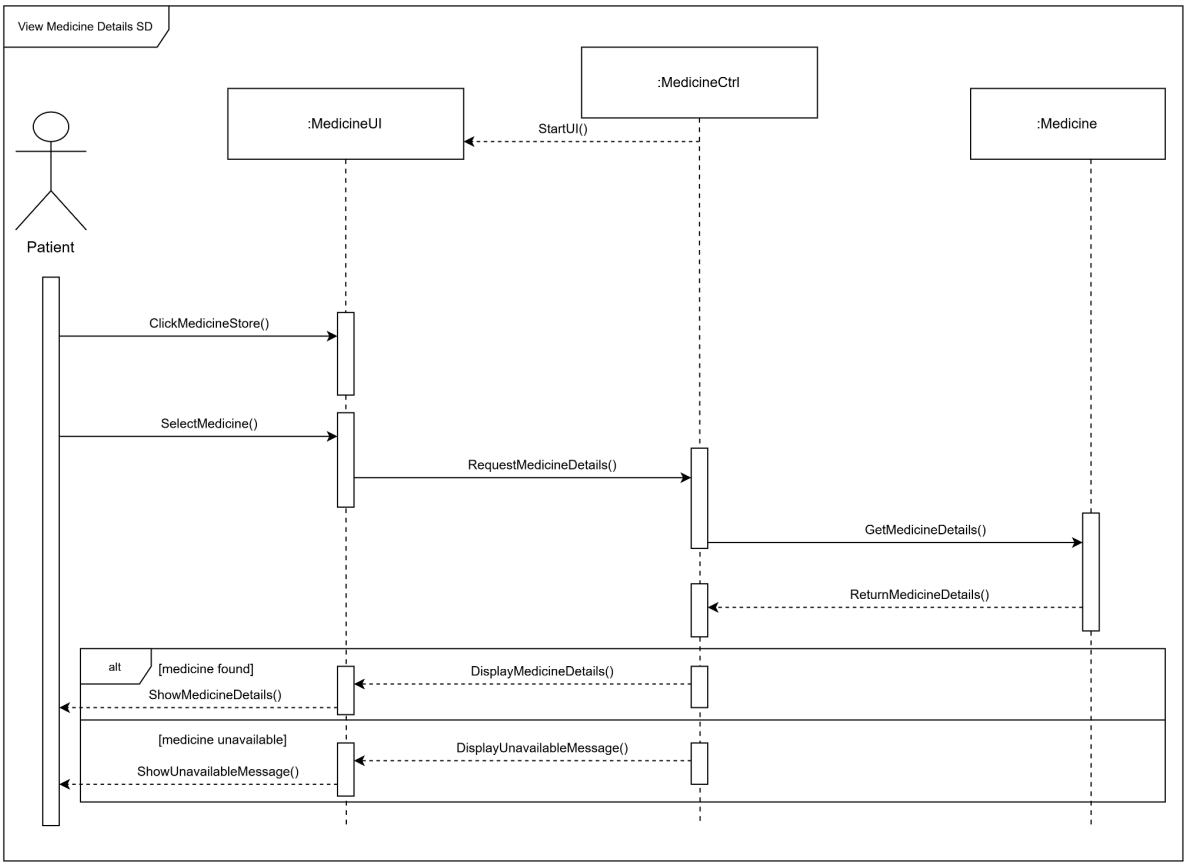


Figure 4.5.23: Sequence Diagram for View Medicine Details

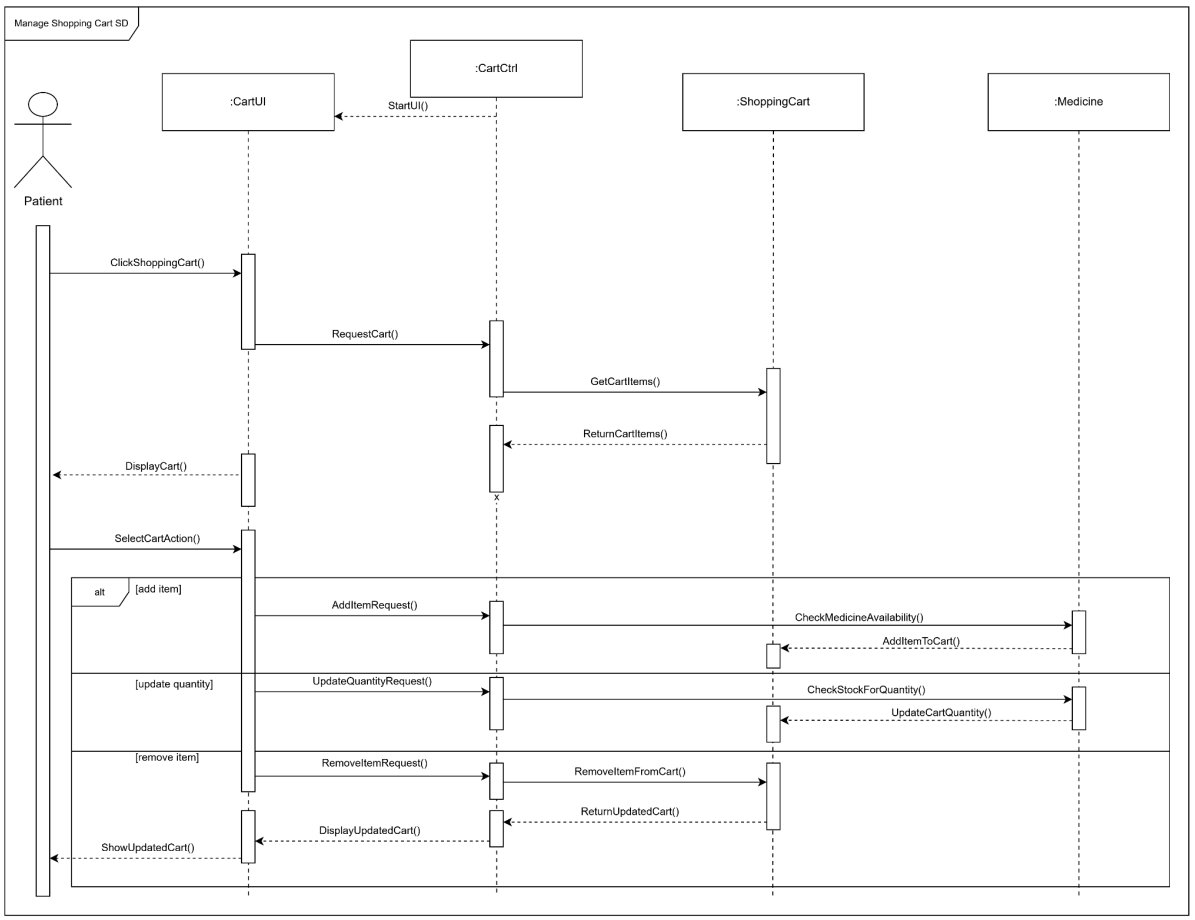


Figure 4.5.24: Sequence Diagram for Manage Shopping Cart

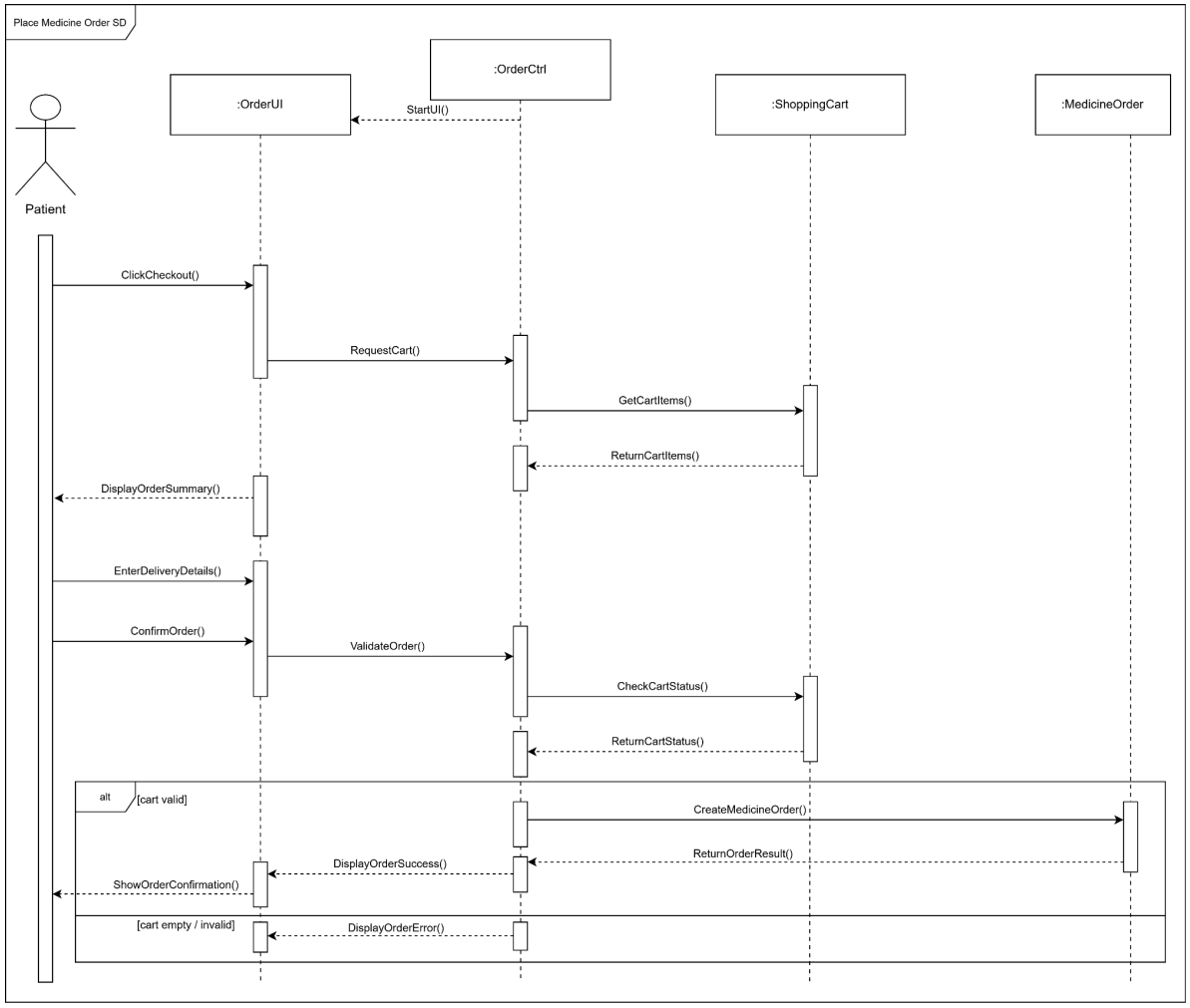


Figure 4.5.25: Sequence Diagram for Place Medicine Order

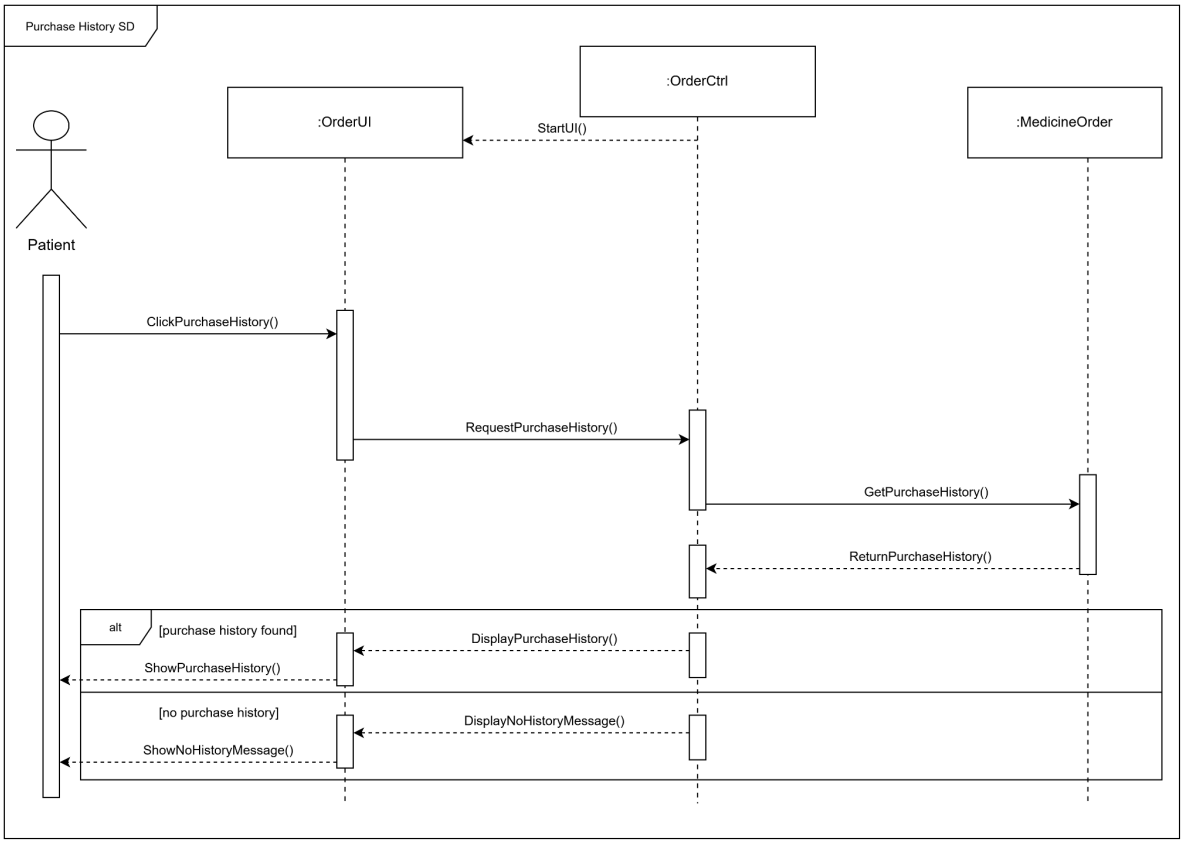


Figure 4.5.26: Sequence Diagram for View Purchase History

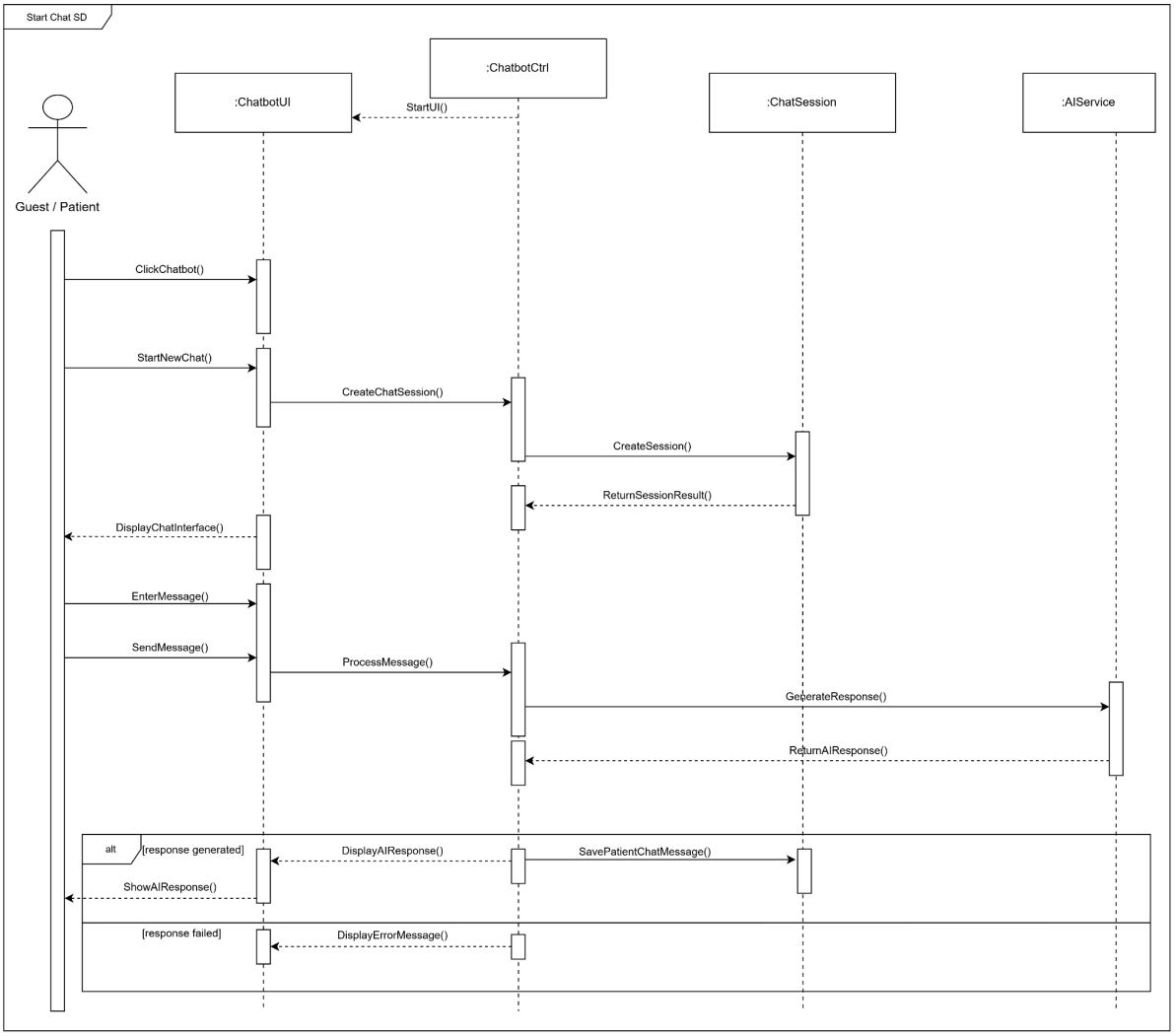


Figure 4.5.27: Sequence Diagram for Start Chat

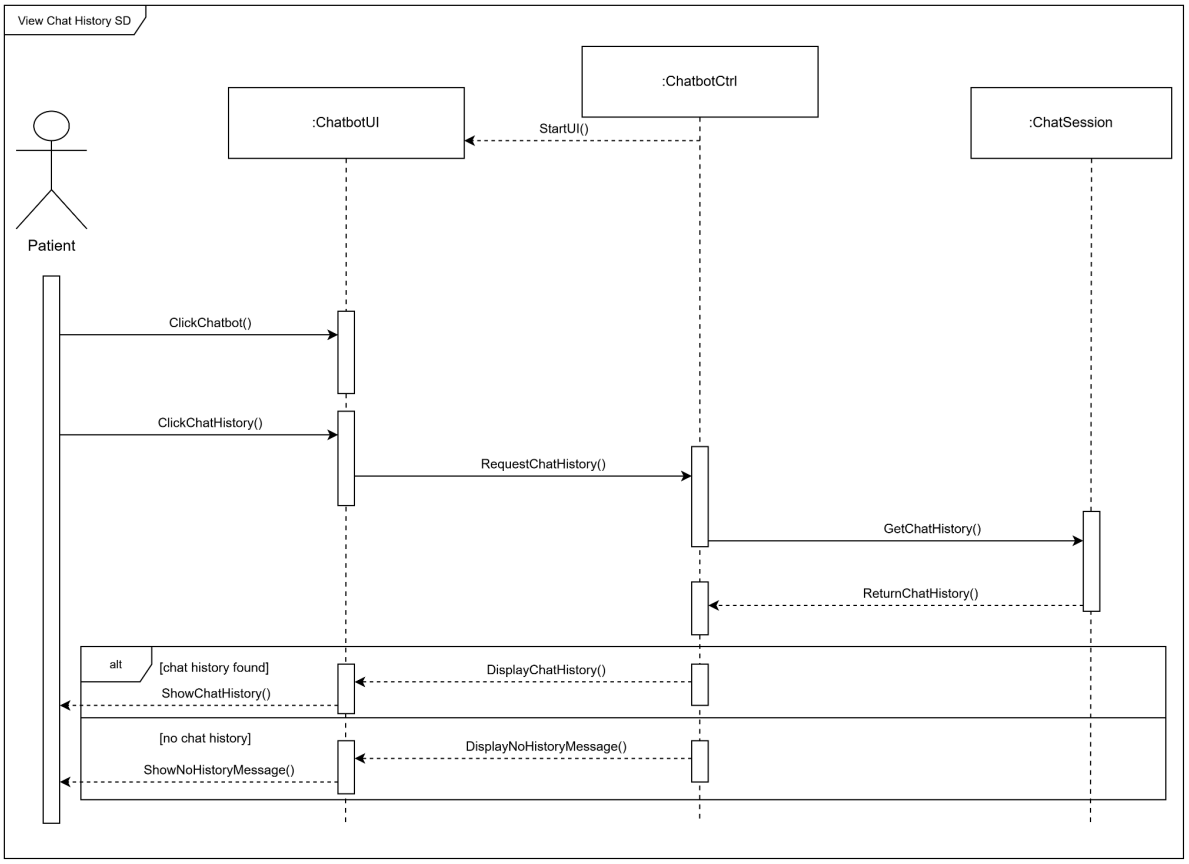


Figure 4.5.28: Sequence Diagram for View Chat History

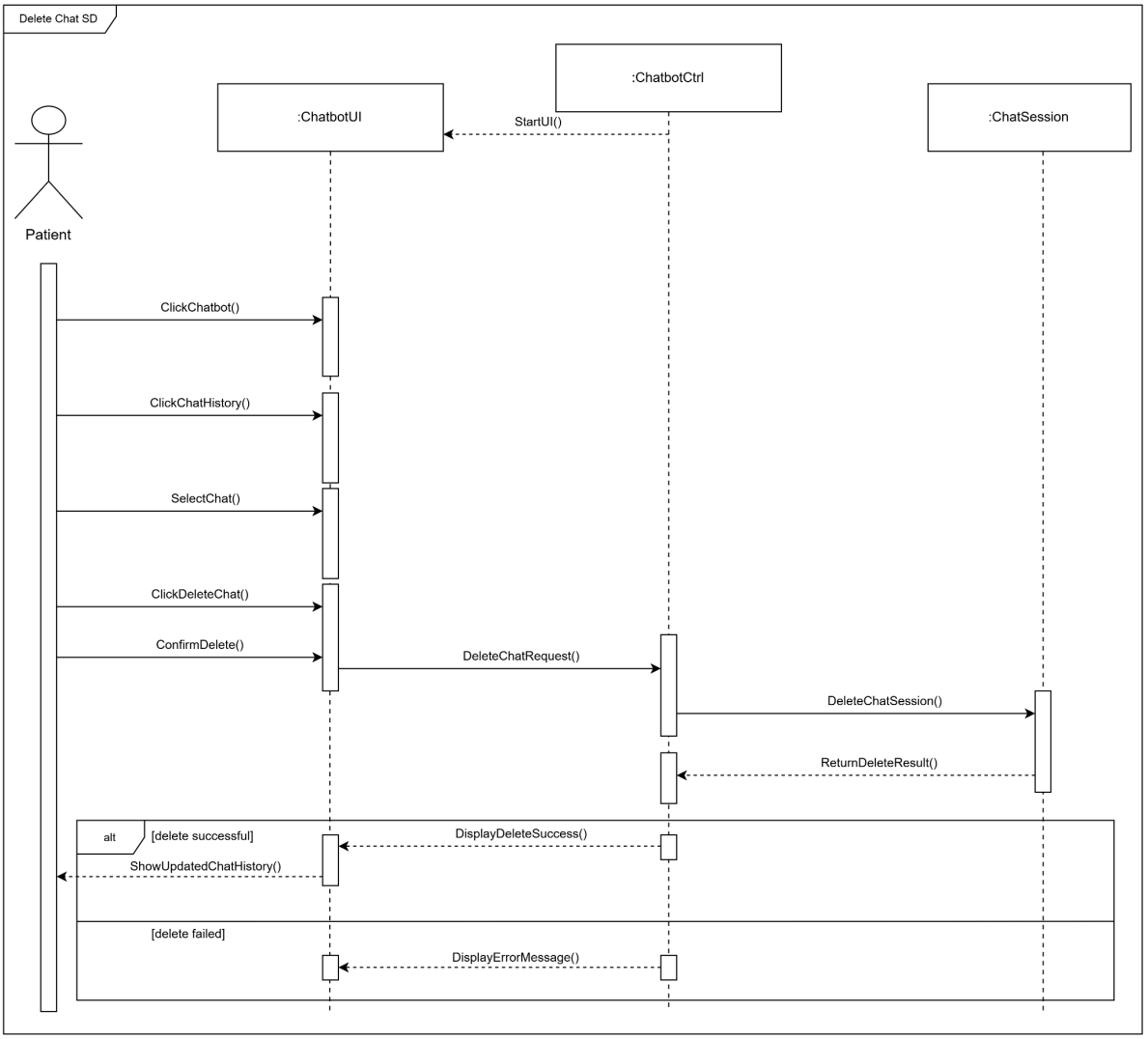


Figure 4.5.29: Sequence Diagram for Delete Chat

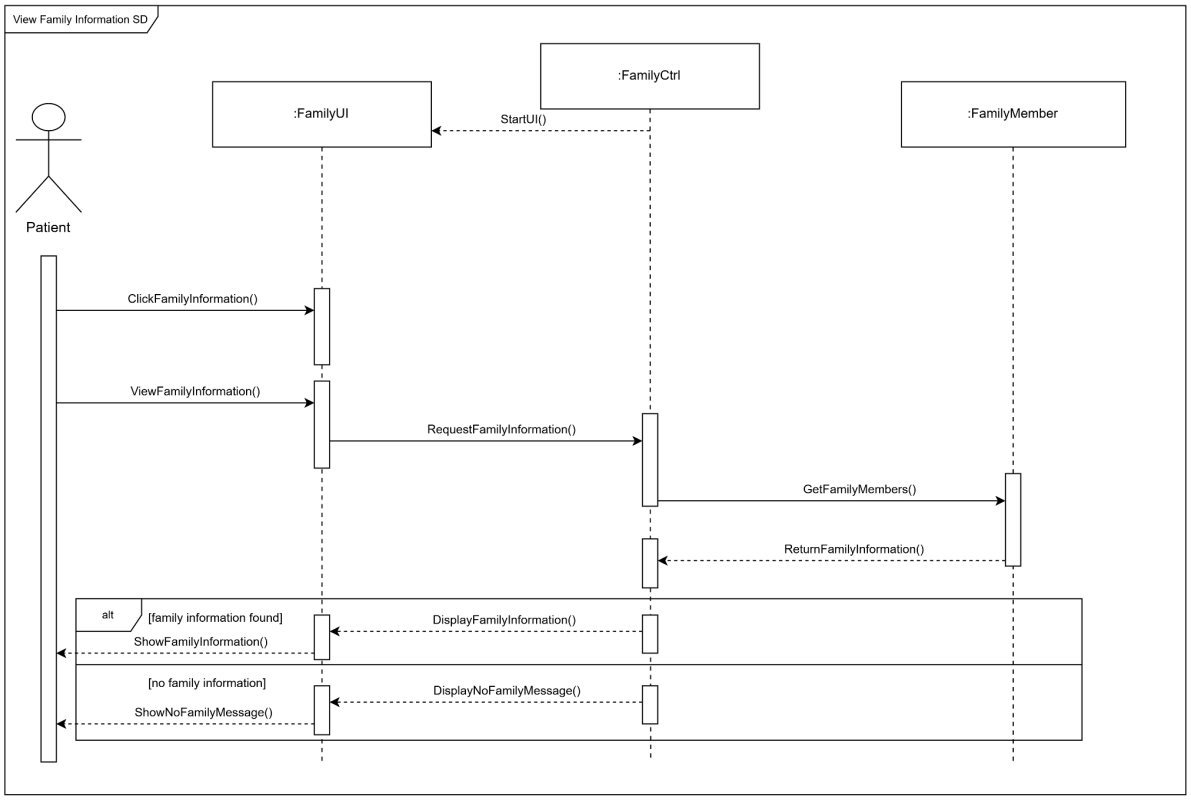


Figure 4.5.30: Sequence Diagram for View Family Information

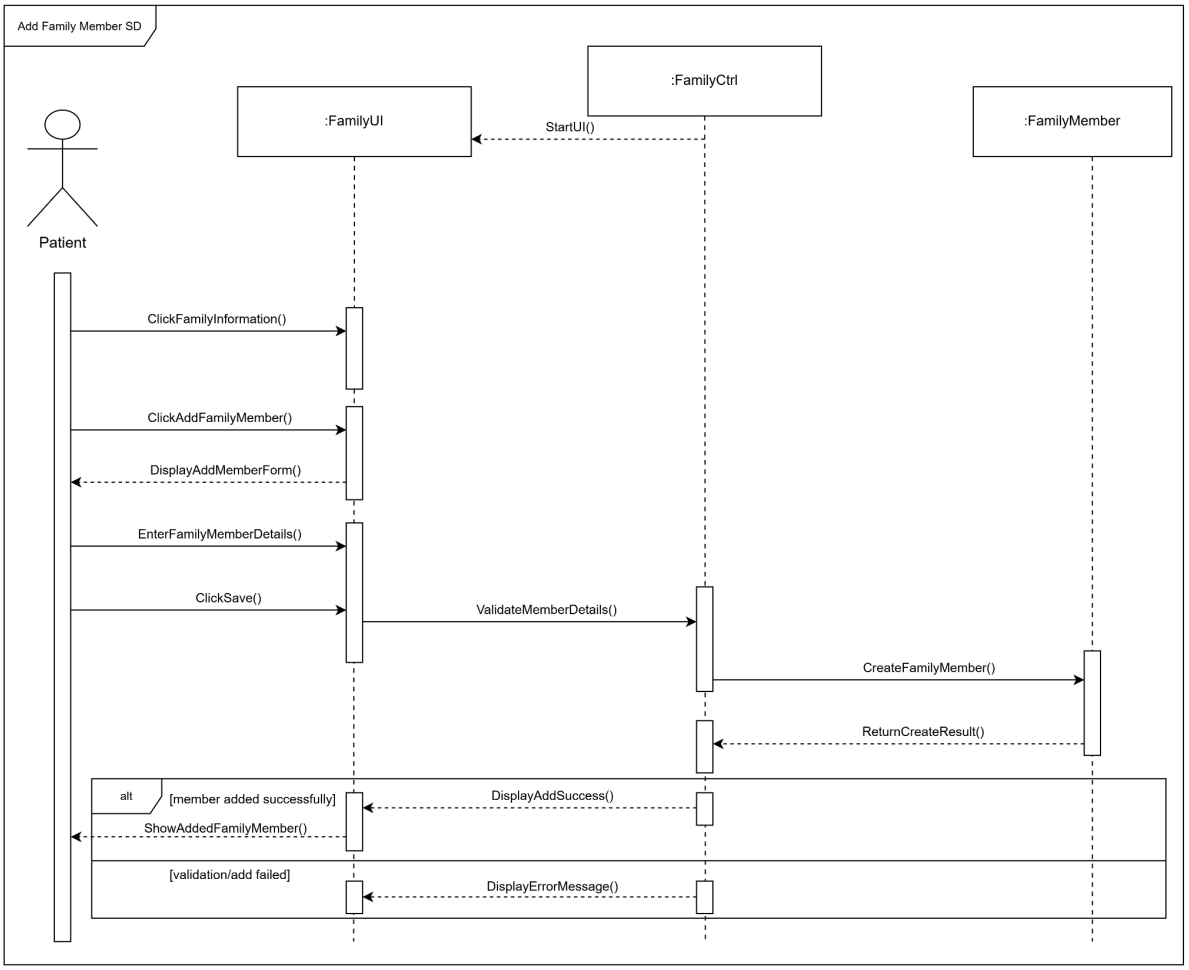


Figure 4.5.31: Sequence Diagram for Add Family Member

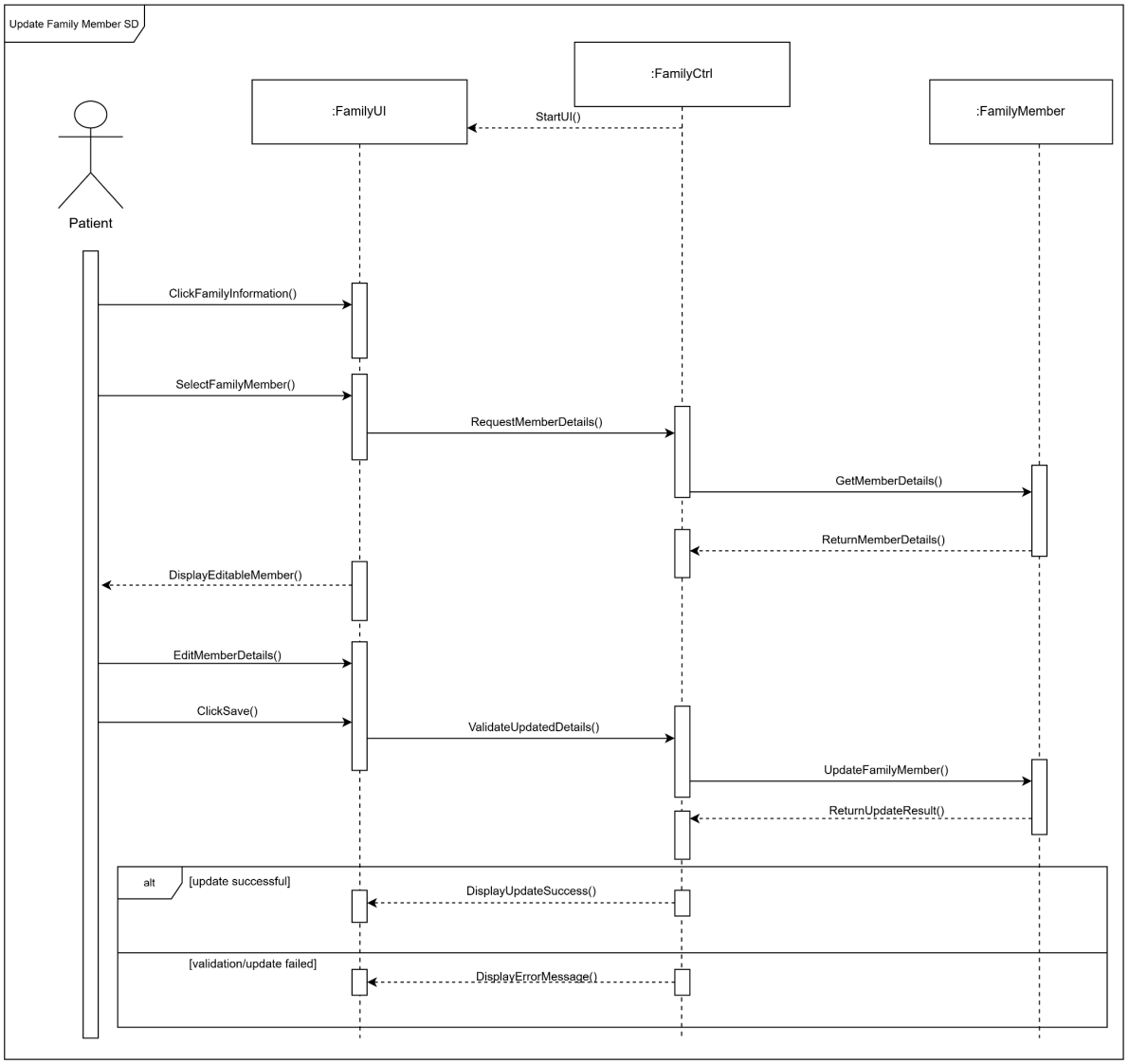


Figure 4.5.32: Sequence Diagram for Update Family Member

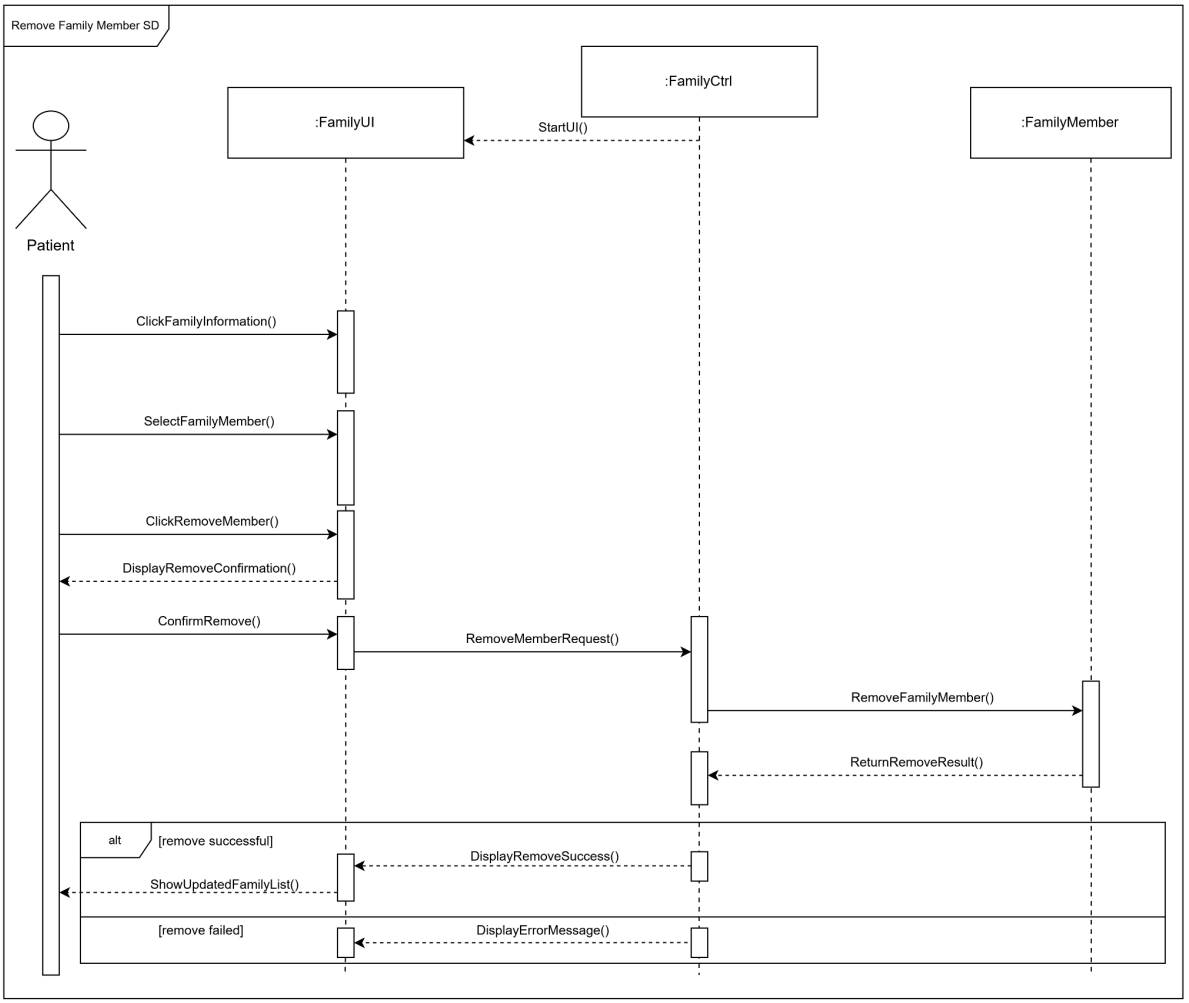


Figure 4.5.33: Sequence Diagram for Remove Family Member

4.6 Collaboration Diagram

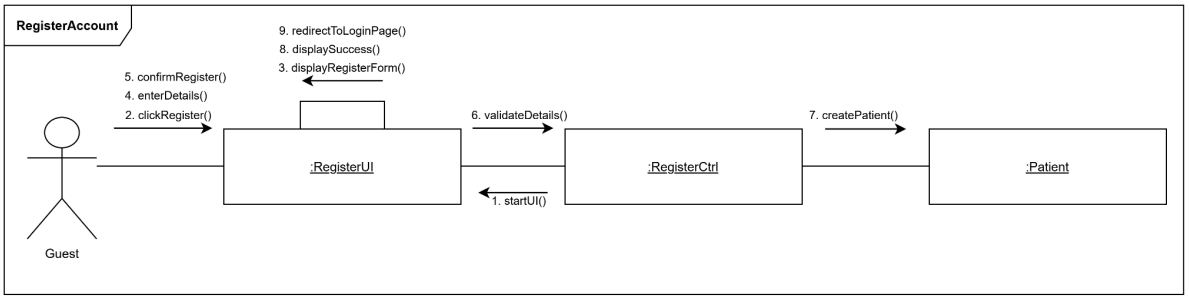


Figure 4.6.1: Collaboration Diagram for Register Account

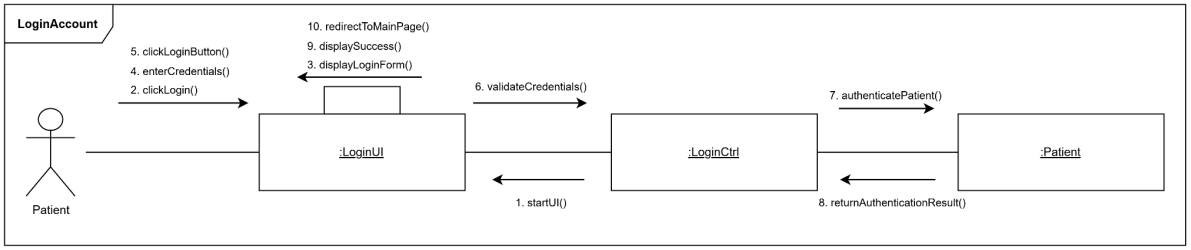


Figure 4.6.2: Collaboration Diagram for Login Account

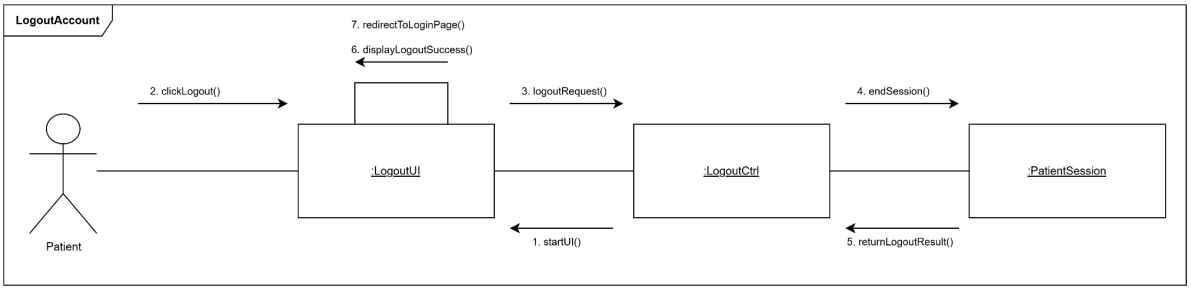


Figure 4.6.3: Collaboration Diagram for Logout Account

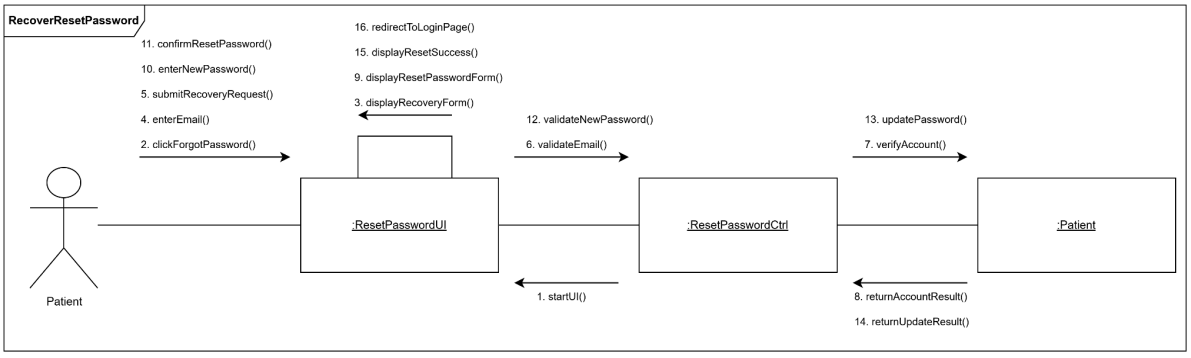


Figure 4.6.4: Collaboration Diagram for Recovery/Reset Password

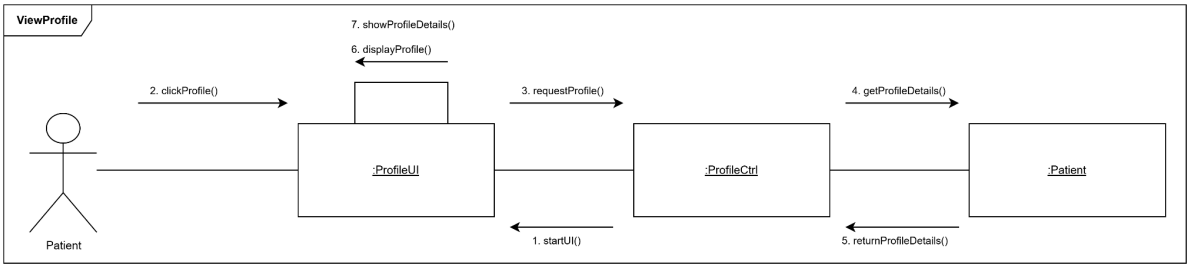


Figure 4.6.5: Collaboration Diagram for View Profile

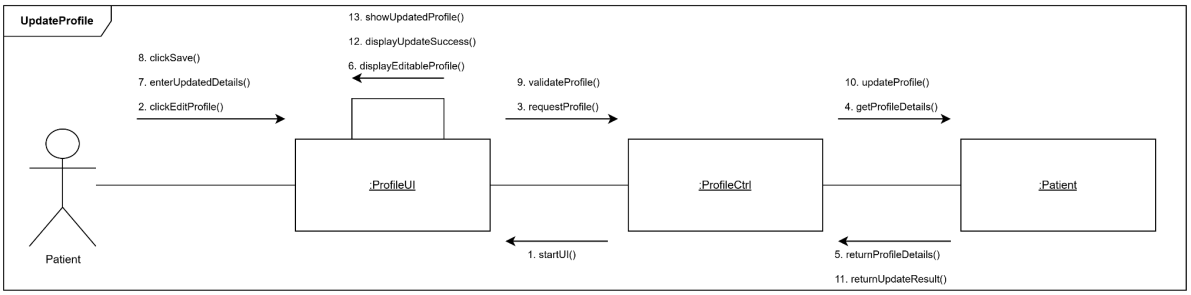


Figure 4.6.6: Collaboration Diagram for Update Profile

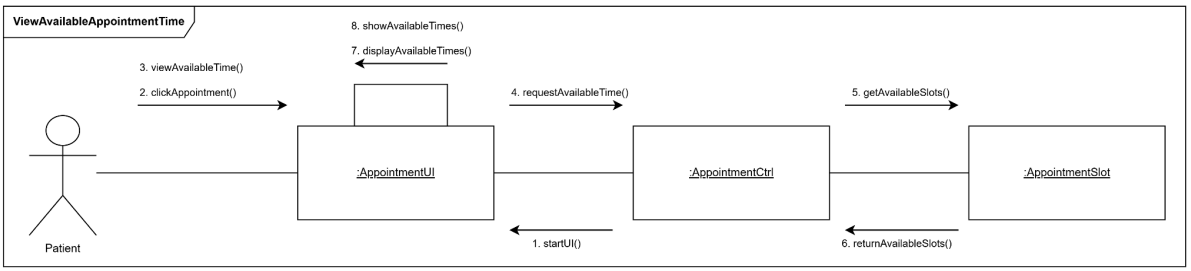


Figure 4.6.7: Collaboration Diagram for View Available Appointment Time

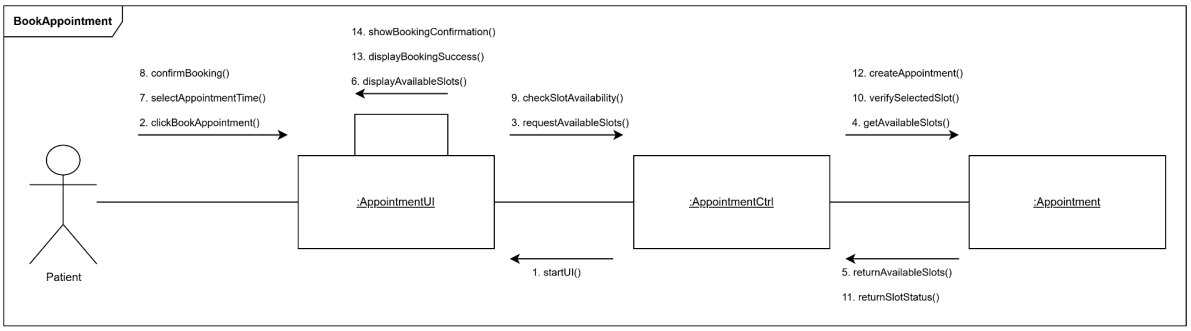


Figure 4.6.8: Collaboration Diagram for Book Appointment

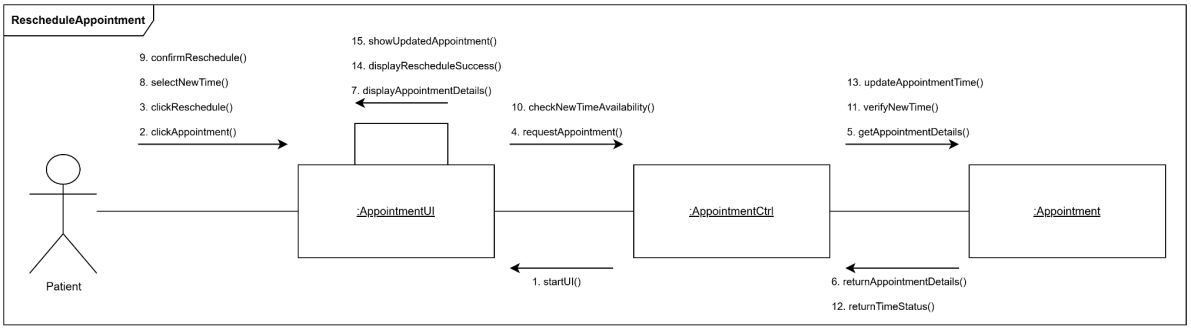


Figure 4.6.9: Collaboration Diagram for Reschedule Appointment

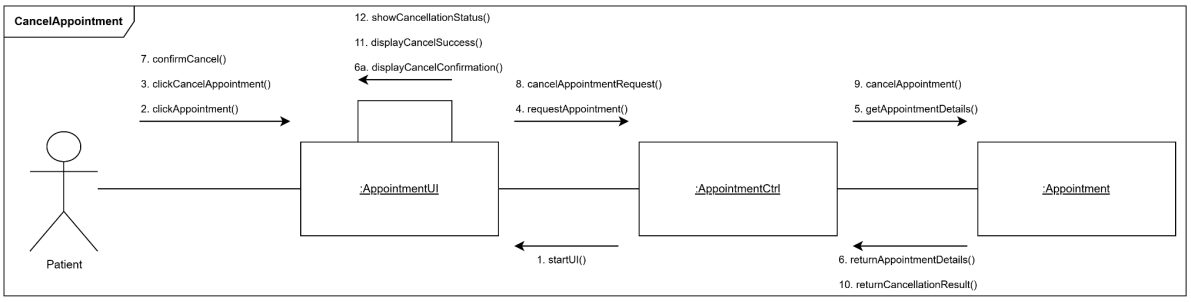


Figure 4.6.10: Collaboration Diagram for Cancel Appointment

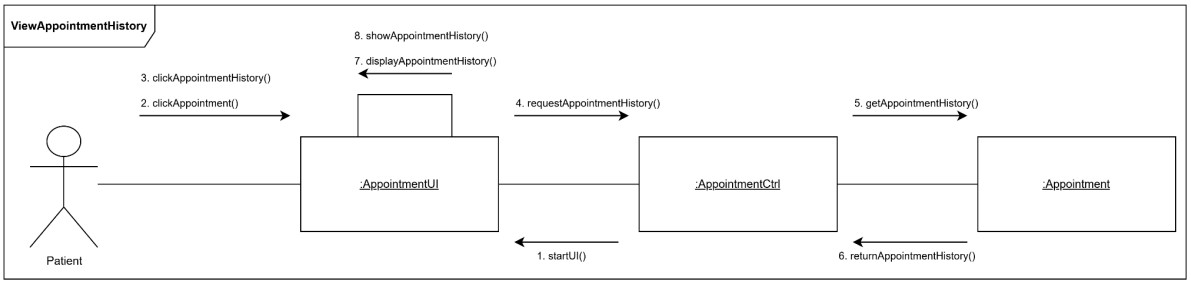


Figure 4.6.11: Collaboration Diagram for View Appointment History

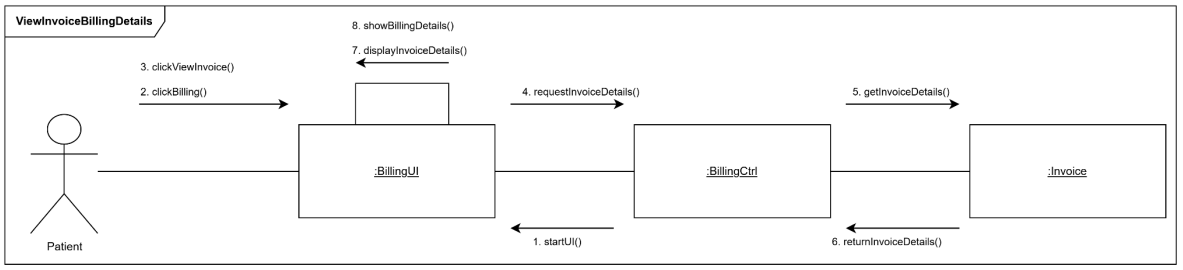


Figure 4.6.12: Collaboration Diagram for View Invoice Billing Details

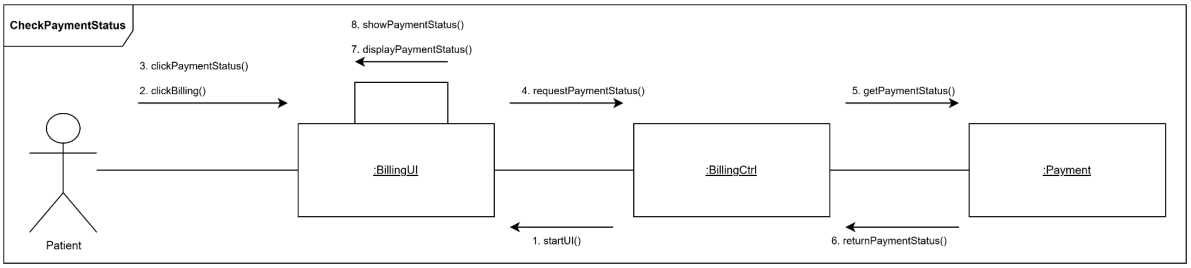


Figure 4.6.14: Collaboration Diagram for Check Payment Status

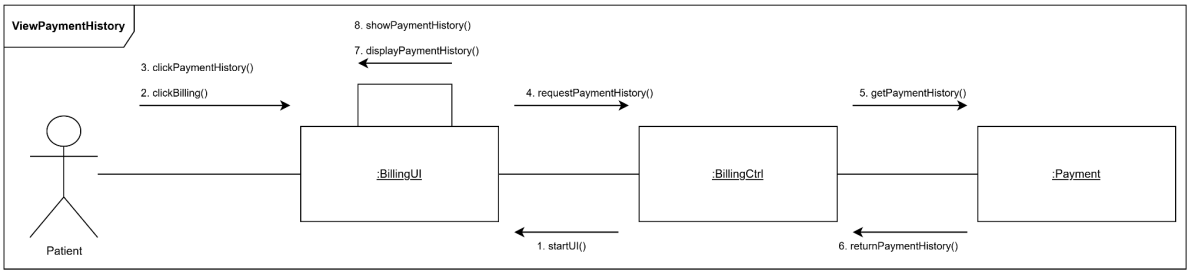


Figure 4.6.15: Collaboration Diagram for View Payment History

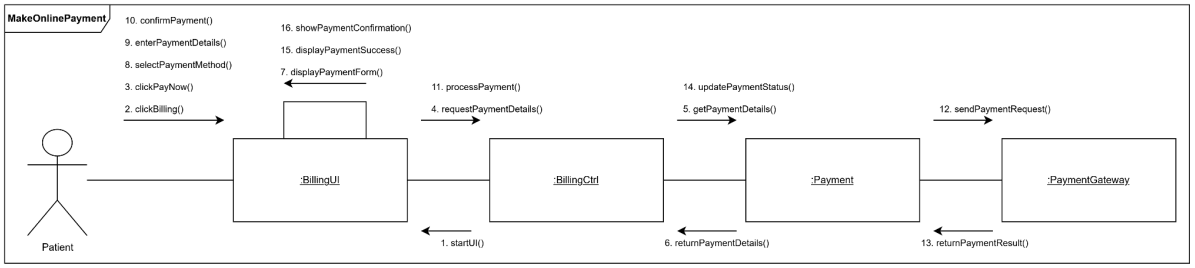


Figure 4.6.16: Collaboration Diagram for Make Online Payment

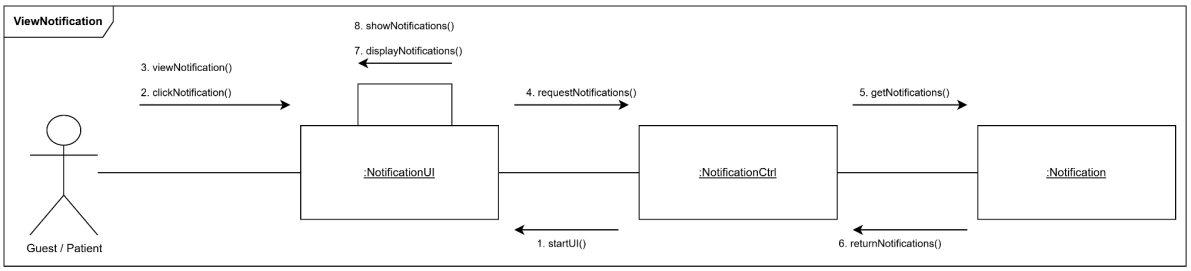


Figure 4.6.17: Collaboration Diagram for View Notification

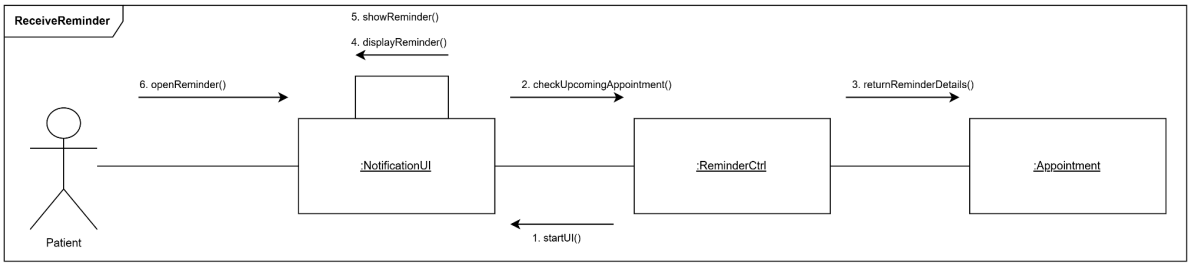


Figure 4.6.17: Collaboration Diagram for Receive Reminder

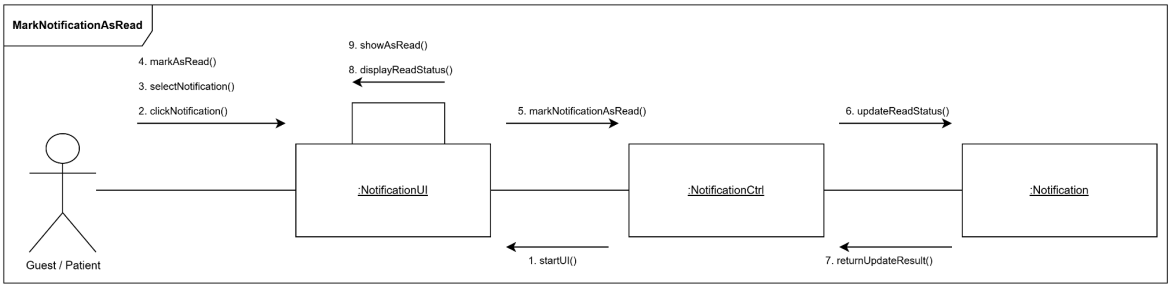


Figure 4.6.18: Collaboration Diagram for Mark Notification As Read

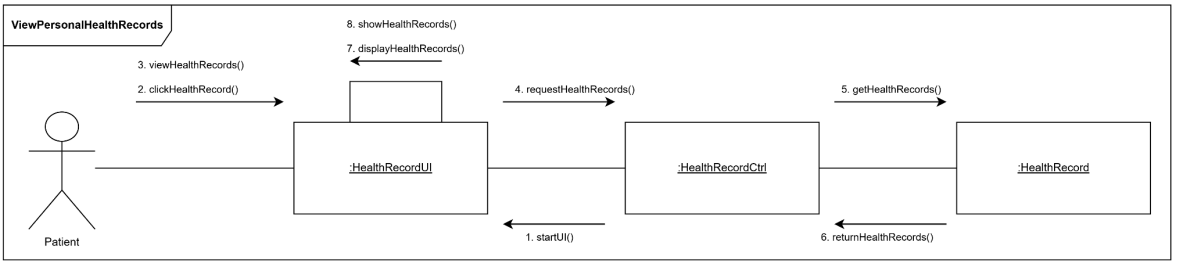


Figure 4.6.19: Collaboration Diagram for View Personal Health Records

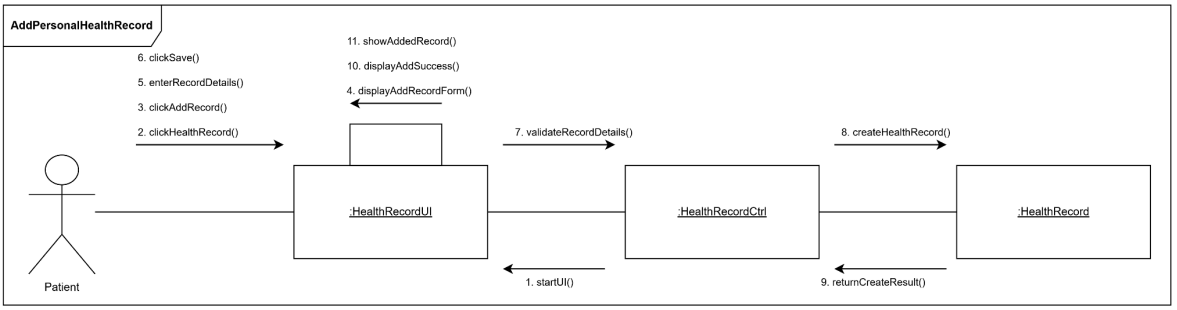


Figure 4.6.20: Collaboration Diagram for Add Personal Health Record

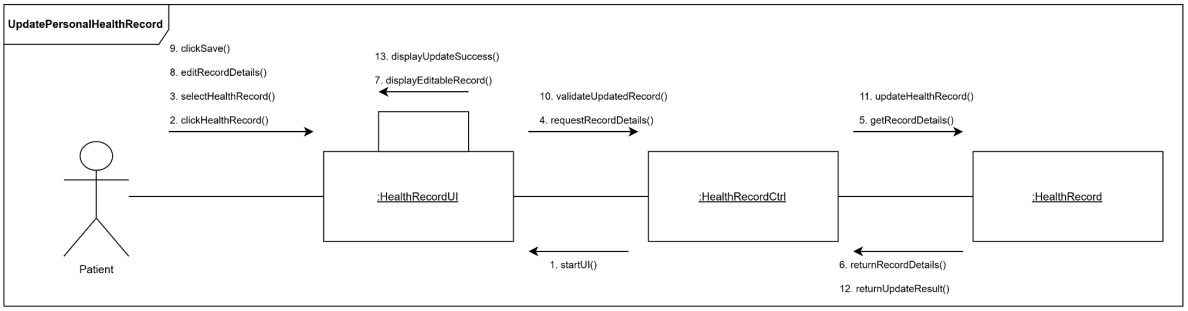


Figure 4.6.21: Collaboration Diagram for Update Personal Health Record

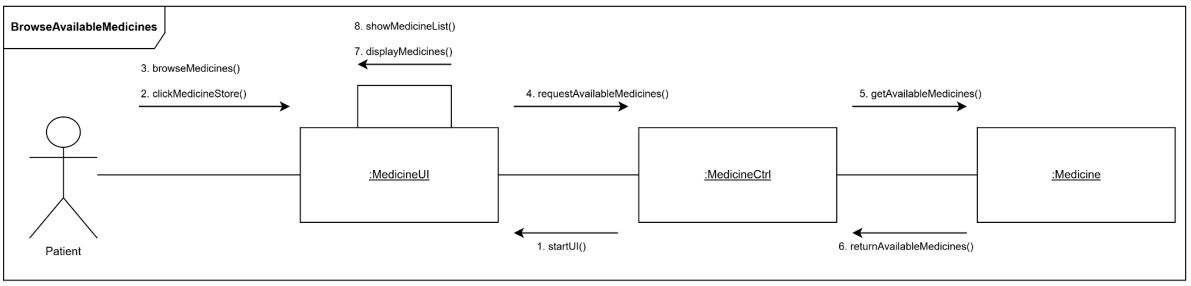


Figure 4.6.22: Collaboration Diagram for Browse Available Medicines

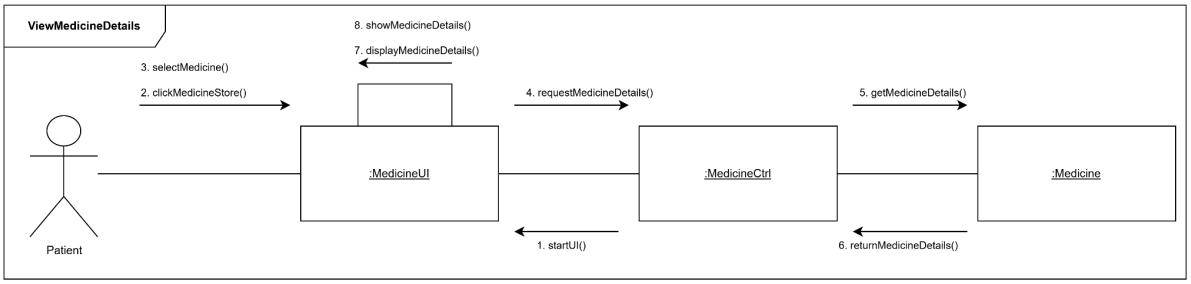


Figure 4.6.23: Collaboration Diagram for View Medicine Details

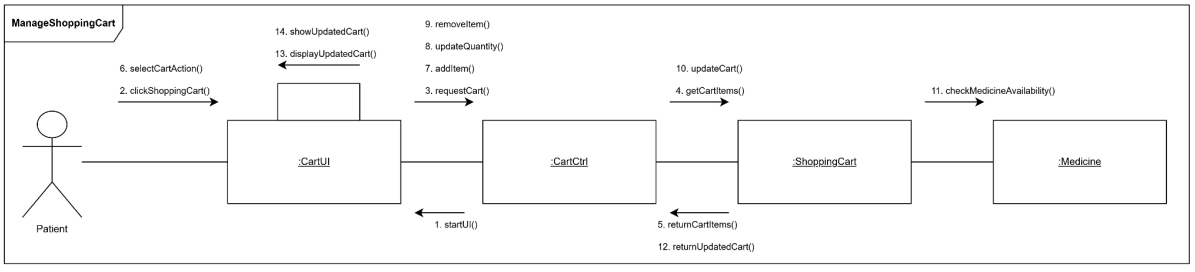


Figure 4.6.24: Collaboration Diagram for Manage Shopping Cart

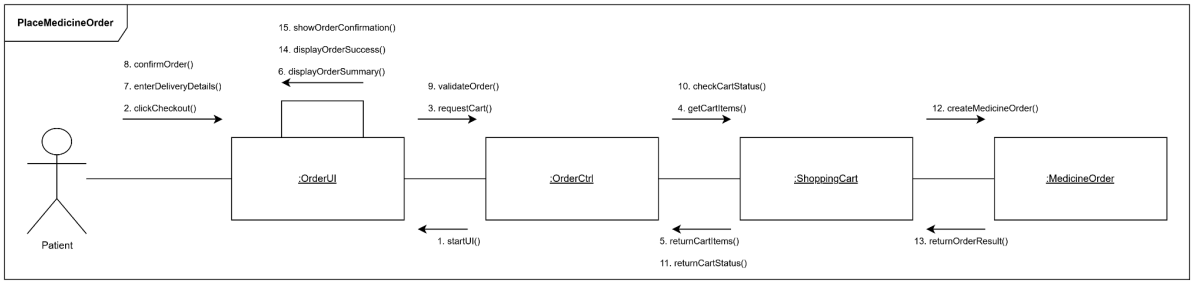


Figure 4.6.25: Collaboration Diagram for Place Medicine Order

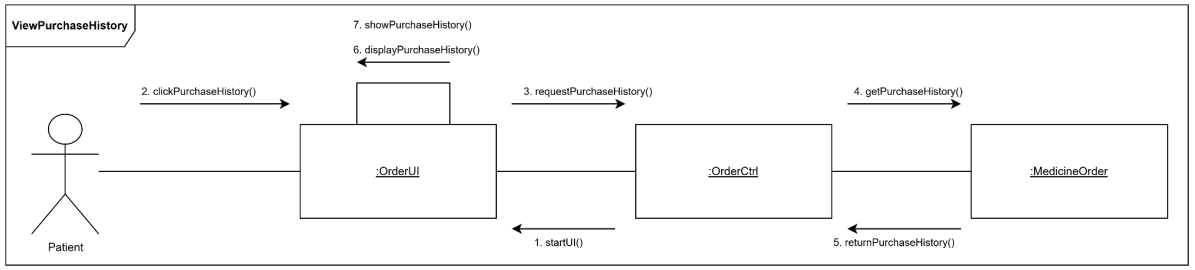


Figure 4.6.26: Collaboration Diagram for View Purchase History

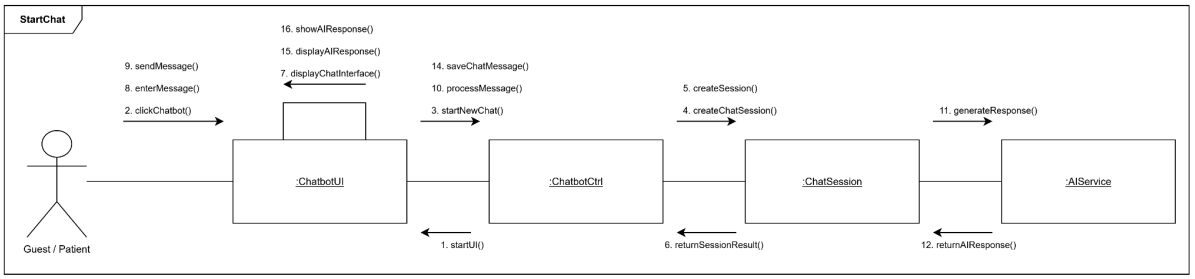


Figure 4.6.27: Collaboration Diagram for Start Chat

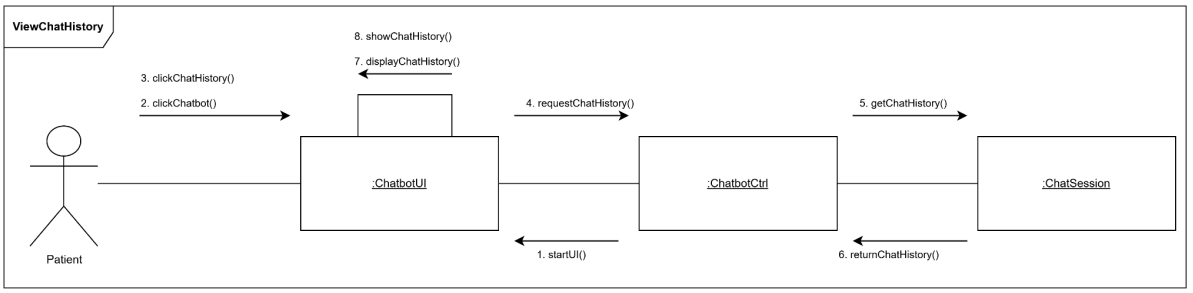


Figure 4.6.28: Collaboration Diagram for View Chat History

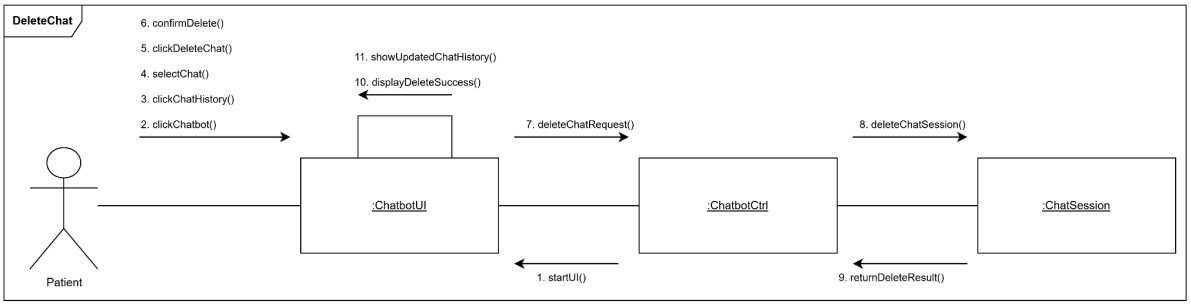


Figure 4.6.29: Collaboration Diagram for Delete Chat

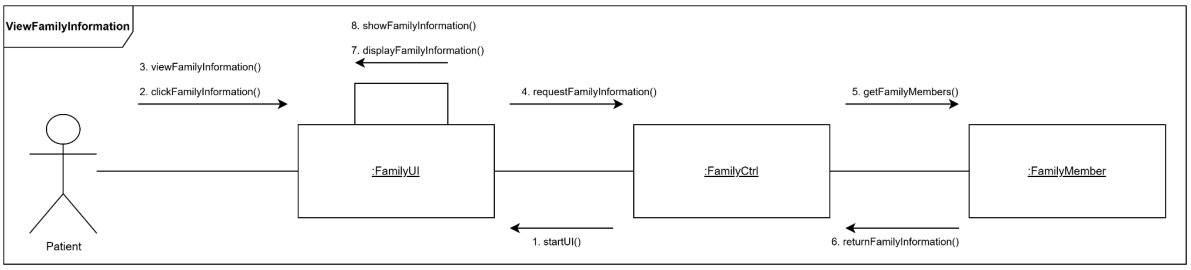


Figure 4.6.30: Collaboration Diagram for View Family Information

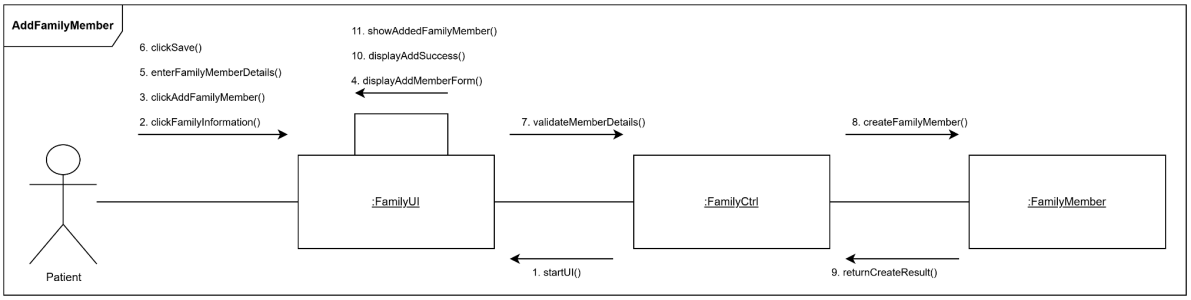


Figure 4.6.31: Collaboration Diagram for Add Family Member

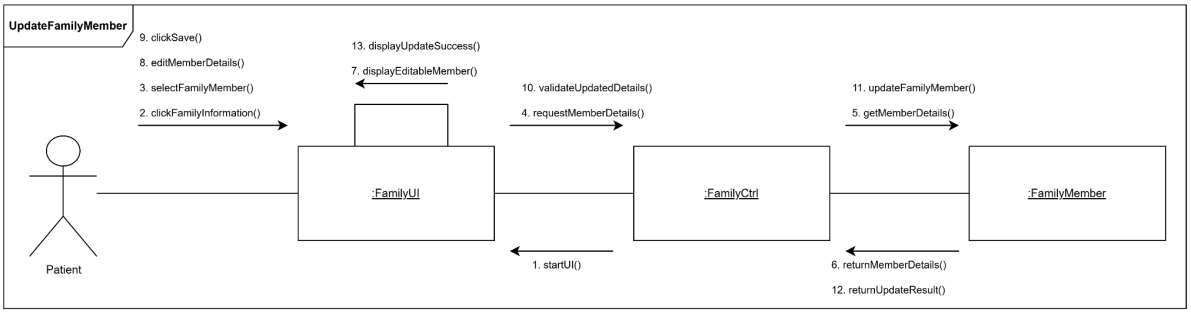


Figure 4.6.32: Collaboration Diagram for Update Family Member

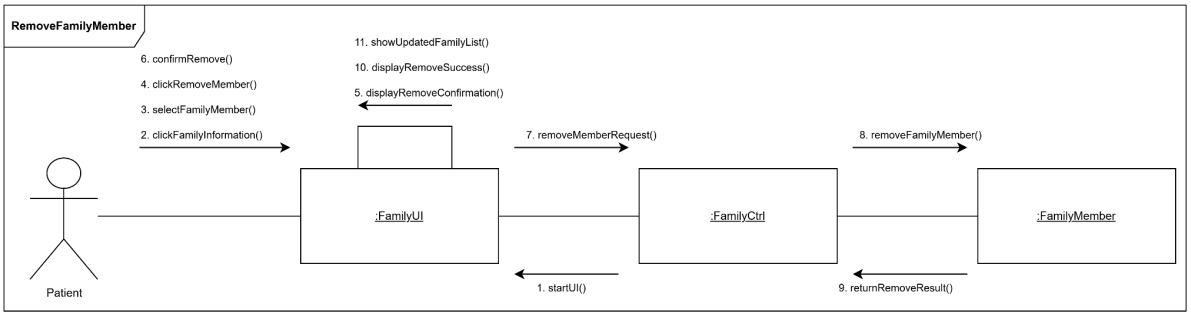


Figure 4.6.33: Collaboration Diagram for Remove Family Member

4.7 System Design

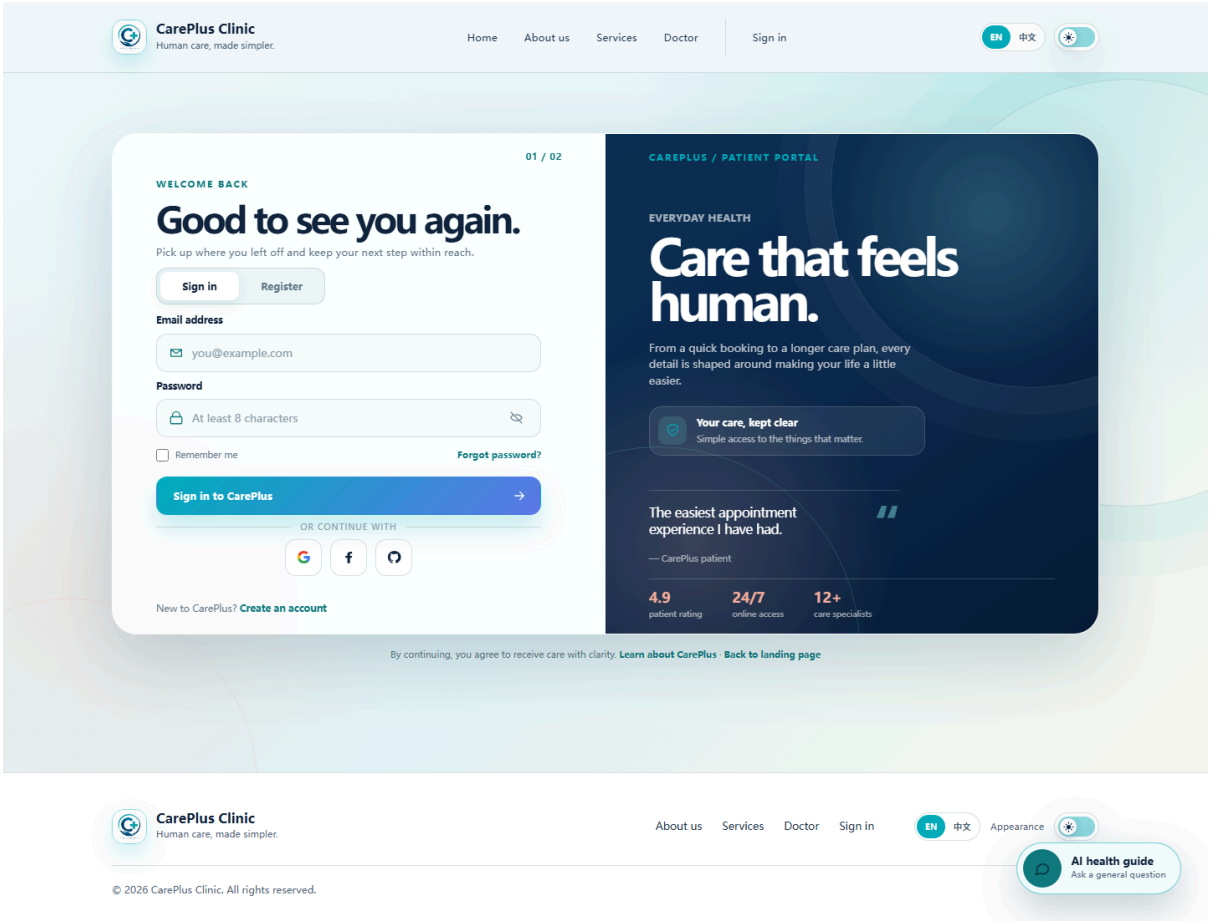


Figure 4.7.1: Authentication and account access page

The page in Figure 4.7.1 shows the authentication interface for the CarePlus patient portal. The user can switch between the Sign in and Register forms, enter an email address and password, select the Remember me option, or open the Forgot password flow. The system validates the submitted credentials before creating a session. New users can register an account, while existing users can use the recovery flow when they cannot remember their password. The social sign-in buttons are also presented as alternative access options in the prototype.

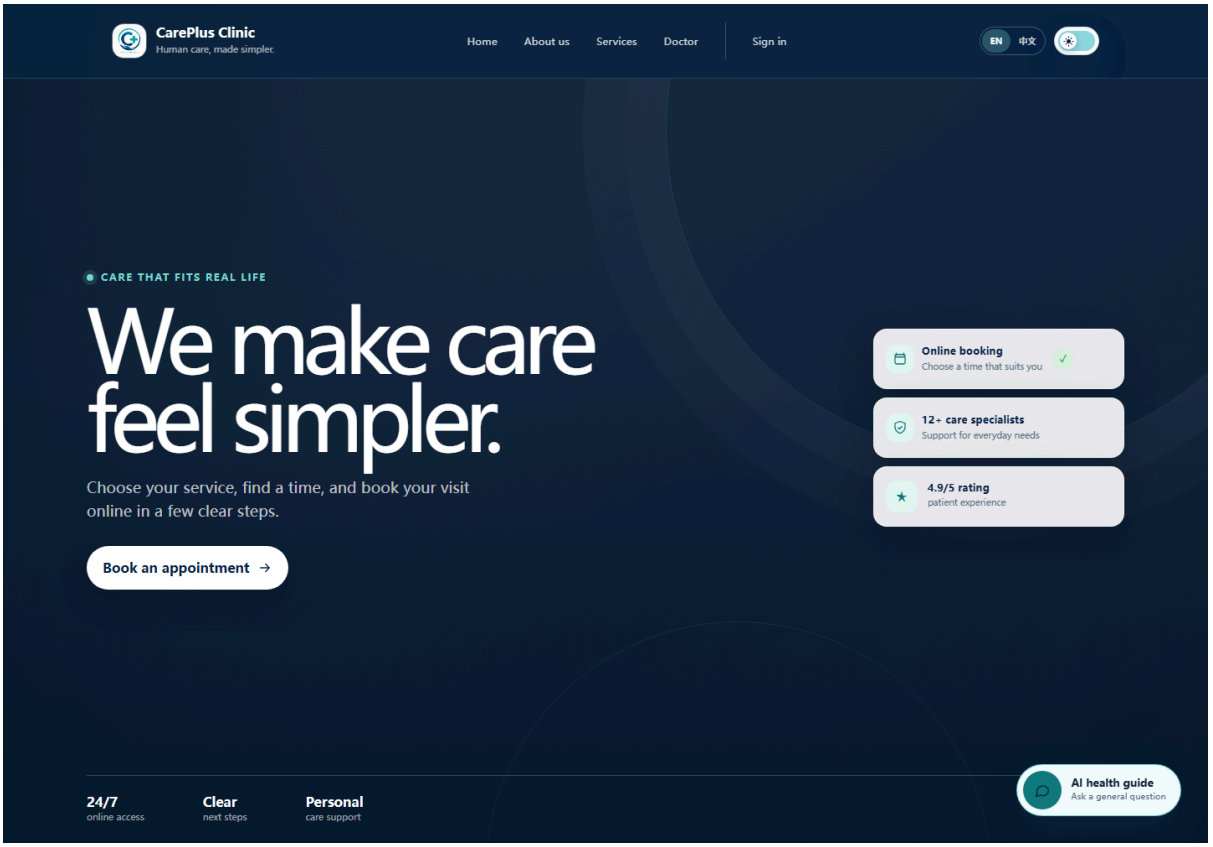


Figure 4.7.2: Public clinic landing page

The page in Figure 4.7.2 shows the public landing page that introduces the clinic before authentication. It presents the clinic identity, online booking call to action, care specialist information, patient rating, language control, appearance control, and the AI health guide entry point. A guest can review the main services and proceed to appointment booking or the sign-in page without navigating through the patient dashboard. This design reduces the number of steps needed to reach the most frequently used clinic functions.

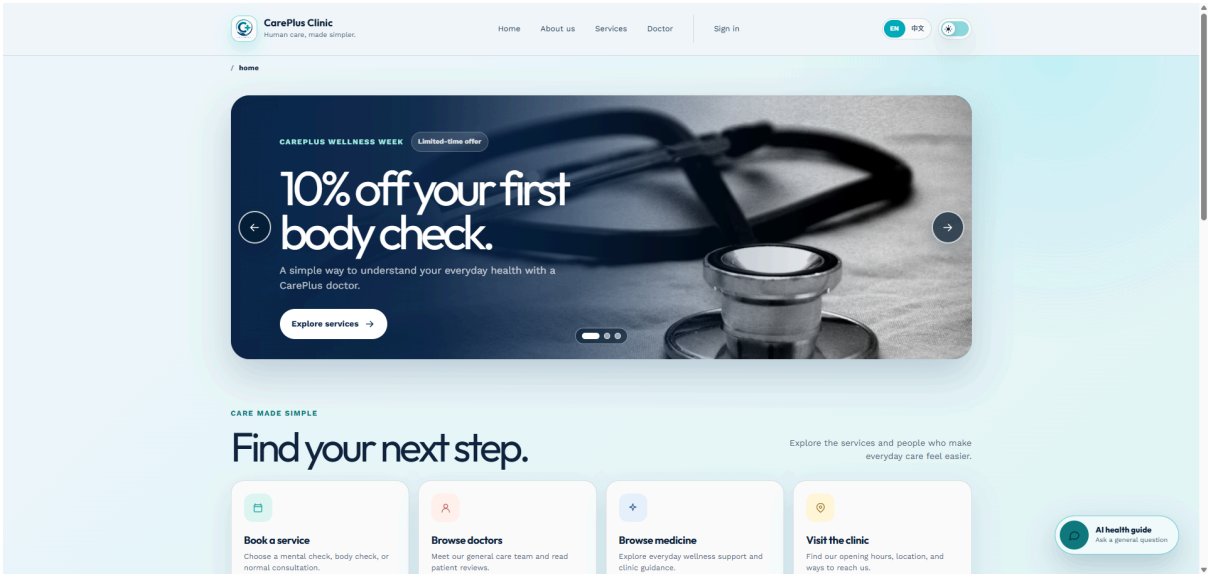


Figure 4.7.3: Patient portal homepage

The page in Figure 4.7.3 shows the patient portal dashboard. It provides a central starting point for browsing doctors, booking a service, viewing prescribed medicine, and checking clinic information. The dashboard also includes a featured doctor panel and a persistent AI health guide button. The shared header and breadcrumb area help the user understand the current location and move between related functions. This layout supports the requirement for a simple and consistent navigation experience across the system.

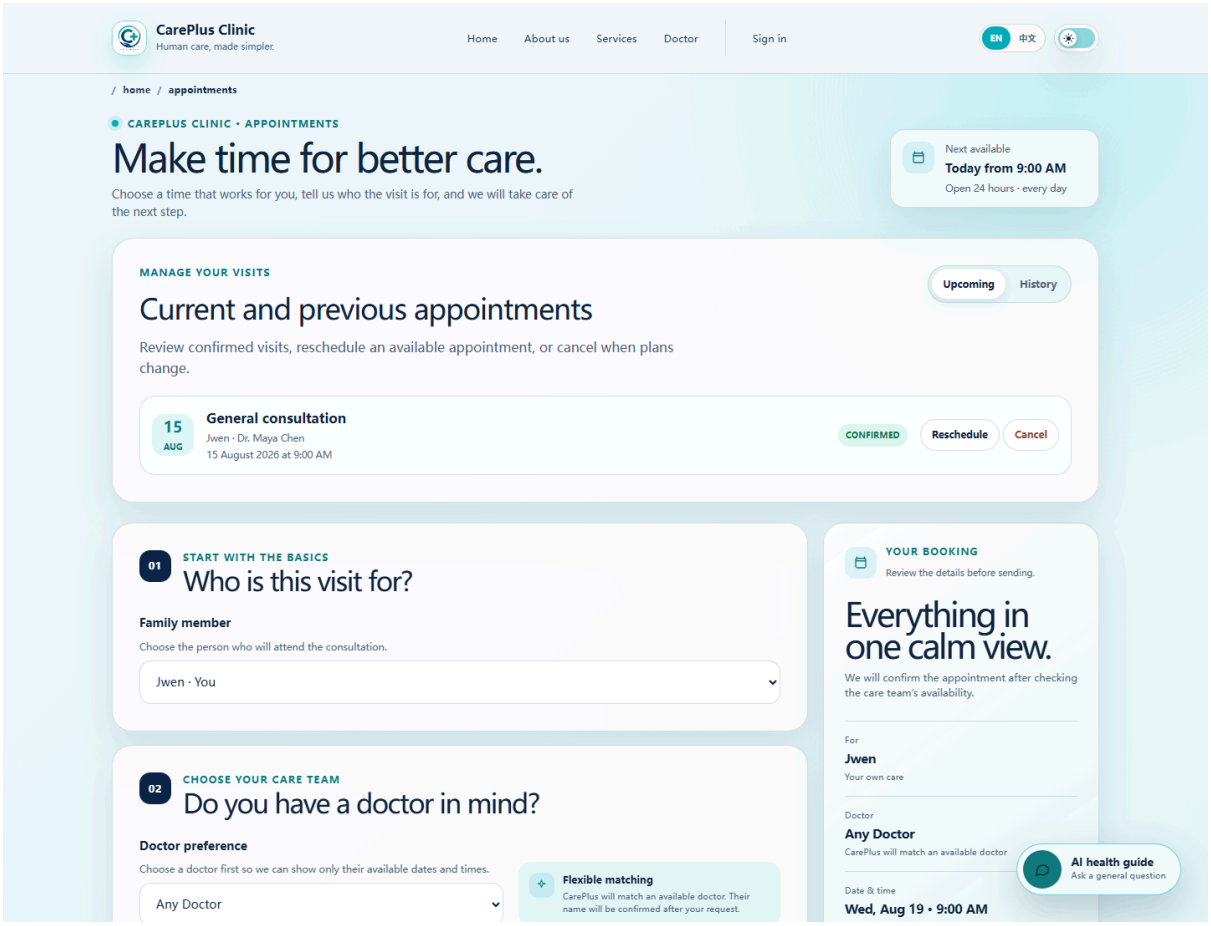


Figure 4.7.4: Appointment booking and appointment management page

The page in Figure 4.7.4 supports the appointment management use cases. The upper section separates upcoming appointments from appointment history and provides actions for rescheduling or cancelling an existing appointment. The booking form then guides the patient through selecting the family member, doctor preference, date, available time, consultation type, and optional notes. A summary panel displays the selected information before submission. The system is designed to show available slots only and to validate the booking details before confirming the appointment.

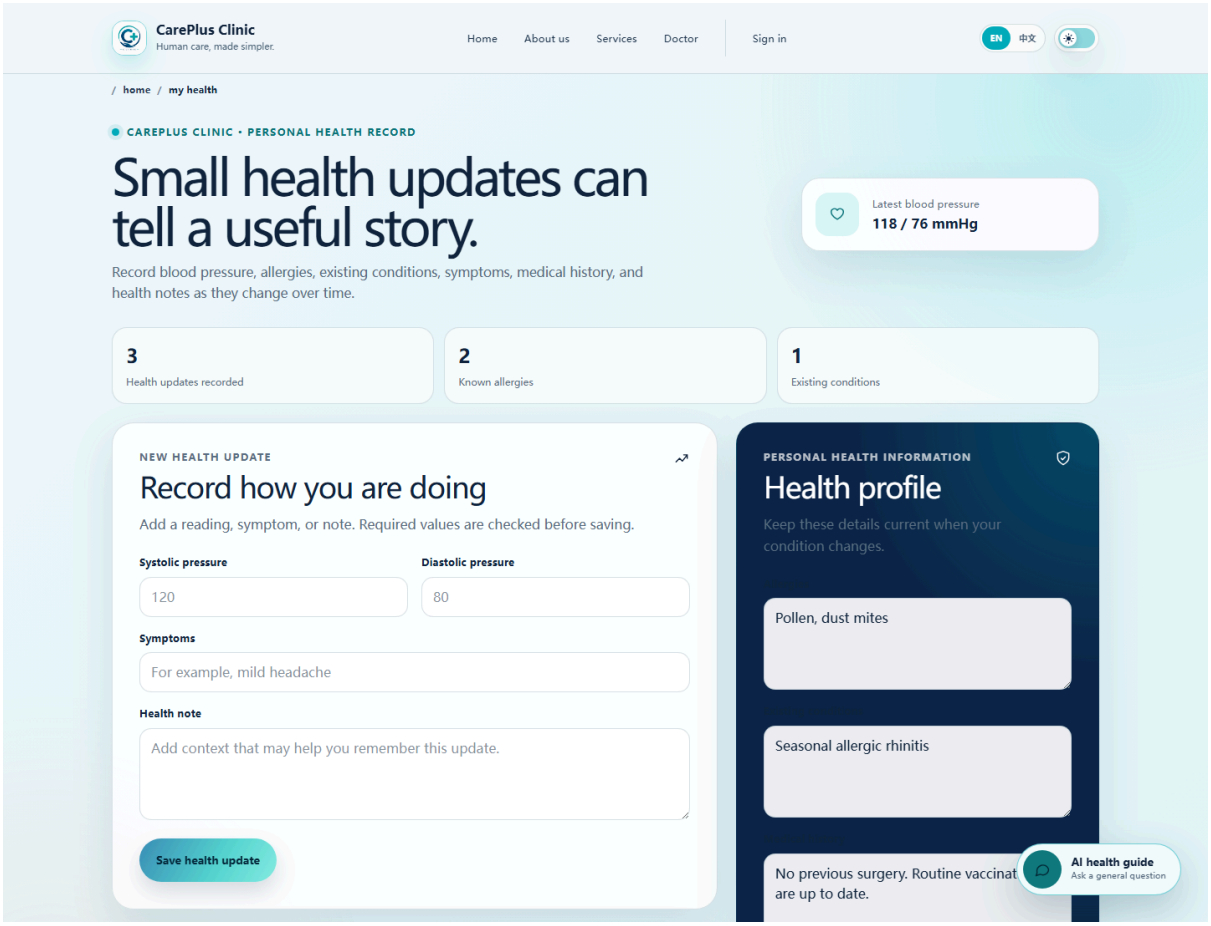


Figure 4.7.5: Personal health record page

The page in Figure 4.7.5 shows the personal health record interface. The patient can record a new health update by entering systolic and diastolic blood pressure, symptoms, and a supporting note. Summary cards show the number of recorded updates, known allergies, and existing conditions. The health profile panel stores longer-term information such as allergies, existing conditions, and medical history. The save action validates the required values and gives the user feedback after the record is updated. A privacy notice reinforces that the information is personal health data.

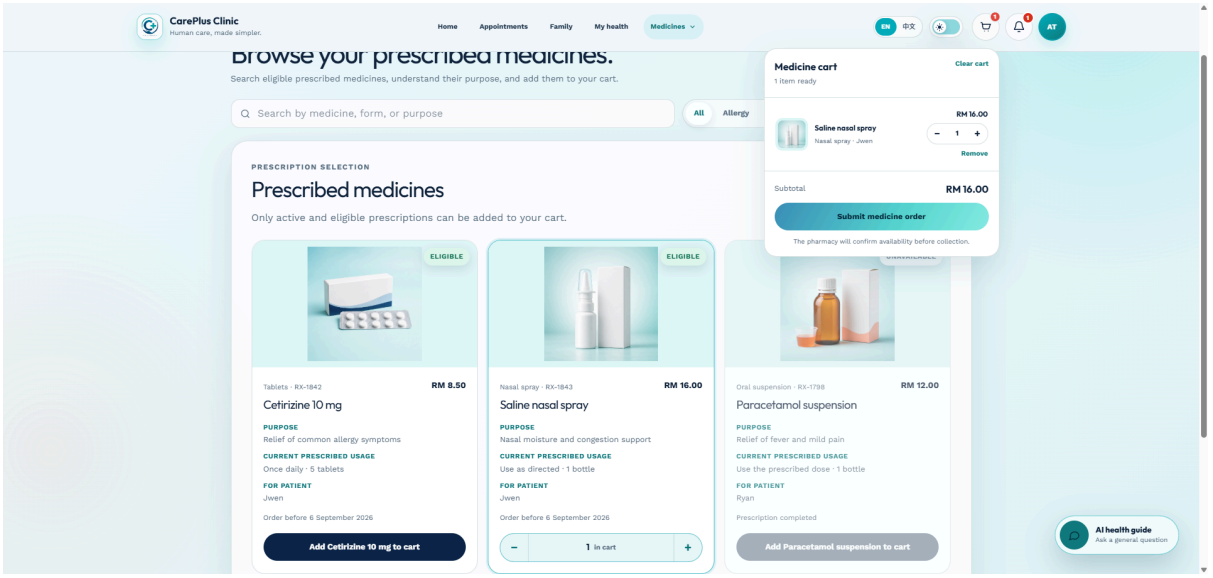


Figure 4.7.6: Prescribed medicine catalogue

The page in Figure 4.7.6 supports the medicine purchase use case. It provides a search field, category filters, prescription eligibility labels, medicine cards, usage information, the patient name, price, and ordering deadline. Eligible prescriptions can be added to the cart, while unavailable or completed prescriptions are clearly disabled. This prevents the user from selecting medicine that is not currently available for purchase and makes the prescription status visible before checkout.

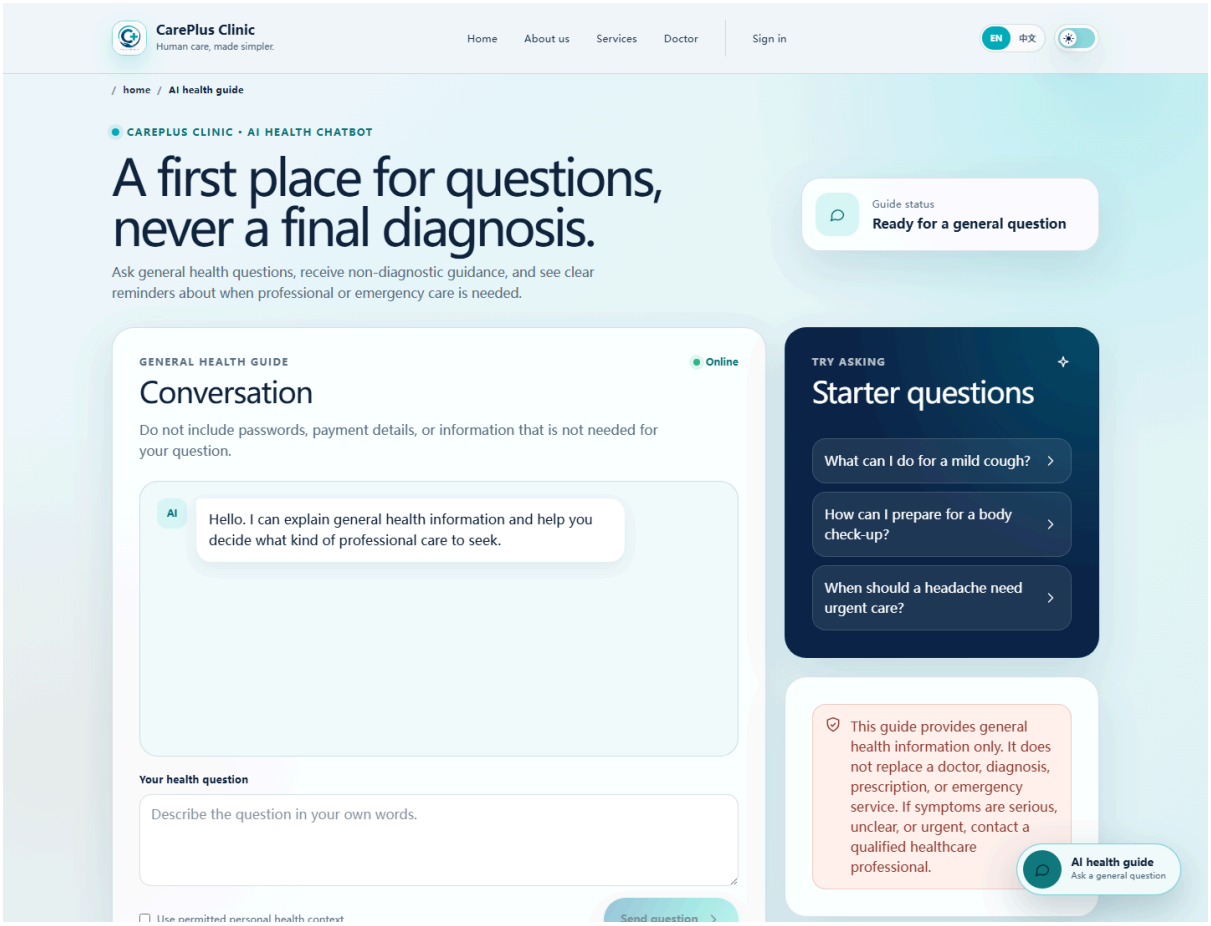


Figure 4.7.7: AI health chatbot page

The page in Figure 4.7.7 shows the AI health chatbot interface. The conversation panel displays the assistant response, while the starter-question panel helps users begin with common health topics. The patient can enter a question and optionally allow permitted personal health context to be used. A prominent disclaimer explains that the guide provides general health information only and does not replace a doctor, diagnosis, prescription, or emergency service. The interface therefore combines convenience with a clear safety boundary for AI-supported health guidance.

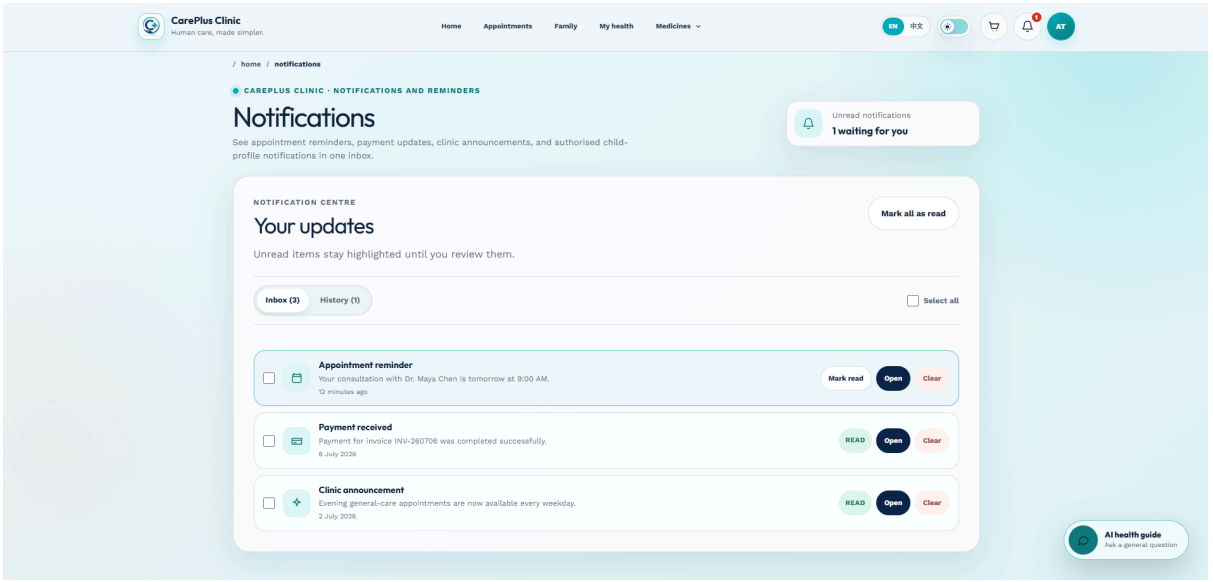


Figure 4.7.8: Notifications and Reminders Page

The page in Figure 4.7.8 shows the Notifications and Reminders page of the CarePlus Clinic Management System. It provides a central inbox for appointment reminders, payment updates, clinic announcements, and authorised child-profile notifications. Unread notifications are highlighted so that the patient can identify important updates easily. The patient can open a notification to view the related information, mark it as read, clear individual notifications, or use the "Mark all as read" function. The Inbox and History tabs separate current notifications from previously reviewed updates, while the notification count in the header indicates the number of unread items.

4.8 Chapter Summary and Evaluation

This chapter presented the design of the CarePlus Clinic Management System through use case, sequence, collaboration, class and entity relationship diagrams. These diagrams explained how patients and guardians interact with the system to manage accounts, appointments, billing and payments, notifications, personal health records, medicine purchases, AI health guidance, and family member information. The user interface prototypes also demonstrated the expected appearance and workflow of the main patient portal pages.

Overall, the proposed design meets the main patient-side requirements by providing centralised healthcare management and easy access to clinic services. It improves usability, organisation, privacy, and communication between patients and the clinic. However, several functions and external integrations, such as online payment, notifications, database connectivity, and AI chatbot services, still require further implementation and testing. The completed design provides a clear foundation for the next development phase.

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